BISITE: Fine Localizations, Smooth and Proper Maps, and the Six-Operations Shape of a Derivator

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No priority is claimed. Status labels distinguish proofs given here, results checked in the literature, refuted assertions, and precisely delimited unresolved questions.

Conventions

We fix Grothendieck universes $\mathbb{U} \in \mathbb{V}$ and take Cat to mean the category of $\mathbb{U}$-small categories. Localizations and categories of coefficients are formed in $\mathbb{V}$. We write $BA = |NA|$, let W_∞ be the functors whose nerves are weak homotopy equivalences, and put $\mathrm{Hot} = \mathrm{Cat}[W_\infty^{-1}]$. A fundamental localizer W contains identities, satisfies two-out-of-three, and satisfies the following weak-saturation clause: if $i : A \to B$, $r : B \to A$, $ri = \mathrm{id}_A$ and $ir \in W$, then $i, r \in W$. It sends every category with a terminal object to the point and satisfies relative Theorem A: for a strictly commuting triangle $A \xrightarrow{u} B \to C$, if $A/c \to B/c$ belongs to W for every $c \in C$, then $u \in W$. We explicitly use the relative form, printed Loc (3 bis) in Chapter XVI; A.1 compares this modern convention with the literal historical variants. A category A is W-aspherical if $A \to e$ belongs to W.

For $u : A \to B$, A/b has objects $(a, u(a) \to b)$ and $b\backslash A$ has objects $(a, b \to u(a))$. The word final denotes ordinary covariant finality (nonempty connected comma categories $b\backslash A$); homotopy finality requires their weak contractibility. Coefficients on S have presheaf variance: $\mathbb{D}_{\mathcal{V}}(S) = \mathrm{HoFun}(NS^{\mathrm{op}}, \mathcal{V})$. Accordingly $f_!$ and f_* mean the left and right adjoints of restriction f^*. A sieve is closed under sources of arrows, and a cosieve under targets. We distinguish categorical equivalence, natural-transformation homotopy, weak homotopy equivalence of nerves, and equivalence of enhanced localizations.

Unless explicitly qualified, pullbacks in Cat and in relative categories are strict pullbacks. Infinity-categorical pullbacks are stated as such. A statement about raw pullback is never inferred solely from an equivalence between localizations. Every numbered statement bears one of the labels Proved, Refuted, Known, or Open. The latter records a limitation of the present audit and does not assert that the problem is open in the literature.

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1 The proposal and its logical premises

1.1 The three relative classes and the two meanings of fibration

We read the supplied transcriptions of *Les Dérivateurs* against the supplied edition PDFs wherever those PDFs are present (Grothendieck, 1991). There is no edition PDF for VI-b or XIIc-a in the supplied archive; we additionally opened the editors' public XIIc PDF and checked ms. 207–208, 217–218 and 222–224, including images of the crucial pages. A correction at a location for which only a transcription was checked is justified by typing or by the mathematical argument, rather than claimed as an independent palaeographical verification. All categories are small in the fixed universe, except that a category of models may be large in a second universe; its slice localizations are assumed to exist in that second universe. No local-smallness conclusion follows from two-out-of-three alone.

A.1. [Proved] The historical and adopted localizer axioms are different formulations. The literal Loc (1) of XVI ms. 1 consists of isomorphisms, two-out-of-three, and the condition that $gf, fg \in W$ imply $f, g \in W$. Grothendieck calls the first two conditions *saturation faible*, and all three *saturation modérée*. Retract closure is added in Loc (1 bis); strong saturation is Loc (1 ter). The literal Loc (3), XVI ms. 2, is absolute Theorem A. Relative Theorem A is Loc (3 bis), ms. 3.

In this paper, a fundamental localizer means the modern relative package specified in the conventions: weak saturation, the terminal-object axiom, and the assertion that a functor $u : A \to B$ over C lies in W whenever $u/c : A/c \to B/c$ lies in W for every $c \in C$. Consequently no argument using relative Theorem A is an argument from the literal absolute Loc (3) alone.

Proof. These are the displayed clauses and terminology of XVI ms. 1–3, checked in both the transcription and the edition PDF. In the relative axiom take $B = C$ and its structure map to be the identity. Since C/c has terminal object 1_c, the terminal-object axiom and two-out-of-three identify its premise with $A/c \to e \in W$. Thus relative Theorem A implies the literal absolute form. The reverse implication is not a formal substitution. □

A.2. [Proved] Intrinsic fibrations and their unconditional closure properties. Let $\mathcal{M}$ have pullbacks and let W be invariant under isomorphism of arrows. For $S \in \mathcal{M}$, let W_S be the inverse image of W under the source functor $\mathcal{M}/S \to \mathcal{M}$. Define

$$\mathrm{Fib}_W = \{p : E \to B \mid p^* : \mathcal{M}/B \to \mathcal{M}/E \text{ sends } W_B \text{ into } W_E\}.$$

Thus every test is an arrow $v : T' \to T$ over B, with arbitrary structure map $T \to B$, and the tested arrow is $E \times_B T' \to E \times_B T$. This class contains isomorphisms and is closed under composition and arbitrary pullback. These assertions require no two-out-of-three axiom. If W is closed under retracts, then so is Fib_W.

Proof. For $E \xrightarrow{p} B \xrightarrow{q} C$, the pullback of an arrow over C along qp is canonically isomorphic to its successive pullbacks along q and p. Both preservation hypotheses apply in that order. For a pullback $p' : E \times_B B' \to B'$, forget a test arrow over B' to an arrow over B and use the canonical associativity isomorphism of pullbacks. An isomorphism induces an equivalence of slices whose underlying arrows are isomorphic to the originals. Finally, if p is a retract of q in the arrow category, the pullback of any fixed test arrow along p is a retract of the corresponding pullback along q, after pulling the latter over the retraction maps of bases. The two maps exhibiting this retract are induced by the maps in the arrow-category retract diagram. Retract closure of W now applies. The additional hypothesis cannot be suppressed merely because XII ms. 208 lists direct factors in a parenthesis: that paragraph has explicitly assumed only isomorphisms and two-out-of-three. □

A.3. [Proved] The coarse, universal, and local relative classes. Suppose W contains isomorphisms and satisfies two-out-of-three. Let $\mathcal{L}$ contain isomorphisms and be closed under composition and pullback. For an arrow $u : X \to Y$ in $\mathcal{M}/S$, put

$$u \in W_S^{\mathcal{L}} \quad \Longleftrightarrow \quad \big(X \times_S T \longrightarrow Y \times_S T\big) \in W \quad \text{for every } (T \to S) \in \mathcal{L}.$$

Let W_S^u be the same class with all $T \to S$ allowed. Then

$$W_S^u \subseteq W_S^{\mathcal{L}} \subseteq W_S.$$

All three classes contain isomorphisms and satisfy two-out-of-three. Their retract closure follows if W has it. These are precisely the relative classes in XII ms. 210 and 220. They must not be confused with the class of universal W-equivalences in $\mathcal{M}$, where the base is the codomain of the arrow being tested.

Proof. Each test is a functor on the slice. The asserted class is the intersection of the inverse images of W under those functors; identities, two-out-of-three, and retracts are preserved by this operation. The identity $S \to S$ is a permitted local test, which proves the second inclusion; the first follows by reducing the family of tests. In particular no descent argument can replace the two-out-of-three premise. A class defined only by full faithfulness of a restriction functor may fail it, as the set-coefficient example $e \to BC_2 \to e$ does: restriction along $BC_2 \to e$ and along their composite is fully faithful, whereas forgetting the C_2-action along $e \to BC_2$ is not full. □

A.4. [Proved] What raw reindexing and postcomposition actually preserve. Under the hypotheses of A.3, a map $f : S' \to S$ in $\mathcal{L}$ induces

$$f^* : (\mathcal{M}/S, W_S^{\mathcal{L}}) \longrightarrow (\mathcal{M}/S', W_{S'}^{\mathcal{L}}).$$

For every $f : S' \to S$, without assuming $f \in \mathcal{L}$, postcomposition induces

$$f_\# : (\mathcal{M}/S', W_{S'}^{\mathcal{L}}) \longrightarrow (\mathcal{M}/S, W_S^{\mathcal{L}}).$$

If f^* also preserves the displayed classes, the ordinary adjunction $f_\# \dashv f^*$ descends to their localizations. For arbitrary f, this calculation alone constructs the descended postcomposition functor; it does not construct its right adjoint.

Proof. For the first assertion, test f^*u along $g : T \to S'$ in $\mathcal{L}$. Its underlying arrow is the pullback of u along $fg \in \mathcal{L}$. For the second, test $f_\#u$ along $h : T \to S$ in $\mathcal{L}$. The projection $S' \times_S T \to S'$ lies in $\mathcal{L}$, and the tested underlying arrow is exactly the pullback of u along this projection. For descent of the adjunction, apply the localization functors to the original unit and counit. Their naturality extends across inverted arrows, and their two triangle identities remain identities. Thus they exhibit the descended adjunction. This also proves the general adjunction lemma at XII ms. 248 and settles the formal postcomposition observation at XII ms. 226–227 with its exact scope. □

A.5. [Proved] The universal saturation of the local test class. Under A.3, define $\widehat{\mathcal{L}}$ by the following condition on $f : S' \to S$: for every $T \to S$, pullback along $f_T : T \times_S S' \to T$ sends $W_T^{\mathcal{L}}$ into $W_{T \times_S S'}^{\mathcal{L}}$. Then $\widehat{\mathcal{L}}$ contains $\mathcal{L}$, contains isomorphisms, and is closed under composition and pullback. Moreover

$$W_S^{\widehat{\mathcal{L}}} = W_S^{\mathcal{L}} \quad \text{for every } S, \qquad \widehat{\widehat{\mathcal{L}}} = \widehat{\mathcal{L}}.$$

It is the largest composition- and pullback-closed class containing isomorphisms that defines this same family of fine relative classes. This is the corrected, second definition at XII ms. 222; the first definition on that page omits the universal quantifier over $T \to S$.

Proof. A.4 proves $\mathcal{L} \subseteq \widehat{\mathcal{L}}$, also after any preliminary pullback. Iterated pullback proves closure under composition; absorbing a preliminary pullback into $T \to S$ proves pullback closure. If $u \in W_S^{\mathcal{L}}$ and $f \in \widehat{\mathcal{L}}$ has target S, its pullback lies in $W_{S'}^{\mathcal{L}}$, hence in $W_{S'}$, which proves $W_S^{\mathcal{L}} \subseteq W_S^{\widehat{\mathcal{L}}}$. The reverse inclusion follows from $\mathcal{L} \subseteq \widehat{\mathcal{L}}$. The defining condition for a hat depends only on this family, proving idempotence. If $\mathcal{K}$ has the stated closure properties and defines the same family, apply A.4 to $\mathcal{K}$ after any preliminary pullback: every member of $\mathcal{K}$ then satisfies the defining condition of $\widehat{\mathcal{L}}$. □

1.2 Transporters, stabilizers, and the cancellation in Chapter II

A.6. [Proved] The corrected transporter Galois connection. Let W contain isomorphisms and be closed under composition in a category $\mathcal{M}$ with pullbacks. Let $A \subseteq W$ contain isomorphisms and be composition-closed, and let L contain isomorphisms and be pullback-closed. Define

$$T_W(A) = \{p : E \to B \mid p^* \text{ sends every arrow of } A \text{ in } \mathcal{M}/B \text{ into } W\},$$
$$K_W(L) = \{u : X \to Y \in W \mid p^*u \in W \text{ for every } (p : Y' \to Y) \in L\}.$$

Then $T_W(A)$ contains isomorphisms and is pullback-closed, $K_W(L)$ contains isomorphisms and is composition-closed, and

$$A \subseteq K_W(L) \quad \Longleftrightarrow \quad L \subseteq T_W(A).$$

Both operations reverse inclusions; their composites are extensive, monotone, and idempotent. The closed pairs are exchanged by inverse order-reversing bijections. Their extreme pairs are

$$(W^u, \operatorname{Mor}\mathcal{M}), \qquad (W, \operatorname{Fib}_W),$$

where W^u means universal equivalences with respect to the codomain of each arrow.

Proof. Pullback stability of $T_W(A)$ uses the quantifier over arrows in a slice: a test over B' may be regarded as a test over B. To show composition stability of $K_W(L)$, pull back a composite vu along $p \in L$. The resulting composite consists of the pullback of v along p and the pullback of u along the pullback of p to the source of v. Both belong to W.

If $L \subseteq T_W(A)$, apply its defining preservation condition to an arrow of A viewed over its own codomain. Conversely, suppose $A \subseteq K_W(L)$, and let $u : X \to Y$ in A have a further structure map $Y \to B$. For $p : E \to B$ in L, the pullback $E \times_B Y \to Y$ lies in L; the kernel test for u therefore gives $p^*u \in W$. This proves the adjunction. Antitonicity is immediate from enlarging the test families. The familiar threefold identities follow directly: extensivity gives $T_W K_W T_W(A) \supseteq T_W(A)$, while applying antitonicity to $A \subseteq K_W T_W(A)$ gives the reverse inclusion. The kernel identity is identical with the roles reversed. These identities imply idempotence. Finally $T_W(\text{Iso})$ is all arrows, $K_W(\operatorname{Mor}\mathcal{M}) = W^u$, $K_W(\text{Iso}) = W$, and $T_W(W) = \operatorname{Fib}_W$. These arguments verify the corrected domains of II (6.7), ms. 86; composition must not be imposed on the right-hand domain in advance. □

A.7. [Proved] The exact composition criterion, II Proposition 6.1. For a closed pair $A = K_W(L)$, $L = T_W(A)$ of A.6, the following conditions are equivalent: L is composition-closed; pullback along every member of L preserves A; and $L = \operatorname{Stab}(A)$, where $\operatorname{Stab}(A) = T_A(A)$. Every stabilizer is composition- and pullback-closed. In particular, a transporter into the larger class W is not generally a stabilizer.

Proof. If L composes, pull an arrow $u \in K_W(L)$ back along $p \in L$, and test the result along $q \in L$. The resulting arrow is a test of u along the composite of q with the appropriate pullback of p, which is again in L. Thus $p^*u \in A$. Conversely, if L preserves A, pulling an A-map successively along two members of L again gives an A-map and hence a W-map. The composite therefore lies in $T_W(A) = L$. Preservation gives $L \subseteq \mathrm{Stab}(A)$; the other inclusion follows from $A \subseteq W$. Finally stabilization under pullback and composition is the same direct argument as A.2 with A in place of W. □

A.8. [Refuted] Transporters need not compose under the hypotheses of II section 6. The claim in the first version of II (6.2)–(6.5) that $T_W(A)$ automatically belongs to a right-hand domain requiring composition closure is false. A counterexample is the chain category $0 < 1 < 2 < 3$, with W consisting of the identities and $0 \to 2, 0 \to 3$, and A consisting of the identities and $0 \to 3$. Here

$$T_W(A) = \mathrm{Mor}([3]) \setminus \{1 \to 3\}, \qquad K_W T_W(A) = A.$$

Consequently $1 \to 2$ and $2 \to 3$ lie in the transporter while their composite does not.

Proof. There is exactly one arrow $i \to j$ when $i \leq j$, and composition is forced by the order. Pullbacks are minima. Both displayed classes contain all isomorphisms and are closed under composition. The only nonidentity test in A is $0 \to 3$, and it can be regarded as an arrow over a base b only for $b = 3$. Pulling it back along $i \to 3$ gives $0 \to i$, which lies in W exactly for $i = 0, 2, 3$. This proves the transporter formula including its slice quantifiers. Of the two nonidentity arrows in W, the arrow $0 \to 2$ fails the kernel test along $1 \to 2$, since its pullback is $0 \to 1 \notin W$. The other survives by the Galois relation. The finite checker `code/a_finite_audit.py` verifies the category, all pullbacks, both computed classes, and the failed composite. We do not claim that this W satisfies two-out-of-three; that axiom is not among the hypotheses of II section 6. □

A.9. [Proved] The meaning of smooth and proper in the manuscript. For a fundamental localizer on Cat, let As_W consist of $u : A \to B$ for which $A/b \to e$ lies in W for every b, and let Coas_W consist of u for which $b\backslash A \to e$ lies in W for every b. The definitions used at II ms. 66 and explicitly reaffirmed at ms. 89 are

$$\mathrm{Liss}_W = \mathrm{Stab}(\mathrm{As}_W), \qquad \mathrm{Prop}_W = \mathrm{Stab}(\mathrm{Coas}_W).$$

The two mixed transporters contain both the corresponding stabilizer and Fib_W. Neither is the definition of smoothness or properness. The proposed inclusion $\mathrm{Liss}_W \cap \mathrm{Prop}_W \subseteq \mathrm{Fib}_W$ at XII ms. 225–226 is explicitly qualified by "peut-être" and is an extra proposed axiom, not a consequence asserted to have been proved there.

Proof. The two definitions are written in II (4.10) and in the paragraph introducing stabilizers after Proposition 6.1. Absolute Theorem A gives $\mathrm{As}_W \subseteq W$; opposition gives the analogous containment for Coas_W. Therefore an As_W-preserving functor sends such maps into W, and a W-preserving functor does so as well. These are the two claimed containments; the coaspherical argument is identical. XII ms. 225 introduces two arbitrary classes, asks whether to assume these closures and the parenthesized inclusion, and then separately proposes the fine-localization axioms. This wording does not identify the inclusion with an earlier theorem. The finite lifting criteria and the actual validity range of the inclusion are treated in Part B. □

1.3 The fibre axioms and the later implication scheme

A.10. [Proved] The three fibre axioms, with their quantifiers. For every $S \in \mathcal{M}$ let $\mathcal{F}_S$ be the full subcategory of $\mathcal{M}/S$ on the intrinsic fibrations. The three assertions at XII ms. 215–218 are

(a) $J_S : \mathcal{F}_S[(W_S^u)^{-1}] \longrightarrow (\mathcal{M}/S)[W_S^{-1}]$ is an equivalence;
(b) $W_S^u = W_S$ on $\mathcal{F}_S$;
(c) $\mathcal{F}_S[W_S^{-1}] \longrightarrow \mathcal{H}[W_S^{-1}] \longrightarrow (\mathcal{M}/S)[W_S^{-1}]$ are equivalences

for every full $\mathcal{F}_S \subseteq \mathcal{H} \subseteq \mathcal{M}/S$. Axiom (a) implies equality of the strong saturations of W_S^u and W_S on $\mathcal{F}_S$. It does not by that argument imply equality of the original classes. Axioms (b) and (c) imply (a).

Proof. The displayed formulations are exactly the three source formulas, with all localizations and restrictions typed. If $u \in W_S$ has source and target in $\mathcal{F}_S$, then J_S sends its image to an isomorphism. An equivalence of categories reflects isomorphisms, so u is inverted by localization at W_S^u. The reverse containment follows from A.3. This is a statement about strong saturation, not about membership in W_S^u. For the last assertion apply (c) to either endpoint and use (b) to identify the source localization with the source of J_S. □

A.11. [Proved] Enough intrinsic fibrations implies the exact fibrewise invariance. Suppose $\mathcal{M}$ has pullbacks, W contains isomorphisms and satisfies two-out-of-three, and every map factors as a W-map followed by a member of Fib_W. Then axiom (b) of A.10 holds, and

$$\mathrm{Fib}_W \cap W = W^u.$$

No terminal object, accessibility, functorial factorization, or strong saturation is required for these conclusions.

Proof. Let $u : X \to Y$ be a W-map over S with $X \to S$ and $Y \to S$ intrinsic fibrations. Factor an arbitrary base change as $T \xrightarrow{i} T_0 \xrightarrow{p} S$, with $i \in W$ and $p \in \mathrm{Fib}_W$. The pullback $u_0 : X \times_S T_0 \to Y \times_S T_0$ lies in W because p is intrinsic. The two vertical comparison maps from pullback over T to pullback over T_0 lie in W, because the pullbacks of $X \to S$ and $Y \to S$ are intrinsic and preserve i. The resulting commutative square and two-out-of-three imply $u_T \in W$. This proves (b).

For $q : X \to S$ in $\mathrm{Fib}_W \cap W$, apply (b) to the arrow from q to 1_S, both fibrant objects of the slice; thus $q \in W^u$. Conversely suppose $q \in W^u$ and let $v : T' \to T$ be a W-map over S. The vertical maps in

$$\begin{array}{ccc} X \times_S T' & \longrightarrow & X \times_S T \\ \downarrow & & \downarrow \\ T' & \xrightarrow{v} & T \end{array}$$

are in W by universal equivalence. Two-out-of-three implies that the upper arrow lies in W. Hence $q \in \mathrm{Fib}_W$, and the identity base change gives $q \in W$. This verifies the formal fibration half of XIII Proposition 4, ms. 10–11, and the argument used in XVI 4.4.1. □

A.12. [Proved] A functorial replacement proves every intermediate-subcategory comparison. Let $\mathcal{A}$ be a category with a class W containing isomorphisms and satisfying two-out-of-three, let $\mathcal{F} \subseteq \mathcal{A}$ be full, and suppose there is $R : \mathcal{A} \to \mathcal{F}$ with a natural transformation

$\eta : 1_{\mathcal{A}} \to iR$ whose components are in W. For every full $\mathcal{F} \subseteq \mathcal{H} \subseteq \mathcal{A}$, both inclusions induce equivalences

$$\mathcal{F}[W^{-1}] \longrightarrow \mathcal{H}[W^{-1}] \longrightarrow \mathcal{A}[W^{-1}].$$

In particular, such slice replacements into $\mathrm{Fib}_W S$, together with enough intrinsic fibrations, verify XII (a), (b), and (c).

Proof. For $u : X \to Y$ in W, the equation $R(u)\eta_X = \eta_Y u$ implies $R(u) \in W$ by two-out-of-three. Thus R is a relative functor. Its restriction to any $\mathcal{H}$ has image in $\mathcal{F} \subseteq \mathcal{H}$, and each component of η at an object of $\mathcal{H}$ is a morphism of $\mathcal{H}$ because that subcategory is full. After localization, these components give a natural isomorphism between the identity and inclusion followed by R. On $\mathcal{F}$ they give the other natural isomorphism. The same proof works with $\mathcal{A}$ in place of $\mathcal{H}$. Finally A.11 identifies the two classes on fibrant objects, so the claimed source in (a) is the localization used here. □

A.13. [Refuted] The implication asked in audit row XIIc B062. The axioms (a) and (b) of A.10, even imposed at every base, do not imply (c). There is a counterexample with six objects, all finite limits and colimits, and strongly saturated W. It also refutes the analogous assertion for enhanced localizations.

Proof. The complete construction and verification are E.12. Its poset has Hasse covers $0 < 1$, $0 < 2$, $1 < 3$, $2 < 3$, $1 < 4$, $3 < 5$, $4 < 5$, and W consists of identities together with $1 \to 3$, $1 \to 4$, $2 \to 3$. At base 5, the intrinsic fibrations form the chain $\{0 < 3 < 5\}$. Their inclusion into the whole slice is a localization equivalence, certified by explicit relative inverse functors and weak natural transformations. The full intermediate category $\{0, 3, 4, 5\}$ is a diamond with only identity weak arrows, so its localization cannot be equivalent to that chain under the inclusion. E.12 checks (a) and (b) at all other bases and proves strong saturation and the enhanced version. The independent finite checker is `code/a_b062_verify.py`. □

The failure is the missing intermediate-subcategory control: the weak natural zigzag proving (a) passes through an object omitted from the chosen full intermediate subcategory. A.12 states sufficient comparison data that prevent this failure. The example has a terminal object but does not require, or satisfy, the extra condition that every terminal map be an intrinsic fibration; that condition is not part of the setup at XII ms. 207 or the question at ms. 218.

A.14. [Proved] The two fine axiom systems in XII ms.225–227. Suppose classes $L, P \subseteq \mathrm{Mor}\,\mathcal{M}$ contain isomorphisms and are closed under composition and pullback. Define $W_S^g = W_S^L$ and $W_S^d = W_S^P$. The clause called (a) on ms. 226, asserting preservation of W^g by L-base change and of W^d by P-base change, follows formally from the closure hypotheses. The clause called (b) on that page is the additional pair of assertions

$$(L/S)[(W_S^u)^{-1}] \simeq (\mathcal{M}/S)[(W_S^d)^{-1}], \quad W_S^u = W_S^d \text{ on } L/S,$$
$$(P/S)[(W_S^u)^{-1}] \simeq (\mathcal{M}/S)[(W_S^g)^{-1}], \quad W_S^u = W_S^g \text{ on } P/S.$$

Here L/S and P/S denote full subcategories on the corresponding structural maps. Under these additional equivalences, each fine theory is a contravariant pseudofunctor on all of $\mathcal{M}$. This conclusion does not require $L \cap P \subseteq \mathrm{Fib}_W$.

Proof. The first assertion is A.4. Pullback preserves the objects of L/S and P/S by the stipulated closure, and preserves W_S^u by its universal quantifier. These relative slice categories therefore form a pseudofunctor, with the canonical associativity and unit isomorphisms of pullback; those isomorphisms satisfy their usual coherence identities before localization and

hence afterwards. Transporting this pseudofunctor across the displayed equivalences gives the asserted fine pseudofunctors. Neither this construction nor its coherence uses the intersection condition. The source calls a further pair of adjunction properties clause (c), refers to VI ms. 108–113, and asks whether its hypotheses can be simplified. This (c) is a different assertion from the intermediate-subcategory (c) at ms. 218. □

A.15. [Proved] The four fibration classes and Loc (7), with reference localizer. For a fundamental localizer W, the four classes in XVI section 4 are distinct definitions. Besides intrinsic Fib_W, one has Grothendieck fibrations; $\mathrm{Fib}_W^{\mathrm{weak}}$, tested by all arrows $T' \to T$ over the base with T' and T contractible in the category-homotopy sense; and $\mathrm{Parf}_W = \mathrm{Liss}_W \cap \mathrm{Prop}_W$. Here category-homotopy contractibility means that the identity functor is connected to a constant functor by a finite zigzag of natural transformations. It implies weak contractibility of the nerve, but replacing the test family by all weakly contractible nerves is a separate assertion, not the literal definition. Let $\mathcal{G}_W$ denote Grothendieck fibrations all of whose transport functors lie in W. For any arrow class $\mathcal{F}$, define

$$\begin{aligned} \mathrm{A}(\mathcal{F}):&\quad \mathcal{F} \subseteq \mathrm{Fib}_W,\\ \mathrm{B}(\mathcal{F}):&\quad W_S \cap \mathrm{Mor}(\mathcal{F}/S) \subseteq W_S^u \quad \text{for every } S,\\ \mathrm{C}(\mathcal{F}):&\quad \mathcal{F} \cap W \subseteq W^u. \end{aligned}$$

Loc (7a), (7b), and (7c) are respectively $\mathrm{A}(\mathcal{G}_W)$, $\mathrm{B}(\mathcal{G}_W)$, and $\mathrm{C}(\mathcal{G}_W)$. For a reference localizer $V \subseteq W$, the conditions $\mathrm{A}_V, \mathrm{B}_V, \mathrm{C}_V$ replace $\mathcal{G}_W$ by $\mathcal{G}_V$. The condition D_V is $\mathrm{A}(\mathrm{Parf}_V)$, or its equivalent B and C forms once the hypotheses of A.17 are verified. Loc (7F) is the factorization condition of A.11, not the inclusion of any one of these reference classes in Fib_W.

Proof. These are the formulas of XVI 4.1–4.4.1, ms. 27–41. The weak condition tests arbitrary maps between category-contractible bases, not every W-map between arbitrary bases. A natural transformation realizes to a homotopy, so a zigzag from the identity to a constant does make the nerve weakly contractible. No converse is used. The class $\mathcal{G}_W$ adds both the cartesian lifting property and the assertion that each induced transport lies in W; neither assertion is part of the definition of intrinsic fibrations. The printed use of "cofibrations" in the (7F) paragraph is inconsistent with the factorization and with the displayed class Fib_W and must read "fibrations"; the edition PDF contains the same word. Thus the separation is mathematical, not merely typographical. □

A.16. [Proved] The implications A to B to C and replacement transfer. Let W and $\mathcal{M}$ satisfy A.11's initial assumptions. If every map factors as a W-map followed by a member of $\mathcal{F}$, then $\mathrm{A}(\mathcal{F})$ implies $\mathrm{B}(\mathcal{F})$. If identities lie in $\mathcal{F}$, then $\mathrm{B}(\mathcal{F})$ implies $\mathrm{C}(\mathcal{F})$.

Suppose that each $p : X \to S$ in $\mathcal{F}$ admits a comparison $j_p : X \to R_p$ over S in W_S^u, with $R_p \to S$ in $\mathcal{G}$. Then $\mathrm{A}(\mathcal{G}) \Rightarrow \mathrm{A}(\mathcal{F})$ and $\mathrm{C}(\mathcal{G}) \Rightarrow \mathrm{C}(\mathcal{F})$. If for every map $h : X \to Y$ over S between $\mathcal{F}$-objects the chosen comparisons extend to a commutative square $R(h)j_X = j_Y h$, then also $\mathrm{B}(\mathcal{G}) \Rightarrow \mathrm{B}(\mathcal{F})$. Functorial replacements suffice; the proof only needs these individual commuting squares.

Proof. The first two assertions follow from A.11 and from taking the target object in the slice to be 1_S. For A-transfer, compare the pullbacks of a W-arrow $T' \to T$ over S along X and along R_p. Both vertical comparison maps are in W by the universal hypothesis on j_p, and the lower horizontal map is in W by $\mathrm{A}(\mathcal{G})$; two-out-of-three gives the upper one. For C-transfer, if $p \in W$, then $R_p \to S$ lies in W by two-out-of-three. It and j_p remain in W after any base change, so their composite does. For B-transfer, if $h \in W$, the commuting square and the two comparisons show $R(h) \in W$. Applying $\mathrm{B}(\mathcal{G})$ makes $R(h)$ universally in W, and the same

square after arbitrary pullback proves the assertion for h. These are the exact content of XVI Lemmas 4.4.1 and 4.4.9 and Corollary 4.4.10, with the compatibility premise stated rather than suppressed. □

A.17. [Proved] The full Brown implication with all its premises. Suppose $\mathcal{M}$ has a terminal object and pullbacks, and W contains isomorphisms and satisfies two-out-of-three. Let $\mathcal{F}$ contain isomorphisms and all terminal maps, be closed under composition and pullback, and admit a factorization of every arrow as a W-map followed by an $\mathcal{F}$-map. Then

$$\mathrm{A}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathrm{B}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathrm{C}(\mathcal{F}).$$

This is the precise abstract assertion of XVI Lemma 4.4.3. The terminal-map premise is part of the statement, not a consequence of an arbitrary intrinsic-fibration factorization.

Proof. Only C ⇒ A remains after A.16. Assume C. Then pullbacks of $\mathcal{F} \cap W$ are again in $\mathcal{F} \cap W$. First observe the elementary functor lemma. If a functor H from a category satisfying the stated factorization and terminal-map conditions sends $\mathcal{F} \cap W$ into a two-out-of-three class V, it sends W into V. For $u : X \to Y$ in W, factor $(1_X, u) : X \to X \times Y$ as $w : X \to Z$ in W followed by $q : Z \to X \times Y$ in $\mathcal{F}$. Both projections $q_X : Z \to X$ and $q_Y : Z \to Y$ lie in $\mathcal{F}$, since projections are pullbacks of terminal maps. The equations $q_X w = 1_X$ and $q_Y w = u$ show that both lie in W. Hence $H(q_X), H(q_Y) \in V$, and the same equations imply $H(w), H(u) \in V$.

For any fixed base A, the full category of $\mathcal{F}$-objects over A satisfies these hypotheses: products are fibre products over A, and an absolute factorization is a relative one because $\mathcal{F}$ composes. Pullback along any $A' \to A$ sends its trivial $\mathcal{F}$-maps into W by C. The functor lemma therefore proves preservation of every W-map between these fibrant objects.

Now let $p : E \to A$ lie in $\mathcal{F}$ and $u : A' \to A$ in W. The factorization just constructed for u writes it as $u = bw$, where $a : Z \to A'$ and $b : Z \to A$ lie in $\mathcal{F} \cap W$, and $aw = 1_{A'}$. Pullback of b is in W by C, so it suffices to handle a base map $w : A' \to A$ admitting a retraction $r : A \to A'$ in $\mathcal{F} \cap W$.

In that case put $E_1 = A \times_{A'} E$, using $r : A \to A'$ and $rp : E \to A'$. Let $l : E_1 \to A$ and $k : E_1 \to E$ be the projections. The map l lies in $\mathcal{F}$, and k lies in $\mathcal{F} \cap W$. There are maps

$$f = (p, 1_E) : E \to E_1, \qquad g = (wrp, 1_E) : E \to E_1.$$

They are defined because $rw = 1_{A'}$; both retract k, so both lie in W. The map f is a map over A between the $\mathcal{F}$-objects p and l. Hence $w^* f \in W$ by the relative functor lemma. The canonical identification $A' \times_A E_1 \cong E$ identifies its comparison into E_1 with g. If $q : A' \times_A E \to E$ is the desired pullback of w, the pullback square gives $fq = g(w^* f)$. The right side and f lie in W, and hence $q \in W$. Applying the argument after a preliminary pullback of p proves the full slice preservation condition $p \in \mathrm{Fib}_W$. This is the mapping-path proof of Brown's lemma, with its relative step made explicit; compare Brown's section 4, Lemmas 1–2 (Brown, 1973). □

A.18. [Refuted] The weak constant-coefficient notion does not compose. The universal weak rational cohomological fibrations of I ms. 117–118 need not be closed under composition. The failure occurs for finite categories. Consequently that notion cannot be silently substituted for the intrinsic class in A.2 or for the class in Loc (7F).

Proof. Let $G = C_2 = \{0, 1\}$ under addition. Define B to have objects a, b, identity only at a, automorphisms G at b, arrows $h_g : a \to b$ for $g \in G$, and no arrows $b \to a$. Composition of automorphisms adds their labels and $k_t h_g = h_{t+g}$. Let $q : B \to [1]$ send a to 0 and b to 1. It is the Grothendieck opfibration obtained from the functor $e \to BG$. Its fibres are e and BG.

For presheaf coefficients, right Kan extension along an opfibration is computed on its strict fibres: the inclusion of the fibre in the over-comma category has a left adjoint given by cocartesian transport. This remains true after any base change. For every complex M of rational vector spaces, the constant-coefficient cohomology of BG is canonically M, since the trivial $\mathbb{Q}[G]$-module $\mathbb{Q}$ is projective, with the augmentation split by $(1 + k_1)/2$. Thus q_* of a constant coefficient is locally constant with the strict-fibre formula, universally under base change.

Define a functor $L : B \to \mathrm{Set}$ by $L(a) = L(b) = G$ and by sending a label g to translation $z \mapsto g + z$. Its category of elements E has objects $(a, z), (b, z)$ for $z \in G$, and arrows $(c, z) \to (d, z')$ given by arrows of B whose assigned translation sends z to z'. Composition is the composition in B. The projection $p : E \to B$ is a discrete fibration and a discrete opfibration, since each translation is bijective. Right direct images of constants are products over its discrete fibres, and every transport permutes the factors, universally. Thus p has the same weak property.

The composite qp has discrete two-object fibre over 0 and the contractible regular action groupoid $G \ltimes G$ over 1. The transport on constant rational cohomology is the diagonal $\mathbb{Q} \to \mathbb{Q}^2$. It is not invertible. Hence the direct image under qp of the constant coefficient is not locally constant. The checker lists all arrows, verifies identities and all associative triples, verifies the covering lifts for p and cocartesian lifts for q, and computes the two fibre component counts. This independently verifies the finite category used in GALOIS D.3; the averaging argument supplies the unbounded-complex assertion that a component count alone would not establish. □

A.19. [Proved] The local-coefficient repair has an asymmetric conclusion. Let $\mathbb{D}$ be a derivator in presheaf variance. Suppose that for $p : E \to B$, after every strict base change, the right direct image of every constant coefficient is locally constant and its value at each base object is canonically the right Kan extension from the strict fibre. If $v : T' \to T$ over B is fully faithful on locally constant $\mathbb{D}$-coefficients, then the pullback $w : E \times_B T' \to E \times_B T$ is a constant-coefficient $\mathbb{D}$-equivalence. This argument does not assert full faithfulness or an equivalence on all locally constant coefficients upstairs.

If the same direct-image and base-change hypothesis is required for every locally constant coefficient on every base-changed total category, then the resulting cohomological-fibration class is closed under composition and pullback. The corresponding statements for left Kan extensions and homology follow by duality.

Proof. Write $p_T : E_T \to T$ and $p_{T'} : E_{T'} \to T'$ for the projections. For a constant coefficient M, put $K = (p_T)_* M_{E_T}$. The canonical mate $v^* K \to (p_{T'})_* M_{E_{T'}}$ is an isomorphism: at each t', the assumed fibre formulas identify it with the identity comparison on the same strict fibre. The object K is locally constant. Full faithfulness of v^* on locally constant objects, applied with source the constant object N_T for every $N \in \mathbb{D}(e)$, gives

$$\mathrm{Hom}_{\mathbb{D}(T)}(N_T, K) \cong \mathrm{Hom}_{\mathbb{D}(T')}(N_{T'}, v^* K).$$

Adjunction to the point and Yoneda identify this with the canonical cohomology comparison; composition of right Kan extensions identifies that comparison with the one induced by w for coefficient M. Only constant M on E_T was available, so this proves exactly the stated conclusion.

Under the stronger hypothesis, start with an arbitrary local system L on E_T. The same calculation applies because $(p_T)_* L$ is now known to be locally constant. If $E \xrightarrow{p} B \xrightarrow{q} S$ are two maps satisfying the stronger hypothesis, then $p_* L$ and $q_* p_* L$ are locally constant, and the composition isomorphism identifies the latter with $(qp)_* L$. Pasting the two universally

invertible base-change mates gives the mate for qp. Pullback stability follows by absorbing an initial base change into the universal quantifier. This verifies the precise repair discussed in I ms. 79–81 and 117–123 without a blanket replacement of the word "equivalence". □

A.20. [Refuted] Properness does not imply local constancy of fibres. The inclusion $j : \{1\} \to [1]$ is proper in Grothendieck's presheaf convention, but its comma transition over $0 \to 1$ is $\varnothing \to e$. Thus the assertion about proper maps at I ms. 102, companion row Ib-C003, is false already for spaces and set coefficients.

Proof. The only arrow issuing from the image object 1 is its identity. Hence j is a Grothendieck opfibration. Its strict-fibre inclusions into the over-comma categories are identities: over 0 both are empty and over 1 both are terminal. The same is true after any base change because the inclusion becomes a cosieve. This is precisely the proper fibre formula. Nevertheless the two comma categories are empty and terminal, so their transition is not a weak homotopy equivalence, and for sets it changes the number of connected components. The category has two objects, one nonidentity arrow, and forced composition; the checker verifies these comma categories. Empty fibres are essential to the counterexample and cannot be discarded. □

1.4 What the finite fibre theory supplies, and what Chapter XIII adds

A.21. [Proved] Finite products and the one-fibrant-leg pullback comparison. Assume the hypotheses of A.11 and slice replacement equivalences as in A.12. In the localized fibrant slice over S, the product of $X \to S$ and $Y \to S$ is represented by $X \times_S Y \to S$. Arbitrary derived reindexing preserves these finite products. Moreover, if $X \to S$ is intrinsic and $Y \to \overline{Y} \to S$ is a replacement with $Y \to \overline{Y} \in W$ and $\overline{Y} \to S$ intrinsic, then $X \times_S Y \to X \times_S \overline{Y}$ lies in W. Thus a strict pullback with one intrinsic leg computes the same relative homotopy object as replacing the other leg first. These are the formal assertions at XII ms. 241–248.

Proof. The fibre product of two intrinsic fibrations is again intrinsic by composition and base-change closure. A map between two such products induced by $u : X \to X'$ and $v : Y \to Y'$ in W factors as

$$X \times_S Y \longrightarrow X' \times_S Y \longrightarrow X' \times_S Y'.$$

The first arrow is a pullback of u along the intrinsic object $Y \to S$, and the second a pullback of v along $X' \to S$, so both are in W. The product functor and diagonal are therefore relative functors, and their ordinary adjunction descends by A.4. This proves the product universal property, not merely an abstract comparison of its objects. The terminal object of the slice is 1_S, and its inclusion from the point is also a relative right adjoint, so the terminal-object assertion descends. Pullback commutes canonically with these products before localization, hence after localization. The last displayed map is a direct test of the definition of $X \to S \in \mathrm{Fib}_W$. □

A.22. [Proved] The intrinsic factorization content of the weak Quillen axioms. The conditions M2, M′, and M″ of XIII ms. 8 imply both intrinsic factorizations: every arrow factors as an intrinsic cofibration followed by a W-map, and as a W-map followed by an intrinsic fibration. They consequently imply the universal-equivalence equalities of XIII Corollary 1, ms. 10, and the fibrewise assertion of XIII Proposition 4, ms. 10–11. They do not identify the designated classes TC, TF with $C \cap W, F \cap W$.

Proof. M2 provides factorizations using (C, TF) and (TC, F). M′ puts TC, TF in W, while M″ puts C and F in the two intrinsic classes. This gives the asserted factorizations with their exact order. A.11 gives $\mathrm{Fib}_W \cap W = W^u$ and the fibrewise assertion; reversing arrows gives the cofibration equality. The hypotheses only contain inclusions $TC \subseteq C \cap W$ and $TF \subseteq F \cap W$. The manuscript later gives separate names to equality on the left and on the right. Its term "exact" for a weak Quillen structure imposes $W = TF \circ TC$, whereas "exact à gauche" imposes $TC = C \cap W$. These are different formulas and none may be substituted for another by terminology alone. □

A.23. [Proved] The variance assertions of XIII section 4 are lifting adjunctions. Let $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$, suppose the ordinary Kan extensions used below exist, and let the two closed lifting pairs be $(C(I), TF(I))$ and $(TC(I), F(I))$, with $C(I), TC(I)$ defined objectwise. For $u : I' \to I$ one has

$$u^*C(I) \subseteq C(I'), \quad u^*TC(I) \subseteq TC(I'), \quad u_*F(I') \subseteq F(I), \quad u_*TF(I') \subseteq TF(I),$$

and

$$u^*F(I) \subseteq F(I') \iff u_!TC(I') \subseteq TC(I),$$
$$u^*TF(I) \subseteq TF(I') \iff u_!C(I') \subseteq C(I).$$

If every component of $i\backslash I'$ has an initial object and the relevant coproducts exist, then all four containments in the last two equivalences hold. In particular injective fibrant diagrams are objectwise fibrant. For a walking-arrow diagram $X_0 \leftarrow X_1$, injective fibrancy is equivalent to X_0 fibrant and $X_1 \to X_0 \in F$; this also implies that X_1 is fibrant.

Proof. Restriction preserves objectwise classes. For any adjunction $L \dashv R$, a commutative square with left edge La and right edge b corresponds bijectively to a square with left edge a and right edge Rb, and the bijection takes fillers to fillers. Thus $La \perp b$ if and only if $a \perp Rb$. Applying this to the two adjunctions and using that the lifting pairs are closed proves every displayed implication and converse. If the comma components have initial objects, the left Kan extension is the coproduct of evaluations at those initial objects, since the diagram being colimited has presheaf variance. A left orthogonal class is closed under these coproducts: lifting out of a coproduct is exactly one independent lifting problem on each summand. Thus $u_!$ preserves C and TC objectwise. Taking $I' = e$ proves the evaluation assertion.

For the walking arrow, lift first at 0 against a trivial cofibration, using fibrancy of X_0, and then lift at 1 against $X_1 \to X_0$. This proves sufficiency. Conversely, test diagrams supported only at 0 to obtain fibrancy of X_0, and test a square whose component at 0 is an identity to obtain the lifting property of $X_1 \to X_0$. This proves the exact arrow criterion. The argument is the actual content of XIII Propositions 1–3, ms. 45–50, and requires no claim that ordinary u_* preserves all weak equivalences. □

A.24. [Refuted] Ordinary Kan extensions do not automatically descend. For (Cat, W_∞), ordinary limits of finite diagrams need not preserve objectwise weak equivalences. Thus an ordinary u_* in XIII ms. 45–50 cannot be identified with a derived u_* without a derivability or fibrancy argument.

Proof. Consider the cospan diagram $e \xrightarrow{0} [1] \xleftarrow{1} e$ and the natural transformation to the constant cospan $e \to e \leftarrow e$, identity at its two outer vertices and the unique functor $[1] \to e$ at the middle vertex. Each component is in W_∞, because $[1]$ has a terminal object. The strict pullbacks are respectively the empty category and the terminal category, and their comparison

is $\varnothing \to e \notin W_\infty$. All objects, arrows, and the empty fibre product are checked by the finite program. This is a counterexample to raw preservation, not a counterexample to the fibrant computation or to the derived adjunction. □

A.25. [Proved] The noetherian sober Alexandrov qualification adds no infinite posets. If a poset with its Alexandrov topology is noetherian and sober, the poset is finite. Hence the apparently more general admissibility qualification in XIII ms. 60–61, requiring the Alexandrov space of I^{op} to be noetherian and sober, introduces no genuinely infinite poset beyond the finite case.

Proof. We use the specialization order, so closed sets are lower sets. Noetherianity forbids an infinite strictly decreasing chain of principal closed sets. It also forbids an infinite antichain: for pairwise incomparable points x_n, the finite unions of their smallest open neighbourhoods form a strictly increasing sequence of open sets. An infinite strictly increasing chain of principal closed sets has a union C that is closed, because arbitrary unions of lower sets are lower. This union is irreducible: if it were contained in $F \cup G$ for two closed sets without being contained in either, choose chain members containing a point outside F and a point outside G; a later principal closed member contains both and is irreducible, a contradiction. Sobriety gives $C = \overline{\{x\}}$. The point x belongs to some member of the chain, which then contains all of C, a contradiction to strict increase. Thus there is no infinite chain in either direction and no infinite antichain. Every infinite poset has a countably infinite subset containing an infinite chain or antichain, by the infinite Ramsey theorem applied to comparability; within an infinite chain, the usual monotone-subsequence argument gives an infinite increasing or decreasing subsequence. Therefore the poset is finite. Reversing the order does not change this conclusion. □

A.26. [Refuted] The intrinsic finite floor does not provide all small Kan extensions. Finite limits, finite colimits, two-out-of-three, and both intrinsic factorization axioms do not imply existence of all small homotopy Kan extensions. A counterexample is the category of finite sets with isomorphisms as W.

Proof. Every map is both an intrinsic fibration and an intrinsic cofibration, since pullback and pushout preserve isomorphisms. Every map therefore has both required factorizations, inserting an identity as the W-factor. All finite limits and colimits exist. Localization changes nothing. But the diagonal functor $\mathrm{FinSet} \to \mathrm{FinSet}^{\mathbb{N}}$ has no right adjoint at the constant family of two-element sets: a representing finite set would have $2^{\aleph_0}$ global elements by the product universal property. It has no left adjoint at the constant singleton family either: a representing finite set L would have $2^{\aleph_0}$ maps to a two-element set, whereas $\mathrm{Hom}(L, 2)$ is finite. Thus XII's finite diagram theory and XIII's finite admissibility statement cannot be promoted to an all-small-shapes derivator without additional hypotheses. □

A.27. [Proved] Closedness upgrades historical exactness to both trivial-class equalities. Let (C, TC, F, TF) satisfy the weak Quillen axioms of XIII ms. 27–28, including two-out-of-three for W, the two factorizations, and $TC \subseteq C \cap W$, $TF \subseteq F \cap W$. If both lifting pairs are closed and $W = TF \circ TC$, then

$$TC = C \cap W, \qquad TF = F \cap W.$$

Moreover W is retract-closed. With the finite limits and colimits of M0 these data are a closed model structure in the original finite-limit convention. The two intrinsic containments M″ make this structure proper. Thus a *closed exact* structure, as required by the uniform

hypothesis at XIII ms. 57, is stronger than the unqualified exact structure distinguished in A.22.

Proof. For $i : A \to B$ in $C \cap W$, exactness supplies $i = pj$ with $j : A \to Z$ in TC and $p : Z \to B$ in TF. Since $i \perp p$, the square with top j and bottom 1_B has a lift $s : B \to Z$. The identities $si = j$, $ps = 1_B$ display i as a retract of j in the arrow category. A left orthogonal class is retract-closed, so $i \in TC$. The dual argument for $p \in F \cap W$ proves the second equality.

For completeness we give the retract argument for W, which does not require functorial factorizations; compare Riehl (2014, Lemma 11.3.3). First, if $f \in F$ is a retract of $w \in W$, factor $w = vj$ with $j \in TC$, $v \in TF$. Lifting j against f extends the retraction of their sources and displays f as a retract of v, hence $f \in TF$. For a general retract $f : A \to B$ of $w : X \to Y$, write the sections as $a : A \to X$, $b : B \to Y$ and retractions as $r : X \to A$, $s : Y \to B$. Factor $f = hg$, with $g : A \to Z$ in TC and $h : Z \to B$ in F. Form $P = X \amalg_A Z$, with structure maps $i : X \to P$ and $x : Z \to P$. Then $i \in TC$. The maps w, bh induce $j : P \to Y$ with $ji = w$, so $j \in W$. The maps $gr, 1_Z$ induce $y : P \to Z$ with $yx = 1_Z$ and $hy = sj$. Hence (x, b) and (y, s) display h as a retract of j. The first case gives $h \in W$; therefore $f = hg \in W$. The remaining model axioms are precisely the stipulated lifting, factorization, and finite-limit axioms. Finally M″ says directly that pushout along C and pullback along F preserve W, which is properness. □

A.28. [Proved] The uniform theorem of XIII and the exact admissibility hypothesis. Assume exactly the uniform-left hypothesis of XIII ms. 57: for every I in the specified diagram domain, the objectwise classes $C(I), TC(I)$ and their right orthogonals $TF(I), F(I)$ form a closed exact weak Quillen structure for the objectwise class $W(I)$. Let $u : I' \to I$ be a functor in the domain for which the ordinary presheaf right Kan extension u_* exists. Then u_* preserves weak equivalences between fibrant diagrams. It consequently induces a right adjoint

$$u^* : \mathcal{M}(I)[W(I)^{-1}] \rightleftarrows \mathcal{M}(I')[W(I')^{-1}] : \mathbf{R}u_*.$$

If the comma categories I'/i and their inclusions into I' belong to the diagram domain, the canonical pointwise formula is

$$i^*\mathbf{R}u_*X \;\cong\; \mathbf{R}\lim_{(I'/i)^{\mathrm{op}}} X|_{I'/i}.$$

In particular every functor between finite posets is admissible under the source's hypotheses. No restriction to finite posets is necessary for admissibility once the relevant ordinary Kan extension and uniform diagram structures exist. In the historical numbering, this proves the right-hand Der 4 and Der 5 assertions of XIII ms. 61; these numbers do not mean modern Der 4 and strongness Der 5.

Proof. A.27 applies separately on every diagram category. A.23 shows that u^* preserves C and TC and u_* preserves F and TF. On the full subcategory of fibrant diagrams, terminal maps and products are fibrant, and factorization of an arrow as TC followed by F stays inside that subcategory. Apply the elementary functor lemma proved in A.17 to u_*: its hypothesis holds because $F \cap W = TF$ and u_* preserves TF. Thus u_* preserves every weak equivalence between fibrant objects. This is Brown's lemma, not an additional infinite-product or transfinite-closure assumption.

We recall explicitly the homotopy-category calculation used to derive the adjunction. For a closed model structure with finite limits and colimits, a cofibrant source A and fibrant target B satisfy

$$\mathrm{Hom}_{\mathcal{M}[W^{-1}]}(A, B) \cong \mathrm{Hom}_{\mathcal{M}}(A, B)/\sim,$$

where $\sim$ is the common left- and right-homotopy relation. Cofibrant and fibrant replacements induce equivalences from their full subcategories after localization. These are the original finite-limit form of Quillen's homotopy-category theorem, I.1, Theorem 1; no functorial replacement is needed (D. G. Quillen, 1967). To apply its derived-adjunction argument, take A cofibrant in $\mathcal{M}(I)$ and B fibrant in $\mathcal{M}(I')$. The ordinary adjunction identifies maps $u^*A \to B$ with maps $A \to u_*B$. It identifies their homotopy classes: a cylinder of A is taken to a cylinder of u^*A, since restriction preserves cofibrations and all objectwise weak equivalences; homotopies out of that cylinder correspond under the original adjunction. The homotopy-category formula therefore gives a natural bijection on localized Hom sets. Replacing arbitrary sources and targets by cofibrant and fibrant ones gives the asserted adjunction. Preservation on fibrant objects, already proved, makes the computation by u_*B independent of the chosen fibrant representative. This is Quillen I.4, Theorem 3, with the comparison maps specified.

For the pointwise assertion, write $q_i : I'/i \to I'$. Its restriction preserves fibrant diagrams by A.23: for $j \in I'$, every component of $j\backslash(I'/i)$ has an initial object, namely the factorization beginning with 1_j of a fixed arrow $u(j) \to i$. Thus a fibrant B restricts to a fibrant diagram on I'/i. The ordinary formula $(u_*B)(i) = \lim_{(I'/i)^{\mathrm{op}}} q_i^*B$ computes both derived sides, and it computes their canonical mate. Finite posets and their relevant comma categories require only finite limits, supplied by M0. A.25 explains why the source's noetherian sober qualification adds no infinite case. □

1.5 Formal closure corrections and the absolute localizer axioms

A.29. [Proved] The polarity calculation in II section 5 requires closure after a join. For any relation $R \subseteq P \times Q$, the maps $A \mapsto A^\perp = \{q \mid pRq \text{ for all } p \in A\}$ and its transpose define an antitone Galois connection on the power sets. Double orthogonality is an extensive idempotent closure; closed subsets are closed under arbitrary intersections, and their join is the closure of their union. If P, Q are complete posets and the principal-polar condition (D_0) of II ms. 69 holds, write c for the induced closure on P. For a subset $A \subseteq P$, its polar closure is the principal lower set

$$A^{\perp\perp} = {\downarrow}c\Big(\bigvee A\Big) = {\downarrow}c\left(\bigvee_{a\in A} c(a)\right).$$

The final closure cannot be omitted from II formula (5.20), ms. 76. One must also distinguish the element $c(\bigvee A)$ of P from its principal lower set, a subset of P.

Proof. The equivalence $B \subseteq A^\perp \iff A \subseteq B^\perp$ is precisely the assertion that every pair in $A \times B$ belongs to R. Extensivity and antitonicity give the threefold identities and hence idempotence, as in A.6. An intersection of closed subsets is closed because it is the polar of the union of their opposite polars. The least closed set containing a union is its closure. Under (D_0), taking all $a \in A$ in the relation amounts to taking their supremum; its polar and double polar are the stated principal lower sets. Extensivity gives $\bigvee A \le \bigvee c(A)$, while monotonicity gives $c(a) \le c(\bigvee A)$ for every a. Applying c to the two inequalities proves the second equality.

For the omitted-closure issue, take the usual annihilator polarity between subsets of $V = \mathbb{F}_2^2$ and subsets of its dual, ordered by inclusion. The induced closure on subsets of V is linear span. The two one-dimensional subspaces $\{0, (1,0)\}$ and $\{0, (0,1)\}$ are closed. Their union has three elements and is not closed; the correct closed join is all four elements of V. The finite checker evaluates all scalar products and double annihilators. This verifies the correction without relying on the companion's wording. □

A.30. [Refuted] Arbitrary unions do not preserve composition closure. The class of arrow classes containing identities and closed under both pullback and composition is not closed under arbitrary unions, contrary to the preliminary lattice description in II ms. 65–67. Intersections have that property; a join requires taking the closure of the union under the specified operations.

Proof. In the chain $0 < 1 < 2$, let L_1 consist of the identities and $0 \to 1$, and let L_2 consist of the identities and $1 \to 2$. Each is pullback-closed: pulling the nonidentity arrow back along an arrow to its target gives either itself or an identity. Each composes, since no two nonidentity arrows in one class are composable. Their union contains $0 \to 1$ and $1 \to 2$ but not $0 \to 2$. The complete category and the closure tests are checked in the attached finite program. More generally, intersections preserve each closure condition because every required output lies in every intersectand; the smallest closed class containing a union is therefore the intersection of all closed classes containing it. □

A.31. [Proved] The bi-site remark fixes a question rather than a general theorem. At VI ms. 81–84 the data $(\mathcal{M}, W)$ specify the coarse slice localization, but the proposed fine classes require additional test data. The two comma tests in Cat have opposite variances: X/s varies covariantly with s, and $s\backslash X$ contravariantly. In XII ms. 222 the class labelled W^g uses the former tests and its universal preservation class is the smooth class discussed in Part B; XII ms. 224's W^d uses the latter. The letters on an earlier page are not a substitute for this variance check.

The remark that weak globular infinity-groupoids might have no useful localization theory is a suggestion at VI ms. 83–84. It is not a nonexistence theorem about fine relative theories. The attached 2026 claims about higher categories and globular groupoids remain "claimed (OpenAI, 2026, unrefereed)" here (OpenAI, 2026a, 2026b); no conclusion of Part A depends on either.

Proof. An arrow $s \to t$ postcomposes $X/s \to X/t$ and precomposes $t\backslash X \to s\backslash X$. The slice projection $S/s \to S$ is a local isomorphism in the discrete-fibration sense and its pullback is X/s; the opposite projection gives the other test. XII ms. 222 and 224 explicitly use these respective classes. The equality with the relevant smooth or proper preservation class needs the extra categorical argument of Part B and does not follow merely from renaming a Galois transporter. VI ms. 83 describes the two slice operations and the involution, and uses speculative language for the groupoid remark; it supplies no axiom forbidding a chosen finer class. Ordinary groupoids provide the unconditional test case in Part E. □

A.32. [Refuted] The literal absolute package does not imply relative Theorem A. The literal axioms Loc (1,2,3) of XVI, with the moderate saturation of A.1, do not imply Loc (3 bis). The failure already occurs in a triangle of finite groupoids over the one-object category BC_2. Thus the absolute package left undecided in POLES A.16 must be distinguished from the relative package used throughout the present paper.

Proof. For a functor $f : A \to B$ between arbitrary small categories define

$$f \in W^\dagger \quad \Longleftrightarrow \quad \pi_0(f) \text{ is bijective, and } \big(|\pi_0(A)| = 1 \Longrightarrow f \in W_\infty\big).$$

Thus on connected categories we retain exactly the weak homotopy equivalences; on categories with at least two components we require only a bijection of components. The empty-to-empty functor also belongs to $W^\dagger$.

We first verify the axioms on all small categories, including infinitely many components. Isomorphisms belong to $W^\dagger$. If two of f, g, gf belong to $W^\dagger$, two-out-of-three for bijections shows that all three maps induce bijections on components, so their three categories have the same component cardinality. If it is not one, no additional condition is required; if it is one, two-out-of-three for W_∞ gives the assertion. For moderate saturation, suppose $f : A \to B$ and $g : B \to A$ have $gf, fg \in W^\dagger$. Bijections of both composites imply that $\pi_0(f)$ and $\pi_0(g)$ are bijections. Again only the connected case requires more. In that case both composites are weak homotopy equivalences, and hence both factors are weak homotopy equivalences: their images in the homotopy category of spaces have invertible composites in both orders, which gives a left and a right inverse for each factor. Classifying spaces are CW complexes, so isomorphism there is equivalent to weak homotopy equivalence. This proves precisely Loc (1), without assuming full arrow-retract closure.

A category is $W^\dagger$-aspherical if and only if its nerve is weakly contractible. Indeed its map to e must have exactly one component, and the remaining condition is exactly membership in W_∞. A category with a terminal object has contractible nerve, proving Loc (2). If every A/b is $W^\dagger$-aspherical, each has weakly contractible classifying space; these CW complexes are contractible. Quillen's Theorem A, in its dual comma formulation, gives $f \in W_\infty \subseteq W^\dagger$, proving the absolute Loc (3) (D. Quillen, 1973, Theorem A, p. 93).

For the failure of the relative clause, use additive notation $G = \mathbb{Z}/2\mathbb{Z}$. Let $C = BG$ have one object c, morphism group G, identity 0, and composition addition. Let E be the groupoid with objects $x \in G$ and exactly one arrow

$$(x, g) : x \longrightarrow x + g \qquad (g \in G).$$

Its identity at x is $(x, 0)$ and $(x + g, h) \circ (x, g) = (x, g + h)$. Define $p : E \to C$ by $p(x) = c$, $p(x, g) = g$. Let $K = BG$, let $A = E \times K$, let $B = E$, and let $u : A \to B$ be projection, with structure maps $pu : A \to C$ and $p : B \to C$. Explicitly, A has the same two object labels as E, an arrow $(x, g, k) : x \to x + g$ for each $g, k \in G$, identities $(x, 0, 0)$ and composition

$$(x + g, h, l) \circ (x, g, k) = (x, g + h, k + l).$$

The functor u forgets k. This gives all objects, arrows, identities, compositions and functors in the triangle.

The groupoid E is connected and has exactly one arrow between any two objects; it is equivalent to e. The groupoid A is connected with automorphism group G at either object, and is equivalent to BG. Consequently u induces $G \to 1$ on fundamental groups. It is not in W_∞, and since both categories are connected it is not in $W^\dagger$.

There is just one object c of the base, so the relative premise has one comma comparison. The objects of E/c are pairs $(x, t) \in G^2$, where $t : p(x) \to c$. An arrow labelled g from (x, t) to $(x + g, t')$ exists precisely when $t' + g = t$. Thus $t + x$ is invariant under arrows. For each of its two values there are two objects and exactly one arrow between every pair. Hence E/c has exactly two connected components, each equivalent to e. There is a strict identification

$$A/c = (E/c) \times K,$$

under which u/c is projection. Its source also has exactly two components, and it is a bijection on those components, indexed on both sides by $t + x$. Therefore $u/c \in W^\dagger$, whereas $u \notin W^\dagger$. This refutes Loc (3 bis).

The saturation boundary is explicit. The nonweak map $BG \to e$ is a retract, in the arrow category, of

$$(BG \amalg e) \longrightarrow (e \amalg e),$$

which collapses the first component and is the identity on the second. This latter arrow is in $W^\dagger$. The sections are the inclusions of the first components; the source retraction is the

identity on BG and sends the added point to its unique object, and the target retraction is the unique map to e. These maps commute and their retraction composites are identities. Thus $W^{\dagger}$ fails Loc (1 bis); the example proves the asserted failure for the actual literal Loc (1), not for an unmentioned stronger saturation convention. The modern split-retraction clause nevertheless follows from the already verified moderate condition by taking $ri = 1$.

The program `code/a_absolute_relative.py` independently verifies all finite category laws, the functors, both comma categories, their components and automorphism groups, and the displayed arrow retract. The general axiom check above does not rest on any finite sampling. For comparison, a base with a terminal object cannot give this kind of counterexample: its terminal-object comma comparison is already the original functor. In particular [1] is not an unresolved test of the relative implication. □

A.33. [Refuted] The converse cocartesian assertion of XII ms.274 needs conservativity. Theorem 3° of XII ms. 274 is false without the conservativity premise introduced in its part 2°. The assertion that for a map of homotopy cocartesian squares $f_0, f_3 \in W$ imply $f_1, f_2 \in W$ fails even when W is strongly saturated and the intrinsic cofibration factorization exists. Corollary 1 on that page is also false without that extra premise.

Proof. Take $\mathcal{M} = \mathrm{FinSet}$ and W the isomorphisms. All maps are intrinsic cofibrations, every map factors as itself followed by an identity, and the W-cocartesian squares are the ordinary pushout squares. Let $D = \{0, 1\}$ and $e = \{*\}$. The two spans

$$X = (D \xrightarrow{1_D} D,\ D \longrightarrow e), \qquad Y = (D \longrightarrow e,\ D \longrightarrow e)$$

both have pushout e. The maps $f_0 = 1_D$, $f_1 : D \to e$, and $f_2 = 1_e$ define a map of spans and induce $f_3 = 1_e$. Thus $f_0, f_3 \in W$ while $f_1 \notin W$. All four component maps and the pushout equivalence relations are checked by the finite program.

Here a W-aspherical set is a singleton. In any pushout square whose initial and final vertices are singletons, both other vertices are singletons: the maps from the initial singleton make both nonempty, and identifying just their chosen basepoints cannot collapse any additional element. Thus both conditions 1 and $1'$ of Corollary 1 hold in this example, although its claimed equivalent implication fails. The counterexample therefore locates the missing hypothesis rather than merely a gap in its proof. The source's Corollary 2, ms. 275, explicitly restores conservativity. In the preceding Cons 2, ms. 273, "cartésien" must read "cocartésien" for the stated reduction; this word is printed that way in the edition, not introduced by transcription. □

A.34. [Refuted] Stabilizers alone do not define the antitone correspondence of the cancelled section. For arbitrary arrow classes A, the operation $A \mapsto \mathrm{Stab}(A)$ is not antitone. Thus the first, cancelled proposal in II section 4 cannot be used in place of the corrected transporter construction in section 6.

Proof. In [2], let A consist of the identities and $0 \to 2$, and let A' be all arrows. Both contain identities and compose, and $A \subseteq A'$. The arrow $1 \to 2$ does not stabilize A, because the pullback of $0 \to 2$ is $0 \to 1 \notin A$. It does stabilize A', as every arrow does. Therefore $\mathrm{Stab}(A') \not\subseteq \mathrm{Stab}(A)$. The checker verifies both stabilizers by all slice tests. By contrast, $T_W(A)$ with fixed target class W is antitone because enlarging A only adds tests; that is the correction proved in A.6. □

2 The two fine theories in categories and simplicial sets

B.1. [Proved] The comma tests and their variance. For a fundamental localizer W and a small category S, put

$$W_S^{\mathrm{pre}} = \{u : X \to Y \text{ over } S : \ s\backslash u \in W \text{ for all } s \in S\},$$
$$W_S^{\mathrm{co}} = \{u : X \to Y \text{ over } S : \ u/s \in W \text{ for all } s \in S\}.$$

Both are weakly saturated classes contained in the total class W_S. Opposition gives an isomorphism of relative categories $(\mathrm{Cat}/S, W_S^{\mathrm{pre}}) \cong (\mathrm{Cat}/S^{\mathrm{op}}, W_{S^{\mathrm{op}}}^{\mathrm{co}})$. For $W = W_\infty$, their enhanced localizations are, respectively, $\mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S})$ and $\mathrm{Fun}(NS, \mathcal{S})$. The value of the first classifier at s is $B(s\backslash X)$; the value of the second is $B(X/s)$. In the notation of XII ms. 222, W_S^{g} is the second class.

Proof. Each comma construction is a functor on Cat/S, so identities, two-out-of-three, and retracts are inherited from W. Relative Theorem A gives $W_S^{\mathrm{co}} \subset W_S$, and its opposite form gives the other inclusion. There are canonical category isomorphisms $(s\backslash X)^{\mathrm{op}} \cong X^{\mathrm{op}}/s$; fundamental localizers are invariant under opposition (Cisinski, 2006, Sorites 3.3.8). These are isomorphisms of specified comma categories, not just equivalences of their nerves. The enhanced identifications are proved in B.3 below. An arrow $s \to t$ acts by precomposition $t\backslash X \to s\backslash X$ and by postcomposition $X/s \to X/t$, which fixes the variance. For $X = \varnothing$ all values are empty; for $S = \varnothing$ the slice has its unique object. □

B.2. [Known] The simplicial slice test, with the necessary base hypothesis. Let S be a small quasicategory and let $u : X \to Y$ be any map of simplicial sets over S; neither X nor Y is assumed fibrant. Then u is a covariant equivalence exactly when

$$X \times_S S_{/s} \longrightarrow Y \times_S S_{/s}$$

is a weak homotopy equivalence for every vertex s. Replacing $S_{/s}$ by $S_{s/}$ gives the contravariant test. For an arbitrary simplicial base these statements hold with a right, respectively left, fibrant replacement $\widetilde{S}_{/s} \to S$, respectively $\widetilde{S}_{s/} \to S$, of the vertex map in place of the literal slice.

Proof. The precise all-base statement is Cisinski (2019, 4.4.29–4.4.30). We recall why the replacement and the word "strict" matter. Choose a left anodyne map $X \to X^{\mathrm{lf}}$ over S with left-fibration target. Pulling it back along the right fibration $\widetilde{S}_{/s} \to S$ remains a covariant equivalence, hence a total weak homotopy equivalence. The same holds for Y. The inclusion of the fibre $(X^{\mathrm{lf}})_s \to X^{\mathrm{lf}} \times_S \widetilde{S}_{/s}$ is a weak equivalence: the chosen vertex is final in $\widetilde{S}_{/s}$ and a left fibration preserves final maps under pullback. Consequently the displayed test is precisely the fibre test for a map of left fibrations. That test detects equivalences by Cisinski (2019, Theorem 4.4.14), or Lurie (2009, Theorem 2.2.1.2 and Corollary 2.2.3.12). When S is a quasicategory, $S_{/s} \to S$ itself is such a right-fibrant replacement, with the vertex 1_s final. This supplies the actual comparison maps used in the reduction. Opposition proves the other assertion. □

B.3. [Proved] Nerve comparison as an equivalence of enhanced localizations. For every small ordinary category S, the ordinary nerve induces equivalences

$$(\mathrm{Cat}/S)[(W_S^{\mathrm{co}})^{-1}]_\infty \simeq (\mathrm{sSet}/NS)[(W^{\mathrm{cov}})^{-1}]_\infty \simeq \mathrm{Fun}(NS, \mathcal{S}),$$

and, by opposition,

$$(\mathrm{Cat}/S)[(W_S^{\mathrm{pre}})^{-1}]_\infty \simeq (\mathrm{sSet}/NS)[(W^{\mathrm{contra}})^{-1}]_\infty \simeq \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S}).$$

Here $W = W_\infty$. These are equivalences of infinity-categories and therefore of homotopy categories. The last comparison is implemented by the straightening–unstraightening Quillen equivalence. No model structure on Cat/S with the ordinary nerve as a right Quillen functor is asserted.

Proof. The nerve preserves strict pullbacks and has $N(S/s) = (NS)_{/s}$ and $N(s\backslash S) = (NS)_{s/}$. Thus B.2 identifies exactly the weak equivalences on nerves. We verify essential surjectivity and all coherent mapping information by giving inverse relative functors. For $K \to NS$, let $T(K) = \Delta/K$ and map it to S by the last vertex. Its objects are simplices $\sigma : \Delta^n \to K$; a morphism $\sigma \to \tau$ is a monotone map $\theta : [n] \to [m]$ with $\sigma = \tau\theta$. Its image in S is the last segment $\tau(\theta(n)) \to \tau(m)$. Composition is composition of monotone maps. There is a natural last-vertex map $\epsilon_K : NT(K) \to K$ over NS. For each s the strict identification

$$T(K)/s \cong \Delta/(K \times_{NS} N(S/s))$$

sends $(\sigma, \alpha : \sigma(n) \to s)$ to the simplex of S/s whose last arrow is α. In the other direction a simplex of S/s is determined by its last such arrow. This also checks the morphisms. Under this identification the pullback of ϵ_K is the usual last-vertex map from the nerve of a simplex category. That map is a weak homotopy equivalence: the simplicial replacement of the simplex decomposition has contractible simplex fibres, and its two augmentations to $N(\Delta/L)$ and L are weak equivalences. Equivalently, this is the simplex-category instance of Cisinski (2019, Proposition 7.3.8 and Examples 7.3.9–7.3.10). It follows from B.2 that every ϵ_K is a covariant equivalence. Two-out-of-three now proves that T preserves covariant equivalences into W_S^{co}.

For a category $X \to S$, evaluation at the last vertex is a functor $T(NX) \to X$ over S, and the preceding calculation shows that it is in W_S^{co}. The two natural transformations $NT \Rightarrow 1$ and $TN \Rightarrow 1$ become natural equivalences after infinity-localization. Thus the two localized functors are inverse, including their mapping spaces. Opposition gives the presheaf comparison. Finally, straightening over the ordinary nerve identifies the covariant localization with coherent S-diagrams of spaces; its Quillen equivalence is Lurie (2009, Theorem 2.2.1.2). For the first classifier the same argument, or the pointwise left Kan formula applied to the Yoneda generators, gives exactly $s \mapsto B(s\backslash X)$. □

B.4. [Refuted] The literal slice recipe over every simplicial base. Lead 1 is false for arbitrary simplicial bases when its tests use literal simplicial slices. Let $S = \Lambda^2_1$, with nondegenerate edges 01 and 12 and no 02 edge. The map $\{0\} \hookrightarrow \Delta^{\{0,1\}}$ over S is a covariant equivalence, but its pullback along $S_{/2} \to S$ is $\varnothing \to \{1\}$ and is not a weak homotopy equivalence.

Proof. The given map is the generating left-anodyne inclusion $\Lambda^1_0 \hookrightarrow \Delta^1$, hence is a covariant equivalence over every base (Lurie, 2009, Proposition 2.1.4.9). The vertices of $S_{/2}$ are the edge 12 and the degenerate edge 22; their projections are 1 and 2. There is an edge between them supplied by the degeneracy of 12, but none projecting to 01. Therefore its pullback over $\{0\}$ is empty and over the edge 01 is the vertex 1. All higher simplices in the latter are degenerate. The two numbers of components are zero and one. The finite simplex data and both intersections are checked in `code/bc_counterexamples.py`. Applying simplicial opposition gives the contravariant counterexample. This refutes the assertion as stated, while B.2 supplies the corrected version. □

B.5. [Known] The exact dictionary for smoothness and properness. For $p : A \to B$ in Cat, Grothendieck's W_∞-properness is equivalent to properness of Np in Cisinski (2019, Definition 4.4.1), and to smoothness of Np in Lurie (2009, Definition 4.1.2.9). Grothendieck's

W_∞-smoothness corresponds to Cisinski-smoothness and to the opposite of Lurie's smoothness. Thus the two cited books use opposite names. The equivalences hold in both directions for all ordinary functors, without a fibration hypothesis.

Proof. For an arrow $\alpha : p(a) \to b$, define the outgoing lift category $L_p^+(a,\alpha)$: an object is $v : a \to a'$ with $p(v) = \alpha$; an arrow $v \to v'$ is $w : a' \to a''$ with $p(w) = 1_b$ and $wv = v'$. It is the comma category $(a,\alpha)\backslash A_b$ for $A_b \to A/b$. Thus Grothendieck-properness says precisely that all these categories have weakly contractible nerves. The incoming version is dual. Every nerve of an ordinary functor is an inner fibration, since the unique inner-horn fillers in nerves are the specified composites. In the ordinary nerve case, Cisinski (2019, Theorem 4.4.36, condition (ii')) identifies the displayed condition with simplicial properness. This particular case is unchanged in the author's July 2026 errata (Cisinski, 2026, pp. 1–4); for arbitrary quasicategories the corresponding fibre-to-comma condition must also be imposed after all base changes. We do not use the incorrect unrestricted version of that theorem. Finally Cisinski-properness is universal preservation of final maps, equivalently of right-anodyne maps (Proposition 4.4.4). HTT Proposition 4.1.2.8 makes this exactly Lurie-smoothness. Opposition supplies the second equivalence. In particular $\{1\} \to [1]$ is Grothendieck-proper and Lurie-smooth, whereas $\{0\} \to [1]$ has the other two names. Neither class is the class of proper maps of geometric realizations. □

B.6. [Proved] The maximal localization class, for every fundamental localizer. Let $\widehat{\mathcal{L}}_{\mathrm{co}}$ consist of those $p : A \to B$ for which every strict base change $p_T : A \times_B T \to T$ preserves W_T^{co} under raw pullback. Then

$$\widehat{\mathcal{L}}_{\mathrm{co}} = \mathrm{Liss}_W, \qquad \widehat{\mathcal{L}}_{\mathrm{pre}} = \mathrm{Prop}_W.$$

No accessibility assumption is needed. In particular the equality left open at XII ms. 222 has an affirmative answer in the manuscript's covariant variance; it is not the assertion that all raw inverse images are homotopical.

Proof. We use the following exact form of Maltsiniotis (2005, Theorem 1.2(c)): a functor is W-smooth if and only if all its base changes preserve W-aspherical functors under pullback. Here "aspherical functor" means that all its over-comma categories are W-aspherical. For $X \to T$, let $R_T X = \int_{t\in T}(X/t)$, with its opfibration to T, and let $j_X : X \to R_T X$ send x to $(p(x), x, 1_{p(x)})$. An object of $R_T X$ is $(t, x, \beta : p(x) \to t)$. The category $j_X/(t,x,\beta)$ is canonically X/x: an arrow from $j_X(y)$ to the given object is determined by $v : y \to x$, since its base arrow must be $\beta p(v)$. Thus j_X is W-aspherical for every fundamental localizer. If $q : T' \to T$ is smooth, its pullback $(R_T X) \times_T T' \to R_T X$ is smooth. The cited characterization implies that $j'_X : X \times_T T' \to (R_T X) \times_T T'$ is aspherical. An aspherical functor $v : U \to V$ over T' is in $W_{T'}^{\mathrm{co}}$: for each t' and $(z,\gamma) \in V/t'$, the comma category $(v/t')/(z,\gamma)$ is v/z, so relative Theorem A first makes v/t' aspherical and then a W-map. For $u : X \to Y$ in W_T^{co}, the map $R_T u$ is fibrewise in W. Its pullback along q is still a fibrewise W-map of opfibrations. Over-comma categories of an opfibration retract, by cocartesian transport, onto its fibres: the inclusion of the fibre over t' has a left adjoint sending $(z, \alpha : pz \to t')$ to $\alpha_! z$. Hence this pulled-back map lies in $W_{T'}^{\mathrm{co}}$. In the natural square with vertical arrows j'_X, j'_Y, three sides are now in $W_{T'}^{\mathrm{co}}$; two-out-of-three gives q^*u in that class. Smoothness is preserved by arbitrary base change by its lifting criterion, which proves $\mathrm{Liss}_W \subset \widehat{\mathcal{L}}_{\mathrm{co}}$.

Conversely, a functor $u : X \to T$ is W-aspherical precisely when the map $(X \to T) \to (T \xrightarrow{1} T)$ is in W_T^{co}, since T/t has a terminal object. If p belongs to the universal-preservation class, every pullback of every such aspherical functor is aspherical. This is exactly condition (c) of the same Theorem 1.2, hence p is smooth. This argument also proves maximality, not

merely containment. Opposition proves the presheaf assertion. It explains why interchanging $s\backslash X$ and X/s while leaving the name "smooth" unchanged gives a false claim. □

B.7. [Proved] Three different fibration classes at the minimal localizer. There are strict inclusions

$$\{\text{Grothendieck bifibrations}\} \subsetneq \mathrm{Liss}_{W_\infty} \cap \mathrm{Prop}_{W_\infty} \subsetneq \mathrm{Fib}_{W_\infty}.$$

For simplicial sets, Cisinski-smooth and Cisinski-proper maps likewise lie in the intrinsic class preserving all weak homotopy equivalences under strict pullback. The analogous inclusion for a larger fundamental localizer is generally false.

Proof. A cartesian lift supplies a terminal object in each incoming lift category; a cocartesian lift supplies an initial object in each outgoing lift category. This proves the first inclusion. Cisinski's Corollary 4.6.5 and Proposition 4.6.2 identify a simplicial map which is both smooth and proper as one whose every strict pullback square is a homotopy pullback. Homotopy pullbacks preserve weak equivalences. B.5 and preservation of pullbacks by the nerve prove the second inclusion for categories.

For strictness of the first inclusion use the four-object poset $x < a < c$, $x < b < c$, with a, b incomparable, mapped to [1] by $x \mapsto 0$ and $a, b, c \mapsto 1$. The only nontrivial outgoing lift poset is $\{a, b < c\}$, which has terminal object c but no initial object. All incoming nonidentity tests are singletons. It is therefore smooth and proper but has no cocartesian lift from x over $0 \to 1$. For strictness of the second inclusion let V be the three-object poset $a < b$, $a < c$, and map a, b to 0 and c to 1. The section s given by $s(0) = a$, $s(1) = c$ satisfies $ps = 1$ and there is a natural transformation $sp \Rightarrow 1_V$ over [1]. It survives every strict pullback. Thus each pullback of p is a weak equivalence, and two-out-of-three in the natural square with any weak equivalence over [1] proves $p \in \mathrm{Fib}_{W_\infty}$. There is no outgoing lift from b to colour 1, so p is not proper. The program checks every finite category law, all diamond lift posets, and the natural-retraction data. These examples also prove the simplicial strictness statements after taking nerves. □

B.8. [Proved] An explicit smooth and proper path resolution. For a small category C and $x \in C$, let $E_x(C)$ have as objects the functors $\gamma : J \to C$, where J has objects $\mathbb{Z}$ and arrows $2i \to 2i - 1$, $2i \to 2i + 1$, which are constantly x with identity arrows at the left end and constantly some object at the right end. Morphisms are natural transformations whose eventual left component is 1_x; composition is componentwise. Then $q : E_x(C) \to C$, taking the eventual right value, is W_∞-smooth and proper, $BE_x(C)$ is weakly contractible, and its strict fibres are the homotopy fibres of $\{x\} \to BC$.

Proof. All objects and morphisms are small: eventually constant diagrams form a small union of finite-diagram sets. Natural transformations have an eventual right component because every sufficiently remote horizontal arrow is an identity. Thus q is an ordinary functor. Fix γ and an arrow $\alpha : q\gamma \to z$. For every sufficiently large even integer k, let γ_k agree with γ through k and be constantly z after k, with the arrow at $k \to k + 1$ equal to α. There is a natural transformation $\eta_k : \gamma \to \gamma_k$, identity through k and equal to α afterwards. In the outgoing lift category, restrict to those $(\eta : \gamma \to \delta)$ whose target is stationary after $k + 1$. This full subcategory has initial object η_k: the unique map $\gamma_k \to \delta$ under γ is η_i for $i \leq k$ and 1_z thereafter. Naturality at the joining edge is exactly the naturality of η. The eventual identity condition forces uniqueness on the entire tail. These subcategories, as k increases through even integers, cover every object and morphism of the outgoing lift category. Their nerves form a filtered union of contractible simplicial sets, hence are weakly contractible: every finite

simplicial sphere and null-homotopy question occurs at a finite stage. For an incoming arrow $z \to q\gamma$, choose k odd, use the joining arrow $k+1 \to k$, and obtain terminal objects in the analogous full subcategories. This proves smoothness and properness directly.

The entire category $E_x(C)$ is the filtered union of the categories of finite-fence diagrams on $[-n, n]$ with leftmost value x, extended constantly outside that interval. Each finite-fence category is contractible. Indeed, forgetting its last vertex has constant extension as a left or right adjoint, according to the orientation of the last edge; repeat until only the fixed object remains. These are actual adjunctions, with components given by the last edge. The same finite-stage argument proves $BE_x(C) \simeq *$. B.7 now says every strict pullback along q is a homotopy pullback. The constant path gives the specified factorization of the chosen point and identifies its fibres with the homotopy fibres of that point map. The construction is an integer-fence version of the categorical cocylinder (Evrard, 1975); the proof here does not assume the inherited PATHS claims. □

B.9. [Proved] The exact localizer threshold for complementarity. For a fundamental localizer W in the relative-Theorem-A convention,

$$\mathrm{Liss}_W \cap \mathrm{Prop}_W \subset \mathrm{Fib}_W \quad \Longleftrightarrow \quad W = W_\infty \text{ or } W = \mathrm{Mor}(\mathrm{Cat}).$$

Accessibility is not required. Thus the tentative complementarity at XII ms. 225–226 is false for every proper enlargement of W_∞ other than the total class.

Proof. Minimality gives $W_\infty \subset W$ (Cisinski, 2006, Corollary 4.2.19). The lift criteria show $\mathrm{Liss}_{W_\infty} \subset \mathrm{Liss}_W$ and $\mathrm{Prop}_{W_\infty} \subset \mathrm{Prop}_W$. Every functor $v : C \to B$ has a factorization $C \xrightarrow{i} P(v) \xrightarrow{p} B$ with $i \in W_\infty$ and p smooth and proper for W_∞: apply the verified model structure of Cisinski (2006, Theorem 5.3.14 and Proposition 5.3.13) only to the accessible minimal localizer. By B.7 every strict pullback along p is a homotopy pullback after realization. If the displayed inclusion holds, p lies in Fib_W. For $u : A \to B$ in W, the map $A \times_B P(v) \to P(v)$ therefore lies in W and realizes the homotopy pullback of Bu along Bv.

This proves that the class induced by W in spaces is closed under arbitrary homotopy pullback. Here the passage between models has specified comparisons: any strict simplicial cospan is replaced by its simplex categories, whose last-vertex comparison maps are weak equivalences; replacing one leg by p then computes its homotopy pullback. Two-out-of-three transfers W across these comparisons. The fundamental-localizer/simplicial-localizer correspondence (Cisinski, 2006, Theorem 4.2.15) ensures this uses the same class, also when it is not accessible.

Suppose $u \in W \setminus W_\infty$. Some homotopy fibre F of Bu is not weakly contractible, and pullback stability makes $F \to *$ a W-map. If $F = \varnothing$, we already have $\varnothing \to * \in W$. Otherwise each point $* \to F$ is in W by two-out-of-three; pulling it back along itself shows $\Omega F \to *$ is in W. Repeating, a nonzero homotopy group or an initial failure of connectedness gives a W-aspherical space Z with two components. The homotopy pullback of two points in different components is empty, again yielding $\varnothing \to * \in W$. Products with arbitrary categories then give $\varnothing \to A \in W$, and two-out-of-three gives every $A \to B$ in W. Conversely the total class satisfies the inclusion trivially and W_∞ satisfies it by B.7. The rigidity agrees with Cisinski (2006, Theorem 6.4.22 and Remark 6.4.23); the argument just given supplies the exact no-accessibility and empty-category qualifications. □

B.10. [Refuted] Complementarity for every finite truncation localizer. For every $n \geq 0$, $\mathrm{Liss}_{W_n} \cap \mathrm{Prop}_{W_n}$ is not contained in Fib_{W_n}. For $n = 0$ a finite witness is the regular covering $EC_2 \to BC_2$. For $n \geq 1$ B.8 supplies a completely specified witness with base the face poset of the boundary of an $(n+2)$-simplex.

Proof. The category EC_2 has objects $0, 1$, a unique arrow between every ordered pair, and forced composition. Send $i \to j$ to $j - i \pmod 2$ in the one-object group category BC_2. Every arrow has a unique outgoing and incoming lift, so the map is a discrete bifibration and smooth and proper for every W. The point map $e \to BC_2$ is in W_0, but its pullback has two discrete objects and maps to the connected category EC_2. It is not in W_0. All laws and component counts are checked by the finite program. For general n, let C_n have the nonempty proper subsets of $\{0, \dots, n+2\}$ as objects, inclusions as arrows, and inclusion composition. Its nerve is the barycentric subdivision of S^{n+1}. Choose any vertex x. The map $e \to C_n$ belongs to W_n. Pullback along $E_x(C_n) \to C_n$ has fibre of type ΩS^{n+1}, and total category weakly contractible. For $n \geq 1$, its π_n is $\mathbb{Z}$, so the resulting map is not in W_n. Every object, arrow and composition of this infinite counterexample was specified in B.8; the finite-base sphere assertion does not assert a finite total category. B.12 gives a finite total-category witness for $n = 1$. □

B.11. [Refuted] A single covering refutes complementarity for every nonzero homology theory. Let E be any nonzero spectrum, with no connectivity assumption, and let W_E mean equivalence on $E \wedge \Sigma^\infty(-)_+$. The three-sheeted action-groupoid covering in the prompt is both W_E-smooth and W_E-proper, but is not in Fib_{W_E}.

Proof. Let $F = \langle x, y \rangle$ be the free group and let it act on $\{1, 2, 3\}$ by $x = (12)$, $y = (13)$. The total category has these three objects and an arrow labelled $g \in F$ from s to $g(s)$; composition is multiplication in F. Its projection to the one-object category BF is a discrete fibration and opfibration. Each lifting category is a singleton, so both W_E conditions hold for the same localizer. Define $\alpha : F \to F$ by $\alpha(x) = x$ and $\alpha(y) = yxyx^{-1}y^{-1}$. On the graph $BF \simeq S^1 \vee S^1$, the stable reduced suspension-spectrum map of $B\alpha$ has matrix I_2, because the exponent sums are $(1, 0)$ and $(0, 1)$. Adjoining the basepoint adds an identity sphere summand. Hence $B\alpha \in W_E$ for every spectrum. The original action is transitive, whereas the pulled-back action sends both generators to (12) and has orbits $\{1, 2\}$ and $\{3\}$. The pullback map therefore identifies two components. To see the failure for nonconnective E as well, choose k with $\pi_k E \neq 0$. Point classes in the two source components give a nonzero difference, split by their component augmentations, which maps to zero because their target points are joined by a path. Thus the map is not an E-equivalence. The program verifies the permutation products, orbit sets and exponent-sum matrix. The infinite group itself is specified by reduced words with free reduction as multiplication, so no finite quotient is being substituted for the stated category. This confirms the inference proposed after PATHS. □

B.12. [Proved] A finite bifibration defeats both W_1 and rational local-system equivalences. Put $W_{\mathbb{Q}}^{\mathrm{lc}} = W_{D(\mathbb{Q})}^{\mathrm{lc}}$, where locally constant diagrams are coherent diagrams of unbounded rational complexes. There is a finite bifibration $p : H \to K$, with 14 objects and 100 arrows upstairs, which belongs to both $\mathrm{Liss}_W \cap \mathrm{Prop}_W$ for every fundamental localizer W, but belongs to neither Fib_{W_1} nor $\mathrm{Fib}_{W_{\mathbb{Q}}^{\mathrm{lc}}}$. The functor p itself belongs to $W_{\mathbb{Q}}^{\mathrm{lc}}$.

Proof. Let K be the poset of nonempty proper subsets of $\{0, 1, 2, 3\}$, with inclusions as arrows. For composable arrows $u : x \subseteq y$, $v : y \subseteq z$, put $c(u, v) = 1 \in C_2$ exactly for

$$(x, y, z) = (\{0\}, \{0, 1\}, \{0, 1, 2\}),$$

and put $c(u, v) = 0$ otherwise. A triple containing an identity satisfies the normalized cocycle equation by cancellation; there is no triple of strict inclusions. Hence c is a normalized 2-cocycle. Define H to have the objects of K, arrows (u, m) for $m \in C_2$, and composition

$$(v, n)(u, m) = (vu, m + n + c(u, v)), \qquad 1_x = (1_x, 0).$$

These formulas specify every identity and composite; the cocycle equation proves associativity. Projection to u defines $p : H \to K$. For a prescribed lift of a base arrow, every cartesian or cocartesian factorization is obtained by solving a unique equation in C_2. Thus p is a bifibration, with fibre BC_2 and identity transports on its automorphism group. Its lifting categories have terminal or initial objects, proving the two assertions about smoothness and properness for the same arbitrary W.

Here is a nonabelian fundamental-group calculation. Write $t_x = (1_x, 1)$ and $h_{xy} = (x \subseteq y, 0)$. Groupoid completion imposes

$$t_x^2 = 1, \qquad t_y h_{xy} = h_{xy} t_x, \qquad h_{yz} h_{xy} = h_{xz} t_x^{c(x,y,z)}.$$

After collapsing a spanning tree, all the t_x become a single central involution t. The nerve of K is the triangulated 2-sphere. Removing the interior of the one triangle supporting c leaves a disk with the same one-skeleton. Its triangle relations kill all remaining horizontal generators. The omitted triangle relation then reads $t = 1$. Consequently BK and BH are connected and simply connected. For the strict pullback $H \times_K H$, replace C_2 by C_2^2 and c by (c, c). The identical calculation leaves $C_2^2/\langle(1,1)\rangle \cong C_2$. The finite program checks the 1,552 associative triples of H, the 12,416 triples of this pullback, all cartesian and cocartesian factorizations, and performs Tietze eliminations in the full group presentations. Its outputs are respectively the trivial group and $\langle t \mid t^2 = 1\rangle$; it does not substitute first homology for a fundamental-group computation.

For W_1, pull back the W_1-map $e \to K$ along p. The resulting map $BC_2 \to H$ fails to induce an isomorphism on fundamental groups. This already proves $p \notin \mathrm{Fib}_{W_1}$. For the rational assertion we specify the coefficient criterion being used. If Z is connected and based, evaluation at the basepoint on $\mathrm{Fun}(Z, D(\mathbb{Q}))$ is conservative and preserves colimits. Its left adjoint has monad $M \mapsto C_*(\Omega Z; \mathbb{Q}) \otimes_{\mathbb{Q}} M$. The usual bar resolution therefore identifies the diagram category with modules over this differential graded loop algebra. For a based map $f : Z \to Z'$, restriction becomes restriction of scalars along $C_*(\Omega f; \mathbb{Q})$. It is an equivalence precisely when that algebra map is a quasi-isomorphism: necessity follows by applying the unit of extension–restriction to the regular module; sufficiency follows by its bar resolution. The criterion is unchanged on homotopy categories, since shifts detect all homotopy groups of mapping spectra. By Rivera, Wierstra, and Zeinalian (2021, Theorem 3), this is equivalent to an isomorphism on π_1 and rational isomorphisms on all higher homotopy groups. Applied componentwise, it also requires a bijection on components: a missing component prevents conservativity, and two components mapping into one prevent essential surjectivity by choosing zero and nonzero values on the two components.

By B.7 the strict fibre BC_2 is the homotopy fibre of $BH \to BK$. Its long exact sequence shows that $\pi_i(p)$ is an isomorphism for $i \geq 3$ and has kernel and cokernel of exponent two in degree two. Both fundamental groups vanish. The preceding rational criterion proves $p \in W_{\mathbb{Q}}^{\mathrm{lc}}$. Pulling this very map back along itself gives $H \times_K H \to H$, whose fundamental groups are $C_2 \to 1$. It is not in $W_{\mathbb{Q}}^{\mathrm{lc}}$; explicitly the sign local system on the source cannot be restricted from the simply connected target. This proves the second intrinsic-fibration failure with finite categories. □

B.13. [Proved] The covering witnesses have different behaviour for local coefficients. Every covering of ordinary categories, meaning a discrete fibration and a discrete opfibration, belongs to Fib_{W_1} and to $\mathrm{Fib}_{W_{\mathbb{Q}}^{\mathrm{lc}}}$. In particular the three-sheeted covering of B.11 belongs to these two intrinsic classes, although it belongs to no Fib_{W_E} for $E \neq 0$. The endpoint map of B.8 with $C = BC_2$ lies in the same two intrinsic classes, while it fails to lie in the rational-homology intrinsic class.

Proof. A covering has discrete strict fibres and bijective transports. Its nerve realizes a homotopy covering, and after any strict base change its strict pullback is the homotopy pullback, by B.7. On each connected component a covering is classified by a π_1-set. A W_1-map is bijective on components and induces isomorphisms of fundamental groups, so its pullback identifies orbit sets and corresponding stabilizer groups. It therefore induces a W_1-map of the total coverings. For $W^{\mathrm{lc}}_{\mathbb{Q}}$, use the criterion proved in B.12 in addition: a covering induces isomorphisms on all homotopy groups in degrees at least two, and hence the corresponding total-space map is a rational isomorphism in those degrees. This proves universal preservation, including disconnected or empty fibres and coverings with infinitely many sheets.

The map $E_x(BC_2) \to BC_2$ has contractible total space and homotopy fibre the discrete group C_2, so it represents the same homotopy covering as the universal two-sheeted cover. All its strict pullback squares compute homotopy pullbacks by B.7. The preceding argument therefore applies to it, without claiming that the actual categorical functor is a covering. The input $e \to BC_2$ is a rational-homology equivalence, since averaging contracts positive-degree rational group homology. Its pullback along the endpoint map is the inclusion of a fibre with two contractible components into a contractible total category; on $H_0(-;\mathbb{Q})$ it is $\mathbb{Q}^2 \to \mathbb{Q}$. This proves the failure for rational homology. In contrast, $e \to BC_2$ is neither a W_1-map nor a $W^{\mathrm{lc}}_{\mathbb{Q}}$-map. Likewise the free-group endomorphism of B.11 is not a W_1-map: it changes the orbit set of the specified action. Restriction of the corresponding rational permutation representation changes its invariant-vector dimension from one to two, so it is not a $W^{\mathrm{lc}}_{\mathbb{Q}}$-map either. The coefficient distinction is therefore essential to each use of the example. □

B.14. [Refuted] Cohomological fibrations are not the proper class. Let $\mathcal{D}$ be the derivator of a nonterminal presentable infinity-category, and use presheaf variance. Define a cohomological $\mathcal{D}$-fibration by requiring, after every strict base change, that right Kan extension sends every locally constant coefficient to a locally constant coefficient and has the canonical strict-fibre formula. This class and $\mathrm{Prop}_{W^{\mathrm{lc}}_{\mathcal{D}}}$ are not contained in one another. Dually, the homological class and $\mathrm{Liss}_{W^{\mathrm{lc}}_{\mathcal{D}}}$ are not contained in one another. In particular both conclusions apply to sets, spaces, and rational complexes.

Proof. Use first the map $p : V \to [1]$ in B.7. Its section s is left adjoint to p: an arrow $s(t) \to v$ is uniquely the arrow of $[1]$ from t to pv. This adjunction has identity unit, and its counit $sp \Rightarrow 1$ lies over the base. The same formulas define an adjunction after every strict base change $p_T : V_T \to T$. If L is locally constant on V_T, the counit induces an equivalence $L \simeq p_T^* s_T^* L$, since its components are arrows of V_T. Put $F = s_T^* L$, a locally constant object on T. The unit $F \to (p_T)_* p_T^* F$ is an equivalence. Indeed, at $t \in T$ its comma formula is induced by

$$V_T/t \longrightarrow T/t.$$

This functor has the left adjoint induced by s_T, so its opposite is initial for limits; the resulting limit is F_t. Equivalently, apply the adjunction contraction to the bar resolution of that limit. Consequently $(p_T)_* L \simeq F$ is locally constant. The strict fibres are $\{a < b\}$ or $\{c\}$, both contractible. Restricting the same unit to a fibre identifies its limit with F_t. These are the canonical comparison maps, natural under any further base change, which proves the required universal property. The outgoing lifting category from b over $0 \to 1$ is empty, so p is not $W^{\mathrm{lc}}_{\mathcal{D}}$-proper. The empty category is not $W^{\mathrm{lc}}_{\mathcal{D}}$-aspherical: restriction from the nonterminal category $\mathcal{D}(e)$ to the terminal category of diagrams on the empty category is not an equivalence.

Conversely, the cosieve $j : \{1\} \to [1]$ is an opfibration and therefore W-proper for every W. Choose a nonterminal coefficient M. Its right Kan extension is the presheaf with values the terminal object at 0 and M at 1; the transition $M \to 1$ is not an equivalence. Thus even this constant input has non-locally-constant right direct image. This proves the other failed

containment. Reversing all indexing categories and exchanging limits with colimits gives the homological assertion, using a noninitial coefficient. A nonterminal presentable category has both a nonterminal and a noninitial object. This concerns the universal condition on all local coefficients, not merely the direct image of one constant coefficient. The two tests must not be interchanged in the manuscript's Chapter I arguments. □

B.15. [Proved] Disposition of the inherited finite coefficient examples. The following table evaluates the maps specified in GALOIS D.3–D.5 for $W = W_1$ and $W = W^{\mathrm{lc}}_{\mathbb{Q}}$. A pair of entries gives these two localizers in that order; a single entry applies to both. Here "smooth" and "proper" always use the same W as the intrinsic column.

specified map	W-smooth	W-proper	Fib_W
D.3 : $p : E \to B$	yes	yes	yes
D.3 : $q : B \to [1]$	no	yes	no
D.3 : qp	no	yes	no
D.4 : $p : E \to B$	yes	yes	no
D.4 : $q : B \to [1]$	no	yes	(yes, no)
D.4 : qp	no	yes	no
D.5 : $H \to K$	yes	yes	no
D.5 : $q : R \to [1]$	no	yes	no

Proof. We give the constructions, so the table does not use the earlier answer as an authority. In D.3, B has objects a, b, $B(a,a) = \{1\}$, $B(b,b) = C_2$, $B(a,b) = C_2$, and $B(b,a) = \varnothing$; postcomposition on $B(a,b)$ is addition in C_2. The projection q is the opfibration of $e \to BC_2$ over [1]. Put $L(a) = L(b) = C_2$, with every labelled arrow acting by addition, and $E = \int_B L$. Thus p is a covering and B.13 applies. For q, the incoming lift category of $0 \to 1$ at b is a discrete two-point category. For qp the corresponding category is again discrete with two objects. Neither is aspherical for either localizer. All three maps are opfibrations and hence proper. The inclusion of the fibre at 0 into $B \simeq BC_2$ fails on π_1, and the inclusion of the fibre C_2^{disc} into $E \simeq EC_2$ fails on components. Pulling back $\{0\} \to [1]$ gives exactly these witnesses for the two failures of intrinsic membership.

For D.4, take K, H from B.12 and $x_0 = \{0\}$. The category B has K and a new object a, arrows $B(a,x) = K(x_0, x)$, no arrows into a except its identity, and the compositions forced by K. Let $r : B \to K$ send a to x_0, let q be the opfibration of $e \xrightarrow{x_0} K$, and put $E = B \times_K H$. The map $p : E \to B$ is a bifibration. Its pullback along $K \hookrightarrow B$ is $H \to K$, so B.12 excludes intrinsic membership. The retraction r and the arrows $a \to x_0$ give $BB \simeq BK \simeq S^2$; their label-zero lifts give $BE \simeq BH$. The fibre of qp at 0 is BC_2, so its inclusion in E fails both localizers on fundamental groups. Objects x of K not containing 0 have no incoming lift from colour 0, excluding smoothness of q and qp. Both are opfibrations. For every $T \to [1]$, the fibres of q_T are a point or the simply connected category K. In the groupoid completion of its Grothendieck construction, every fibre object has a unique path from the chosen section object. Factorization into a cocartesian arrow followed by a fibre arrow proves naturality of these paths. Thus $\Pi_1(q_T)$ is an equivalence, for every T, and two-out-of-three gives $q \in \mathrm{Fib}_{W_1}$. For rational local systems, the pullback of $\{0\} \to [1]$ is $e \to B \simeq S^2$, which fails on $\pi_2 \otimes \mathbb{Q}$.

Finally R has objects b, a, automorphism group C_2 at b, identity only at a, one arrow $b \to a$, and the relation that precomposition by either automorphism fixes that arrow. Its projection is the opfibration of $BC_2 \to e$. The incoming lift category at a is BC_2, excluding smoothness for both localizers. The category R has terminal object a; the fibre inclusion $BC_2 \to R$ fails on π_1 and detects the intrinsic failure. The first D.5 map is precisely B.12. The code specifies these finite categories by tables and checks their units, associativity, all

relevant lift categories and component data. Thus the proofs retain the different roles of weak constant, fully faithful, and equivalent local-coefficient tests. □

B.16. [Proved] Accessible coefficient localizations and the limit of the general statement. If W is an accessible fundamental localizer, let $\mathcal{S}_W$ be the presentable localization of spaces corresponding to W. Then

$$(\mathrm{Cat}/S)[(W_S^{\mathrm{pre}})^{-1}]_\infty \simeq \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S}_W), \qquad (\mathrm{Cat}/S)[(W_S^{\mathrm{co}})^{-1}]_\infty \simeq \mathrm{Fun}(NS, \mathcal{S}_W).$$

For an arbitrary fundamental localizer the relative definitions, the opposition isomorphism, B.6 and B.9 remain valid without accessibility. A presentable coefficient category in the displayed formulas is not asserted without the accessibility hypothesis.

Proof. For a strict diagram $F : S \to \mathrm{Cat}$, write $\int_S F$ for the opfibred Grothendieck construction. The comma functor R_S of B.6 and this construction are relative functors between $(\mathrm{Cat}/S, W_S^{\mathrm{co}})$ and the category of strict S-diagrams with pointwise W-maps. Indeed, R_S detects its specified class by its definition, while $(\int_S F)/s \to F(s)$, given by cocartesian transport, is left adjoint to the fibre inclusion. It is consequently in W, naturally in F and s. The natural unit $X \to \int_S R_S X$ is fine by B.6, and the natural evaluation $R_S \int_S F \to F$ is pointwise in W. Their localizations are inverse. For accessible W, the combinatorial Thomason model category of Cisinski (2006, Theorem 5.2.15) presents $\mathcal{S}_W$. Its category of strict diagrams has the derived diagram comparison: Cisinski (2019, Example 7.9.6 and Theorem 7.9.8) apply using functorial model factorizations and small homotopy colimits. Thus its pointwise localization is $\mathrm{Fun}(NS, \mathcal{S}_W)$. This verifies the strict-diagram step, rather than inferring it from an absolute equivalence of homotopy categories. Opposition proves the first formula. The hypotheses of the purely relative arguments in B.1, B.6 and B.9 do not involve the model structure or a small generating set, which proves the last assertion. For example W_0 gives set-valued diagrams, whose coarse localization need not identify with their locally constant part; Part E treats that separate obstruction. □

B.17. [Known] Intrinsic fibrations for truncation localizers. For $n \geq 0$, a functor $p : T \to B$ lies in Fib_{W_n} if and only if, for every $X \to B$, the canonical map

$$\tau_{\leq n} B(T \times_B X) \longrightarrow \tau_{\leq n} BT \times_{\tau_{\leq n} BB} \tau_{\leq n} BX$$

is an equivalence of n-types. The pullback on the right is a homotopy pullback in $\mathcal{S}_{\leq n}$. For $n = 0$ this is equivalent to the following elementary pair of conditions. For every connected component B_j of B, write $T_j = T \times_B B_j$. Then $\pi_0(T_b) \to \pi_0(T_j)$ must be bijective for each $b \in B_j$; and for every arrow $\beta : b \to b'$ and every component c of T_j there must be an arrow of T over β in component c.

Proof. The general equivalence is precisely the intrinsic weak-fibration criterion of Cisinski (2006, Proposition 6.2.3). Its proper-localizer hypothesis is supplied for W_n by that book's §9.2.1, and Corollary 9.2.15 identifies its weak equivalences with the asserted truncations. These hypotheses matter: the criterion is not being applied to a nonproper homology localizer. For clarity, the elementary specialization can also be proved directly. Pullback of $e_b \to B_j$, a W_0-map, gives the first condition. Pullback of either endpoint of $[1] \xrightarrow{\beta} B$ then forces an arrow joining the two layers in each prescribed component, giving the second. Conversely, for connected $X_a \subset X$, label objects of $T \times_B X_a$ by their component c in T_j. Objects over any fixed $x \in X_a$ with a given label form a nonempty connected category by the first condition.

Every arrow of X_a has a lift connecting the associated labelled categories by the second. Finite zigzags therefore connect exactly the objects with the same label. Hence, naturally in X,

$$\pi_0(T \times_B X) \cong \pi_0(T) \times_{\pi_0(B)} \pi_0(X).$$

This formula preserves W_0-maps and proves sufficiency, including empty fibres. It also explains directly why the regular C_2 covering fails B.10. □

B.18. [Proved] Useful bounds for a homology intrinsic class. For every spectrum E, Fib_{W_E} contains product projections and inclusions of unions of connected components, together with their closure under composition, strict base change and retracts. It is contained in the class of $p : T \to B$ for which every inclusion

$$T_{\sigma(i)} \longrightarrow T \times_B [m], \qquad \sigma : [m] \to B, quad0 \le i \le m,$$

is an E-equivalence. For $E \neq 0$ this upper bound is strict.

Proof. For a product projection the induced map is $u \times 1_K$. The identity $(X \times Y)_+ \cong X_+ \wedge Y_+$ identifies its E-homology comparison with the smash product of the E-equivalence of u and $\Sigma^\infty(BK)_+$. This is an equivalence. For a component inclusion the nerve is a disjoint union of components; restriction selects summands of a componentwise equivalence. The three closure assertions follow from iterated pullbacks and retracts. The inclusion $\{i\} \to [m]$ is a weak homotopy equivalence, so its pullback gives each necessary test in the display. The covering of B.11 passes every one of these tests: over a simplex it is a constant covering, each fibre inclusion being a weak homotopy equivalence. B.11 proves that it nevertheless fails intrinsic membership for every nonzero E. This proves strictness without identifying model fibrations of a nonproper localization with intrinsic fibrations. □

B.19. [Proved] The full fence construction and its mapping-path factorization. Let $\mathcal{P}(C)$ be the category of all eventually constant diagrams $J \to C$ on the integer fence of B.8, with all natural transformations. Write σ, β for the eventual left and right values. Then $(\sigma, \beta) : \mathcal{P}(C) \to C \times C$ is smooth and proper for W_∞, and therefore for every fundamental localizer. For every $f : A \to C$, the functors

$$E(f) = A \times_{C,\sigma} \mathcal{P}(C), \qquad i_f(a) = (a, \mathrm{const}_{f(a)}), \qquad p_f(a, \gamma) = \beta(\gamma)$$

give $f = p_f i_f$, with $i_f \in W_\infty$ and p_f smooth and proper. All formulas are natural in strictly commuting squares. For $C = [1]$, the endpoint functor is neither a Grothendieck fibration nor an opfibration. Thus the verified fence construction in PATHS has exactly the asserted homotopical properties, without being a bifibration.

Proof. For an outgoing endpoint pair (α, β) from a path γ, choose a sufficiently negative odd integer a and sufficiently positive even integer b. Replace the left tail through a by the source-target of α and join it at $a + 1 \to a$ using α; replace the right tail after b by the target of β and join it at $b \to b + 1$ using β. There is a natural transformation from γ to this new path, identity between the joins and the prescribed endpoint arrows outside. Among outgoing lifts stationary outside this interval, it is initial: the comparison to any other lift is that lift's transformation inside, and the endpoint identities outside. Naturality at the two joins is precisely the two original commutative squares. Uniqueness follows from the identity endpoint components and stationary tails. Letting the interval expand gives a filtered union of these contractible categories; every simplex and every finite homotopy in the nerve belongs to a stage. Thus the entire lifting category is weakly contractible. For incoming lifts use an even

left boundary and an odd right boundary; the same argument gives terminal stage objects. This proves smoothness and properness by the exact lifting tests. It is the two-ended version of B.8, with the parity choices specified.

The functor σ is the composite of this endpoint map and a product projection, so is also smooth and proper. Its strict fibre at x, with identity left endpoint on morphisms, is exactly $E_x(C)$ of B.8 and is weakly contractible. B.7 now shows that every strict base change of σ is a weak equivalence. In particular $E(f) \to A$ is one; its section i_f is a weak equivalence by two-out-of-three. The map $E(f) \to A \times C$ is a base change of the full endpoint map, and projection $A \times C \to C$ is both smooth and proper. Stability under composition proves the assertion for p_f. B.7 also identifies its strict fibres with the homotopy fibres of Bf. Applying a commuting square pointwise to these diagrams proves strict naturality.

For the failure of a fibration, incoming lifts into the constant 1 path over $(0 \to 1, 1)$ are the eventual [1]-paths from 0 to 1, ordered pointwise. A proposed terminal path is still 0 sufficiently far left; a single-transition path changing to 1 earlier has no map to it. Dually, in the outgoing lift category from constant 0 over $(1, 0 \to 1)$, a proposed initial path changes to 1 before some later single-transition path and has no map to that later path. Single transitions are taken across an even-to-odd forward edge, so all these diagrams are legitimate fence functors. The obstructions hold at arbitrarily large supports, which is why a finite cut-off cannot establish existence of universal lifts. This proof verifies the exact convention of PATHS E.4–E.5; it does not identify it with every convention called an Evrard cocylinder (Evrard, 1975). □

B.20. [Refuted] Reversing path words does not repair the all-type endpoint functor. Use the all-type category $\mathrm{Ch}(C)$ with objects finite oriented interval diagrams and morphisms (h, η), where h is an endpoint-preserving, weakly monotone functor of their interval types and $\eta : \gamma \Rightarrow \delta h$. Its endpoint map is smooth for every C, but for $C = [1]$ it is not proper and is neither a Grothendieck opfibration nor a fibration. The conclusion cannot be transferred from B.19 by calling both objects "path categories" or by reversing the path word.

Proof. First fix a path γ of type τ and incoming endpoint arrows $\alpha : a' \to a$, $\beta : b' \to b$. Adjoin these arrows to obtain the path δ of type $(+, \tau, -)$ from a' to b', with its collapse map $d : \delta \to \gamma$. Let L be the incoming lifting category and let $R = \mathrm{Ch}(C; a', b')/\delta$. Composition with d defines $i : R \to L$, and R has terminal object 1_δ. For a lift $(h, \eta) : \zeta \to \gamma$, pad ζ with an identity edge at each end. The extended type map $(+, h, -)$ and the transformation whose middle is η and whose new endpoints are identities define an arrow of this padded path to δ. The joining naturality equations are precisely $\eta_{\mathrm{left}} = \alpha$ and $\eta_{\mathrm{right}} = \beta$. This defines a functor $F : L \to R$. Collapsing the two added identity edges gives a natural transformation $iF \Rightarrow 1_L$ over γ. Hence BL is homotopy dominated by the contractible space BR, and is itself contractible. This proves smoothness, with the exact type convention retained.

Let e be the empty interval type. An endpoint-preserving type map $e \to \tau$ forces the two endpoints of τ to coincide, so $\tau = e$. For the empty path at 0, the base arrow $(1_0, 0 \to 1) : (0, 0) \to (0, 1)$ consequently has no outgoing lift. Its outgoing lift category is empty. This proves failure of properness and of an opfibration, for every nontrivial component-detecting localizer. For failure of a fibration use the empty path at 1 and the incoming base arrow $(0 \to 1, 1_1)$. Its incoming lift category is $\mathrm{Ch}([1]; 0, 1)$, since collapse to e and the transformation to constant 1 are unique. Suppose a terminal path c had nonempty type τ. Form d on the concatenated type $\tau\tau$, constant at 0 on the first block and equal to c on the second. There are two distinct type maps $\tau\tau \to \tau$, collapsing respectively the first and the second block. The first makes d exactly the pullback of c. For the second pullback, the first block is c and the second constant at 1; the pointwise inequalities define a natural transformation from d. Both transformations have identity endpoints. Hence there are two distinct arrows $d \to c$, contradicting terminality. All identities and composites in this category are respectively $(1, 1)$ and $(kh, (\theta h)\eta)$, so the

argument specifies the actual infinite counterexample and its type-map distinction. Taking the opposite of a Grothendieck construction reverses its indexing category of types. Reversing each interval word within the unchanged indexing category does not perform that operation. This locates the exact variance error in the proposed deduction of properness at VII ms. 161–166. The two distinct constructions therefore confirm the classification in PATHS E.3–E.5. The factorization used here is the independently verified fence factorization of B.19. □

B.21. [Open] Replacing a smooth and proper representative by a bifibration. For the one-object category of the monoid $R = \{1, e_{00}, e_{01}, e_{10}, e_{11}\}$ with $e_{ij}e_{kl} = e_{il}$ and $1r = r1 = r$, it remains unsettled here whether the local system of path spaces has a representative by a strict Grothendieck bifibration with contractible total category. The eventual fence endpoint functor is not such a representative.

Exact remaining issue and verified obstruction. Put $e = e_{00}$. Both eR and Re consist of nonunits. For incoming lifts to a constant fence path over endpoint pair $(1, e)$, a natural transformation has eventual components 1 on the left and e on the right. A candidate terminal lift has some component equal to e. Move the single change from 1 to e farther right, across an odd-boundary incoming edge, to obtain another lift whose component there is still 1. A map to the candidate would require $e\theta = 1$, impossible. For outgoing lifts the same construction across an even boundary would require $\theta e = 1$, again impossible. These equations verify the failure in the precise monoid, whose 125 associativity equations and nonunit ideals are checked by the program. They do not obstruct a different total category equipped with coherent cartesian and cocartesian universal arrows. Constructing that representative, or an invariant forbidding every such representative, is the remaining statement. No bifibration-existence assertion is used in Parts B or E. □

B.22. [Proved] The maximal simplicial classes and the all-base recipe. For every simplicial base S, let W_S^{cov} and W_S^{contra} denote the actual covariant and contravariant equivalences. The maps whose every base change preserves these classes under raw pullback are, respectively, Cisinski-smooth and Cisinski-proper maps. Moreover the manuscript's universal test recipe gives exactly

$$W_S^{\mathrm{Liss}_{\mathrm{Cis}}} = W_S^{\mathrm{cov}}, \qquad W_S^{\mathrm{Prop}_{\mathrm{Cis}}} = W_S^{\mathrm{contra}}.$$

Thus opposition gives a perfect duality over S^{op} for every simplicial base, although B.4 refutes the version using literal slices at an arbitrary base. For accessible W, replace fibrewise equivalences of spaces by fibrewise W-equivalences after left or right fibrant replacement. These two accessible localizations classify diagrams with values in $\mathcal{S}_W$ on the directed infinity-category presented by S or its opposite, and their restrictions to ordinary nerves are B.16. Their universal raw-pullback classes contain the minimal smooth and proper classes. Complementarity with intrinsic total W-equivalences holds for these classes only for W_∞ and the total localizer.

Proof. Pullback along a Cisinski-smooth map is left Quillen for the covariant model structures (Cisinski, 2019, Proposition 4.4.18); this remains true after every base change. Conversely, suppose a simplicial map p has the stated universal preservation property. If $i : X \hookrightarrow Y$ is left anodyne and Y maps to the base of p, regard i as a covariant weak equivalence over its own target Y. The pullback i' is a monomorphism and a covariant weak equivalence over its own target. Such a map is left anodyne: the trivial-cofibration lifting property in that slice applies to every left fibration over the target, which is precisely the defining left anodyne lifting property. Consequently all pullbacks of left-anodyne maps along p are left anodyne. The opposite of Cisinski (2019, Proposition 4.4.4) identifies this condition with Cisinski-smoothness.

This proves maximality. A covariant equivalence passes every smooth pullback test and then is a total weak equivalence. Conversely the tests include the right-fibrant replacements of each vertex, which are smooth right fibrations. B.2's all-base formulation detects covariant equivalences on precisely those tests. Opposition proves the entire contravariant statement.

For accessible W, compose the covariant classifier with pointwise localization $\mathcal{S} \to \mathcal{S}_W$. This is an accessible reflective localization, computed pointwise, and its inverse image specifies the claimed weak equivalences. A choice of a categorical fibrant replacement of S identifies its target with diagrams on the infinity-category presented by S; no literal $\mathrm{Fun}(S, \mathcal{S}_W)$ with a nonfibrant simplicial set as infinity-category is used. If p is minimally smooth, replace an arbitrary $X \to S$ by a left fibration $X^{\mathrm{lf}} \to S$. The unit remains a minimal covariant equivalence after pullback along p. For left fibrations the new weak equivalences are fibrewise W-equivalences, and arbitrary pullback preserves these tests exactly. Two-out-of-three proves preservation for arbitrary X, Y. This proves containment of the minimal smooth class, and its dual. If their intersection universally preserved total W-equivalences, the nerves of the minimal smooth and proper factorizations in B.9 would imply homotopy-pullback stability of W. The proof of B.9 then forces W_∞ or the total class. The two converse cases follow from B.7 or vacuity. In particular every finite and homological counterexample above gives a simplicial counterexample simply by taking nerves. This argument does not rename a general W-smooth map "Lurie-smooth": Lurie's unqualified term uses the minimal space-valued theory. □

3 Presheaf models, test categories, and strict higher categories

C.1. [Proved] What a presheaf model supplies before choosing a direction. Let A be any small category, $\widehat{A} = \mathrm{Set}^{A^{\mathrm{op}}}$, and $i_A X = A/X$ its category of elements. For a fundamental localizer W put $W_A = i_A^{-1} W$. For each $S \in \widehat{A}$ there is a canonical isomorphism

$$\widehat{A}/S \cong \mathrm{Set}^{(A/S)^{\mathrm{op}}}.$$

There are two specified, weakly saturated classes on this slice obtained from B.1:

$$W_S^{\mathrm{pre,el}} = i_A^{-1} W_{A/S}^{\mathrm{pre}}, \qquad W_S^{\mathrm{co,el}} = i_A^{-1} W_{A/S}^{\mathrm{co}}.$$

Both are contained in the total class W_A over S. These define localizations for any A, but test-category theory alone does not identify them with the two simplicial theories of Part B.

Proof. The displayed isomorphism sends $X \to S$ to the presheaf $(a, s) \mapsto \{x \in X(a) : p(x) = s\}$. Its inverse takes the disjoint union of these fibres over $s \in S(a)$, with the specified restriction maps. The two constructions are inverse on objects and arrows, and the resulting category of elements over A/S is canonically $A/X \to A/S$. Pulling back the two functorial comma tests of B.1 consequently gives the asserted classes, their weak saturation and their containment in the total class. This constructs actual candidates without assuming an opposition functor on $\widehat{A}$. If W is accessible and A a W-local test category, the absolute class W_A supports the monomorphism model structure of Cisinski (2006, Corollary 4.2.18). If A is a W-test category, its absolute homotopy theory presents $\mathcal{S}_W$ by that book's Theorem 4.2.15 and the test-category comparison. Neither assertion mentions or chooses directed slices. C.2–C.3 give an explicit reason that the latter choice cannot be silently inferred from the absolute comparison. □

C.2. [Refuted] The two category-of-elements candidates need not be dual, even on a test category. Assume $W \subseteq W_0$. For every small A and $S \in \widehat{A}$, $W_S^{\mathrm{pre,el}}$ consists exactly of isomorphisms. For $A = \Delta$, $S = 1$, and $W = W_\infty$, $W_1^{\mathrm{co,el}}$ is instead the ordinary class of

weak homotopy equivalences of simplicial sets. Their enhanced localizations are not equivalent. Thus transporting both fine tests through the category of elements does not give an automatic extension of Part B to presheaves.

Proof. Fix $(a, s) \in A/S$. An object of $(a, s)\backslash(A/X)$ is $(b, x, h : a \to b)$ with $S(h)p(x) = s$. Its invariant $X(h)x \in X(a)_s$ is preserved by every morphism. In the subcategory with invariant x_0, the object $(a, x_0, 1_a)$ is initial: its unique arrow to the displayed object has underlying arrow h. Hence this comma category is the disjoint union of contractible categories indexed by $X(a)_s$. A map on such categories is in $W \subseteq W_0$ if and only if it is bijective on these indices. Necessity follows from components; sufficiency follows because a bijection of the contractible components is already a W_∞-map. Requiring the condition for all (a, s) says exactly that $X \to Y$ is objectwise bijective, hence an isomorphism of presheaves over S.

For the other comma orientation there is a strict isomorphism

$$(A/X)/(a, s) \cong A/(X \times_S h_a),$$

where $h_a \to S$ is the map represented by s. An object on either side is $(b, x, h : b \to a)$ satisfying $p(x) = S(h)s$, with the same triangle condition on arrows. For $A = \Delta$ and $S = 1$, these tests are the weak equivalences $X \times \Delta^n \to Y \times \Delta^n$. The last-vertex comparison identifies the classifying space of their simplex categories with their realizations. The case $n = 0$ is $X \to Y$, and projection from a product with Δ^n is a weak homotopy equivalence, proving the asserted characterization. Localization at isomorphisms is the nerve of the ordinary category of simplicial sets, with discrete mapping spaces. Localization at weak homotopy equivalences is $\mathcal{S}$, which has, for example, $\mathrm{Map}(*, S^1) \simeq S^1$. An equivalence preserves mapping spaces and cannot identify these two infinity-categories. This is a counterexample to the specified category-of-elements extension, not a denial of the simplicial opposition in B.22: that opposition uses the directed simplex structure, which this transport of the two tests has failed to preserve. □

C.3. [Refuted] Elementwise smoothness and properness do not admit a general exchange involution. Define a map $u : X \to Y$ in $\widehat{A}$ to be elementwise W-smooth or W-proper when $i_A u : A/X \to A/Y$ has that categorical property. Every presheaf map is elementwise W-smooth, for every fundamental localizer. If $W \subseteq W_0$, a map is elementwise W-proper exactly when all its naturality squares are pullbacks of sets. Already for the test category $A = \Delta$, no autoequivalence of $\widehat{A}$ can exchange these two classes.

Proof. The functor $A/X \to A/Y$ is a discrete fibration. Given $h : (b, y') \to (a, y)$ and a lift (a, x) of (a, y), its unique cartesian lift has source $(b, X(h)x)$. Thus its incoming lift categories have terminal objects and it is smooth for every W. An outgoing lifting category, for $x \in X(a)$ and $h : (a, u(x)) \to (b, y)$ in A/Y, is the discrete set

$$\{x' \in X(b) : X(h)x' = x,\ u(x') = y\}.$$

There are no nonidentity vertical arrows, since the category-of-elements map is a discrete fibration. This set is W-aspherical for $W \subseteq W_0$ exactly when it is a singleton. For all choices this is exactly bijectivity of $X(b) \to X(a) \times_{Y(a)} Y(b)$, or the cartesian naturality condition. For $A = [1]$, the presheaf $X(0) = \{*\}$, $X(1) = \varnothing$ mapping to the terminal presheaf gives a nonsingleton, indeed empty, outgoing lifting set over $0 \to 1$. For the test category Δ, take $\partial\Delta^1 \hookrightarrow \Delta^1$, the vertex 0, and the arrow in the element category from that vertex to the nondegenerate edge. It has no outgoing lift in the boundary. Thus the proper class is strictly smaller than all maps, whereas the smooth class is all maps. An autoequivalence takes all maps to all maps and cannot exchange them. This proves a precise negative answer for a natural pair on a presheaf category and identifies the missing directional input. □

C.4. [Proved] The extra structure that does transport a duality. Let M have pullbacks and a weakly saturated class W, and let $\mathcal{L}_+, \mathcal{L}_-$ be two pullback-stable classes. Suppose $D : M \to M$ is an involutive equivalence preserving W, together with its canonical pullback comparisons, and carrying $\mathcal{L}_+$ to $\mathcal{L}_-$. Then the universal-test classes satisfy

$$D(W_S^{\mathcal{L}_+}) = W_{DS}^{\mathcal{L}_-},$$

and D induces an equivalence of the corresponding enhanced slice localizations. It also exchanges their universal raw-pullback-preserving classes. Such data are sufficient; being a test category supplies none of the required exchange isomorphisms by itself.

Proof. For $u : X \to Y$ over S and $a : T \to S$ in $\mathcal{L}_+$, the pullback comparison of the equivalence D identifies $D(a^*u)$ with $(Da)^*Du$. Membership in W is equivalent on the two sides. Every map in $\mathcal{L}_-$ to DS is, up to the equivalence, of this form. This proves the displayed equality in both directions. An equivalence of relative categories induces an equivalence of their infinity-localizations. Applying the same calculation after a further arbitrary base change proves the last exchange assertion. This proof uses no model structure. For simplicial sets the involution is induced by the automorphism $\rho : \Delta \to \Delta$, $\rho(h)(i) = n - h(m - i)$ for $h : [m] \to [n]$; it is an automorphism of Δ, not a claimed equivalence $\Delta \simeq \Delta^{\mathrm{op}}$. B.22 supplies the exchange of its smooth and proper tests. For a general indexing category an involution must be chosen and its exchange property proved. Replacing A by A^{op} instead changes the presheaf category to copresheaves and need not give an equivalence: for the fork $a \leftarrow b \to c$, the ordinary presheaf category recovers the idempotent completion of A as its noninitial indecomposable projectives. Indeed a projective is a retract of a coproduct of representables and an indecomposable one is a retract of one representable; a finite poset has no nonsplit idempotents. The fork and its opposite have different numbers of minimal and maximal objects, so the two reconstructed categories are not equivalent. The analogous infinity-categorical distinction is used in Part G. □

C.5. [Known] Theorem A for the required strict higher slices. For $1 \le n \le \omega$, write $V^{(n)}$ for strict n-functors whose Street nerves are weak homotopy equivalences; this notation is distinct from the truncation localizer W_n of B.10. Use the under-slices of Ara and Maltsiniotis (2018, §4) and their total-dual over-slices. For a strictly commuting triangle $X \xrightarrow{u} Y \to S$, either of the conditions

$$s\backslash u \in V^{(n)} \quad (s \in S_0), \qquad \text{or} \qquad u/s \in V^{(n)} \quad (s \in S_0)$$

implies $u \in V^{(n)}$. The same result allows the specified oplax triangle in the cited theorem.

Exact reference and conventions. The under-slice assertion is Ara and Maltsiniotis (2018, Theorem 5.2 and Remark 5.3), *Un théorème A de Quillen pour les ∞-catégories strictes I: la preuve simpliciale*. The over-slice assertion is Ara and Maltsiniotis (2022, Corollary 6.8), with the total-dual slice isomorphism in its proof. The constructions restrict to strict n-categories; they do not require a model structure. This resolves the titles and variance of the result cited from memory in the prompt. The 2-categorical predecessors include Chiche's *Théories homotopiques des 2-catégories*; the all-dimensional statement used here is the stated Ara–Maltsiniotis theorem, rather than an inference from the existence of a Street nerve. □

C.6. [Proved] Two strict higher relative theories and a definite duality. For a strict small n-category S, define V_S^{pre} and V_S^{co} by the two slice tests in C.5. They are weakly

saturated and contained in the total class $V^{(n)}$. Let D_{tot} reverse every positive-dimensional cell. Then

$$(n\mathrm{Cat}/S, V_S^{\mathrm{pre}}) \cong (n\mathrm{Cat}/D_{\mathrm{tot}}S, V_{D_{\mathrm{tot}}S}^{\mathrm{co}})$$

as relative categories, and hence their enhanced localizations are equivalent. This conclusion holds for every $1 \leq n \leq \omega$ and uses none of the claimed September 2026 model structures.

Proof. For each object s, the slice construction is a functor on the strict slice category, so identities, retracts and two-out-of-three are inherited from Street weak equivalences. C.5 proves containment in the total class. There is a canonical strict isomorphism

$$D_{\mathrm{tot}}(X/s) \cong s\backslash D_{\mathrm{tot}}X.$$

Theorem 6.6 of Ara and Maltsiniotis (2022) proves invariance of Street weak equivalences under every cell duality. The displayed identity and this invariance prove the relative-category isomorphism, including both directions; applying infinity-localization proves the claim. These are the particular lax and colax conventions fixed in C.5. Changing which higher coherence cell is reversed changes the slice convention and must be accompanied by the corresponding cell duality. □

C.7. [Proved] What replaces the single opposite in strict dimension n. For each subset $J \subseteq \{1, \ldots, n\}$, or $J \subseteq \mathbb{N}_{>0}$ when $n = \omega$, reversing the j-cells for $j \in J$ defines an involutive automorphism D_J of strict n-categories and preserves $V^{(n)}$. The generators commute and compose by symmetric difference. Define $\mathrm{Liss}^{(n)}$ to be the maps whose every strict base change preserves V^{co} under raw pullback, and $\mathrm{Prop}^{(n)}$ using V^{pre}. Then both classes contain isomorphisms and are stable under composition and strict base change; D_{tot} exchanges them. These definitions alone do not identify them with a higher lifting-category criterion or with smooth/proper Street-nerve maps.

Proof. Reversal exchanges the selected source and target operations and the corresponding order of composition. The globular identities and interchange law are invariant under these simultaneous exchanges. Applying the same reversal twice is identity, and independent exchanges commute. This proves the automorphism and symmetric-difference assertions. Its preservation of $V^{(n)}$ is the verified theorem cited in C.6. For the preservation classes, composition of raw pullbacks is raw pullback of the composite, up to its canonical isomorphism. A further base change is another member of the universally quantified family. These two facts prove the closure assertions. The slice isomorphism of C.6 identifies the preservation test for p with the opposite test for $D_{\mathrm{tot}}p$, proving exchange. For $n = 2$ the generators are usually called op and co, and their product coop is the total dual. The formal result concerns those specified tests; an unproved higher analogue of B.6 is not hidden in the notation $\mathrm{Liss}^{(n)}$. □

C.8. [Known] The verified absolute model structure in dimension two. The category of strict 2-categories has the proper combinatorial Thomason model structure with weak equivalences detected by the Street nerve, and $c_2\,\mathrm{Sd}^2 \dashv \mathrm{Ex}^2\,N_2$ is a Quillen equivalence with simplicial sets in the Kan–Quillen structure.

Exact opened theorem. This is Ara and Maltsiniotis (2014, Theorem 6.27 and Corollary 6.32), *Vers une structure de catégorie de modèles à la Thomason sur la catégorie des n-catégories strictes*. The distinction between the Quillen adjunction in Theorem 6.27 and the Quillen equivalence in Corollary 6.32 is retained. Neither theorem identifies the two fine relative classes over a higher base. The ordinary nerve in Part B and the twice-extended Street nerve here are different comparison functors. □

C.9. [Proved] The exact coarse consequence of an absolute higher model theorem. Suppose, for a fixed n, that the model structure and Quillen equivalence claimed in family 317 are valid (OpenAI, 2026b, claimed, unrefereed). Then the total-weak-equivalence localization of $n\mathrm{Cat}/S$ presents $\mathcal{S}_{/|N_nS|}$, the slice of spaces over the classifying space of the Street nerve. It has the usual derived postcomposition and pullback adjunctions. This is a coarse statement. It does not imply recognition of C.6's fine localizations or any universal base-change property for the classes in C.7. For $n = 2$ the premise is already verified by C.8.

Proof. The assumed Quillen equivalence identifies the absolute enhanced localization with spaces and the image of S with $|N_nS|$, since the natural extension map $N_nS \to \mathrm{Ex}^2 N_nS$ is a weak homotopy equivalence. The slice model category presents the corresponding infinity-categorical slice. One may check a nonfibrant base by choosing a fibrant replacement $S \to S^{\mathrm{f}}$: right properness makes the postcomposition–pullback adjunction between these two model slices a Quillen equivalence. For a fibrant base the ordinary slice-model comparison computes the derived slice directly. This supplies the needed comparison for every S, not just fibrant higher bases. The resulting adjunctions are those in the slice of spaces. The fine classes require tests by higher comma categories and retain directed information; none of these tests occurs in the absolute model hypothesis. Already for $n = 1$ and $S = [1]$, the coarse category is $\mathcal{S}$ whereas the fine category is the arrow infinity-category of spaces. Thus the absolute theorem cannot, by this slice argument, prove the requested fine recognition. This conditional argument does not certify the unreviewed 2026 manuscript or its formalization. □

C.10. [Open] The strict higher fine recognition and base-change problem. For a general strict n-category S, $2 \le n \le \omega$, the following specific extension is not proved here: the Street nerve induces equivalences from the two enhanced localizations in C.6 to the actual covariant and contravariant localizations over N_nS, with the slice conventions matched as in C.5; moreover C.7's preservation classes admit a local higher comma-lifting criterion and satisfy the universal Kan base-change axioms of Part E. These are required before declaring every such $(n\mathrm{Cat}, V^{(n)}, \mathrm{Liss}^{(n)}, \mathrm{Prop}^{(n)})$ an instance of the abstract theorem. Accepting family 317 does not remove these requirements.

The missing comparisons. C.5 proves only that the fine tests imply total weak equivalence. It does not identify a higher comma nerve with a right or left fibrant replacement of a vertex, nor provide inverse relative functors as in B.3. For an ordinary nerve the strict simplex-category last-vertex functor lands in the category because composites are equalities. For a general strict higher target the relevant simplex data include nonidentity higher cells, so that construction cannot simply be reused as a strict functor. Furthermore N_nS need not be a quasicategory; B.4 shows why literal simplicial slices cannot be substituted without their own proof. What is needed is a relative straightening comparison respecting those higher slice tests, followed by the universal base-change comparison for its specified pullback functors. The walking noninvertible 2-cell is already a base outside the ordinary-category proof. C.6–C.7 settle the formal duality and its weak-equivalence invariance; C.8–C.9 settle the verified or conditional absolute consequence. No full higher fine theorem, and no derivator calculus indexed by all strict n-categories, is asserted on the strength of an absolute model structure alone. □

4 Topological spaces: open tests, coefficients, and support

In this section Top denotes all small topological spaces; in particular we do not impose the Hausdorff condition on a space over a finite Alexandrov base. We write $\Pi X = \mathrm{Sing}(X) \in \mathcal{S}$ for its weak homotopy type. An infinity-categorical localization is used whenever the target is

an infinity-category. Its ordinary homotopy category is the ordinary localization occurring in the manuscript.

D.1. [Known] The actual comparison supplied by open tests. For a space Y, put

$$W_Y^{\mathrm{opn}} = \{u : X \longrightarrow X' \text{ over } Y : \ \Pi(u^{-1}U) : \Pi(X \times_Y U) \longrightarrow \Pi(X' \times_Y U) \text{ is an equivalence for every open } U \subseteq Y\}.$$

Here $u^{-1}U$ means the displayed restricted map, not an inverse-image operation on morphisms. This class contains the isomorphisms and satisfies two-out-of-three and retract closure. There is a canonical functor

$$C_Y : (\mathrm{Top}/Y)[(W_Y^{\mathrm{opn}})^{-1}] \longrightarrow \mathrm{CoShv}(Y;\mathcal{S}), \qquad C_Y(X)(U) = \Pi(X \times_Y U). \tag{D.1}$$

Its values are hypercomplete cosheaves. This statement does not assert that (D.1) is an equivalence for every Y.

Proof. For each open U, inverse image gives a functor from the slice to Top. The asserted closure properties follow by taking the intersection of the inverse images of the corresponding properties of weak homotopy equivalences. The higher Seifert–van Kampen theorem says that $U \mapsto \Pi U$ is a cosheaf: an open cover presents its union as the homotopy colimit of its Cech diagram after applying Π. Its precise forms are Lurie (2017, Proposition A.3.2, Remark A.3.8, and Lemma A.3.10). The last lemma supplies hyperdescent. Pulling an open cover or open hypercover of $U \subseteq Y$ back to X gives one of $X \times_Y U$. Consequently the same assertions hold for $C_Y(X)$. A map u belongs to the specified class exactly when $C_Y(u)$ is an objectwise equivalence. The universal property of localization therefore supplies (D.1). For the empty open the value is the initial space, as required for a cosheaf; no nonemptiness or local connectedness assumption was used. □

D.2. [Proved] A complete realization theorem for finite Alexandrov bases. Let P be a finite poset with the topology of downsets. There are equivalences

$$(\mathrm{Top}/P)[(W_P^{\mathrm{opn}})^{-1}] \simeq \mathrm{Fun}(P,\mathcal{S}) \simeq \mathrm{CoShv}(P;\mathcal{S}). \tag{D.2}$$

The first equivalence sends $f : X \to P$ to the covariant diagram $p \mapsto \Pi f^{-1}(P_{\leq p})$. It includes arbitrary, possibly empty fibres and non-Hausdorff total spaces. For a discrete, possibly infinite base P the same statement holds by decomposition into its open fibres.

Proof. We first work before localization. Define

$$D : \mathrm{Top}/P \longrightarrow \mathrm{Top}^P, \qquad D(X)(p) = X_{\leq p} = f^{-1}(P_{\leq p}).$$

For a strict covariant diagram $F : P \to \mathrm{Top}$ define $E(F)$ to have underlying set $\coprod_{r\in P} F(r)$ and projection $(r,z) \mapsto r$. Give it the topology generated by the subsets

$$V(p,O) = \{(r,z) : r \leq p,\ F(r \leq p)(z) \in O\}, \qquad O \subseteq F(p) \text{ open}. \tag{D.2a}$$

The projection is continuous since its inverse image of $P_{\leq p}$ is $V(p,F(p))$. A natural transformation $D(X) \to F$ gives the map $x \mapsto (f(x), \eta_{f(x)}(x))$ to $E(F)$. The inverse image of (D.2a) is $\eta_p^{-1}(O)$, open in $X_{\leq p}$ and therefore in X. Conversely, a map $X \to E(F)$ over P defines on $X_{\leq p}$ the continuous function obtained by transporting its value in $F(f(x))$ along $f(x) \leq p$. These constructions are mutually inverse and natural. Thus $D \dashv E$ as an adjunction of ordinary categories.

The counit at p is

$$\epsilon_{F,p} : DE(F)(p) \longrightarrow F(p), \qquad (r, z) \longmapsto F(r \leq p)(z).$$

It has the continuous section $s_p(z) = (p, z)$. On $DE(F)(p)$ the formula $H(z, t) = z$ for $0 \leq t < 1$, and $H(z, 1) = s_p\epsilon_{F,p}(z)$, gives a strong deformation retraction. For a generator $V(q, O)$ with $p \leq q$, membership is unchanged during this homotopy, by composition in F. For a generator with $p \not\leq q$, its inverse image is $(V(q, O) \cap DE(F)(p)) \times [0, 1)$. Both sets are open in the product. This verifies continuity even though the deformation need not be over P. Hence ϵ_F is a pointwise weak equivalence. The triangle identity $\epsilon_{DX} \circ D(\eta_X) = 1_{DX}$ shows that the unit is a weak equivalence on every principal open.

For a finite P, equivalence on principal opens implies equivalence on all opens. To see this, use induction on a downset U. Choose a maximal element p of U and write $U = (U \setminus \{p\}) \cup P_{\leq p}$; their intersection is $P_{<p}$. All three inverse images are open. The open-cover homotopy pushout theorem of D.1 and induction give the result for U. This also proves that E sends pointwise weak equivalences to W_P^{opn}: use naturality of its counit, then the induction. Thus the relative adjunction has unit and counit in the specified weak-equivalence classes and induces an equivalence of localizations.

The pointwise singular–realization adjunction identifies the localization of Top^P with that of sSet^P; its unit and counit are pointwise weak homotopy equivalences. Strict diagrams of simplicial sets with their projective weak equivalences present $\mathrm{Fun}(P, \mathcal{S})$ by Lurie (2009, Proposition 4.2.4.4 and Remark 4.2.4.5). This states the exact rectification input; it is not inferred merely from $\mathrm{Top}[W^{-1}] \simeq \mathcal{S}$.

Finally, restriction of a cosheaf to principal opens gives a diagram $P \to \mathcal{S}$. Its inverse is $F \mapsto (U \mapsto \mathrm{colim}_{p \in U} F(p))$. The same finite-open induction proves codescent and proves that the unit and counit are equivalences. For discrete P, a space over P is the disjoint union of its open fibres, and both sides of (D.2) are $\prod_{p \in P} \mathcal{S}$. The empty base gives the terminal category. □

D.3. [Refuted] The two variances are not automatically equivalent unstably. For finite Alexandrov P the sheaf theory is $\mathrm{Shv}(P; \mathcal{S}) \simeq \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{S})$, whereas D.2 gives the covariant diagram category for open inverse-image tests. These categories need not be equivalent. In particular an unstable sheaf–cosheaf duality for all topological spaces is false.

Proof. For a finite poset, a sheaf is recovered from its values on principal opens by $F(U) = \lim_{p \in U^{\mathrm{op}}} F(P_{\leq p})$. Finite gluing, with the induction in D.2 reversed, proves the formula and its inverse. Take $P = \{b, a, c\}$ with precisely the nonidentity relations $b < a$ and $b < c$. This poset is not isomorphic to its opposite: it has one minimal and two maximal elements.

An equivalence between the two space-valued diagram categories would identify their completely compact objects, meaning objects K for which $\mathrm{Map}(K, -)$ preserves every small colimit. In a presheaf category these are exactly retracts of representables. Indeed write K as its canonical colimit of representables. Preservation of this colimit makes 1_K factor through one representable; this exhibits K as its retract. Conversely the property is preserved by retracts and holds for representables. In a poset there are no nontrivial idempotent endomorphisms, so its category of such retracts is the poset itself. The two categories would therefore imply $P \simeq P^{\mathrm{op}}$, which is impossible. The six candidate bijections and their order tests are exhausted in `code/dg_finite_checks.py`. This counterexample concerns a finite non-Hausdorff space. It is not presented as a counterexample under the stronger Hausdorff hypothesis. □

D.4. [Refuted] Proper maps do not preserve open-test equivalences. Even for locally compact Hausdorff spaces, strict pullback along a proper map need not preserve W^{opn}. Thus

geometric proper base change for sheaves cannot be read as this assertion about spaces over a base.

Proof. Set $Y = [0,1]$, let $j : (0,1] \hookrightarrow Y$ be a map over Y, and let $i : \{0\} \hookrightarrow Y$. The map i is proper. For any open $U \subseteq Y$, every connected component of U is an interval. The component containing 0, if there is one, stays nonempty and contractible after 0 is deleted, and all other components are unchanged. Consequently $U \cap (0,1] \to U$ induces a bijection on components and isomorphisms on all based homotopy groups. Thus $j \in W_Y^{\mathrm{opn}}$. Its strict pullback along i is $\varnothing \to \{0\}$, which is not a weak homotopy equivalence. All maps and spaces in this example have been specified; the failure is on π_0, not on a choice of derived replacement. Testing instead on all closed subsets would detect $\{0\}$ and would not invert j. That alternative is therefore a different recipe. □

D.5. [Proved] Local product maps preserve the open-test class. Suppose $g : Z \to Y$ is locally a projection: each point of Z has an open neighbourhood isomorphic over an open $V \subseteq Y$ to $V \times K \to V$, with K a topological space. Then strict pullback along g sends W_Y^{opn} to W_Z^{opn}.

Proof. Let $u : X \to X'$ be in W_Y^{opn}. On a product rectangle $V' \times K' \subseteq V \times K$, with both factors open, its pullback is

$$(X \times_Y V') \times K' \longrightarrow (X' \times_Y V') \times K'.$$

The singular functor preserves products, and products of spaces preserve weak equivalences, including the case $K' = \varnothing$. Hence this map is a weak equivalence. Such rectangles form a basis locally on Z. On the basis and its intersection refinements the two inverse-image homotopy-type functors agree. Hyperdescent in D.1, or equivalently the basis form of the open-cover homotopy-colimit theorem, then proves their agreement on every open of Z. The argument uses no compactness or contractibility of K. □

D.6. [Known] Proper base change is a coefficient theorem. Consider a cartesian square of locally compact Hausdorff spaces

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y. \end{array}$$

If f is proper, the canonical mate $g^* f_* \to f'_* g'^*$ is an equivalence on sheaves of spaces. It is also an equivalence on sheaves with values in a stable bicomplete infinity-category. These are ordinary infinity-sheaves; hypercompletion is not silently substituted.

Proof. The space-valued statement is exactly Lurie (2009, Corollary 7.3.1.18). Its proof first establishes properness of the induced geometric morphism and then the appropriate pullback of infinity-topoi; the displayed comparison is the adjointability mate in that theorem. For stable coefficients the extension and its hypotheses are given by Volpe (2025, Section 6.1 and Proposition 6.9), using $f_! = f_*$ for proper f. This identifies the specified map, not merely two abstractly equivalent functors. D.4 explains why the theorem cannot be transferred to strict fibres of arbitrary spaces over Y. □

D.7. [Known] A precise smooth topological comparison. A space K is essential when the constant-sheaf functor $\mathcal{S} \to \mathrm{Shv}(K;\mathcal{S})$ has a left adjoint. A shape submersion $g : Z \to Y$ is locally, on its source, a projection $V \times K \to V$ with V open in Y and K essential. Assume either Y is locally compact or all these K are locally compact. Then for any presentable coefficient infinity-category $\mathcal{C}$ there is an adjunction $g_\sharp \dashv g^*$ on sheaves with these coefficients. In a cartesian square as in D.6 with g a shape submersion and with the same local compactness condition after pullback, the canonical maps

$$g'_\sharp f'^* \longrightarrow f^* g_\sharp, \qquad g^* f_* \longrightarrow f'_* g'^*$$

are equivalences.

Proof. These are Definition 3.23, the hypothesis immediately preceding Lemma 3.26, and Lemmas 3.26–3.27 of Volpe (2025), with the square relabelled. Their proof reduces to the indicated product charts, where essentiality supplies the left adjoint, and then uses descent. The second map is the comparison between the full right adjoints of the first. This is a specified geometric sufficient condition, not a definition of "locally aspherical" by the homotopy types of individual fibres. D.5 additionally establishes preservation of open-test equivalences for this class. □

D.8. [Known] The stable sheaf–cosheaf equivalence and the shriek operation. For a locally compact Hausdorff space Y and a stable bicomplete infinity-category $\mathcal{C}$, there is an equivalence

$$\mathbb{V}_Y : \mathrm{Shv}(Y;\mathcal{C}) \xrightarrow{\simeq} \mathrm{CoShv}(Y;\mathcal{C}), \qquad \mathbb{V}_Y(F)(U) = \operatorname*{colim}_{K \subseteq U \ mcompact} \mathrm{fib}\big(F(Y) \to F(Y \setminus K)\big). \tag{D.8}$$

For a continuous map $f : X \to Y$ between such spaces, cosheaf pushforward f_+ is $(f_+G)(U) = G(f^{-1}U)$. The topological operation with proper support is $f_! = \mathbb{V}_Y^{-1} f_+ \mathbb{V}_X$ and has a right adjoint $f^!$. For proper f it agrees with f_*. This does not identify $f_!$ with the left adjoint $f_\sharp$ of D.7.

Proof. Equation (D.8) is Lurie (2017, Theorem 5.5.5.1 and Proposition 5.5.5.10), with the opposite categories in that theorem taken on both sides. Existence of the indicated adjoints and the proper identification are established in Volpe (2025, Section 6.1). They depend on compact support, absent from the definition of an ordinary Kan extension on a category. For example, for $a : \mathbb{R} \to *$ and constant coefficients k, $a_\sharp k \simeq k$ while $a_! k \simeq k[-1]$ in cohomological grading. The first equality is homotopy invariance on the contractible line; the second follows by one-point compactification, whose reduced cohomology is k in degree one. Thus even on a manifold the two left adjoints have different right adjoints. Composing (D.8) with coefficient dualization gives the usual contravariant Verdier duality only on subcategories where that coefficient dualization is an equivalence. □

D.9. [Refuted] Realization does not turn every Kan extension into a geometric direct image. Under the least-vertex stratification, categorical properness and properness of a realized finite-poset map are different. Ambient topological direct image need not preserve the least-vertex constructible sheaves and need not equal categorical right Kan extension.

Proof. Let $P = [1]$, so $|NP| = [0,1]$ has strata $[0,1)$ and $\{1\}$. Let $i : * \to P$ select 0. Its realization is a proper closed inclusion, while i is not categorically proper: the arrow $0 \to 1$ has no outgoing lift. A presheaf on P is an arrow $F_1 \to F_0$. The right Kan extension of a

two-point set M from 0 is the constant arrow $M \xrightarrow{1} M$, since both comma categories $*/0$ and $*/1$ are terminal. On the interval, direct image from the endpoint is the skyscraper: its stalk is M at 0 and the terminal space at every other point. It is not locally constant on $[0, 1)$. The categorical sheaf would instead be constant on that stratum. The finite category, its comma categories, and the value cardinalities are verified in `code/dg_finite_checks.py`. □

D.10. [Proved] A geometric class on which the categorical comparison is literal. For finite posets equipped with their least-vertex models, closed cosieve inclusions and product projections have topological direct images equal to their categorical right Kan extensions for space, spectrum, and derived-ring coefficients. A sieve is represented by the open complement of the complementary cosieve, rather than in general by the inclusion of its own nerve. These assertions use the finite-poset exodromy comparison, as verified in G.10 below.

Proof. If $i : B \hookrightarrow P$ is a cosieve, $|NB|$ is a closed subcomplex and is a union of least-vertex strata: a chain whose minimum belongs to B lies entirely in B. Topological closed direct image is terminal off this subcomplex and agrees with the given sheaf on it. It preserves constructibility and is right adjoint to the specified restriction, so uniqueness of right adjoints identifies it with i_*. This also identifies units, counits, and base-change mates.

If $A = P \setminus B$ is the complementary sieve, let $O = |NP| \setminus |NB|$. A point of O has positive total barycentric coordinate on vertices of A. Delete the other coordinates and renormalize; straight-line interpolation with this operation gives a deformation retraction $O \to |NA|$. Its minimum vertex is unchanged throughout. Constructible exodromy therefore identifies coefficients on O with coefficients on A, including restriction and its adjoints. The inclusion $|NA| \hookrightarrow |NP|$ itself need not be open, as $\{0\} \hookrightarrow [0, 1]$ demonstrates.

Finally the canonical homeomorphism $|N(P \times Q)| \cong |NP| \times |NQ|$ respects the minimum strata. Projection has compact fibre $|NP|$. Over a stratum of Q, the given sheaf is a family of fixed constructible sheaves on $|NP|$ with locally constant transport; taking its sections leaves that transport locally constant. This follows as well by the finite exodromy diagram calculation, since the sections are a finite homotopy limit. The direct image thus preserves constructibility and is the unique right adjoint to restriction. D.6 applies to its geometric pullback squares. For the three stated coefficient categories the finite exodromy comparison is available, and finite limits commute with their locally constant transport. □

D.11. [Proved] The realization comparison for every proper map of finite posets. Let $f : P \to Q$ be a W_∞-proper map of finite posets. Equivalently, for every $p \in P$ and $q \geq f(p)$ the outgoing lifting poset

$$L(p, q) = \{r \in P : r \geq p,\ f(r) = q\} \tag{D.11a}$$

is weakly contractible. With the minimum stratifications, the topological direct image $|Nf|_*$ preserves constructibility and identifies with the categorical f_*, including its adjunction data. This holds for spaces, spectra, and the derived category of modules over any ring. Thus the requested geometric class contains the realizations of all such proper functors, with the specified constructibility property, although mere topological properness does not suffice by D.9.

Proof. Write $g = |Nf|$. We must compare the fibres at interior simplex points as well as at vertices. Fix a point $x \in |NQ|$ whose positive barycentric support is the chain $\sigma = (q_0 < \cdots < q_n)$, with weights $\lambda_i > 0$. The fibre $Z_x = g^{-1}(x)$ is a finite polytopal complex. Its cells are

indexed by nonempty chains κ of P with $f(\kappa) = \sigma$. The cell indexed by κ is the product of open simplices

$$\prod_{i=0}^{n} \operatorname{int} \Delta^{\kappa \cap P_{q_i}} : \quad (t_p)_{p \in \kappa \cap P_{q_i}} \text{ has sum } \lambda_i. \tag{D.11b}$$

The closure is the corresponding product of closed simplices. Its faces are precisely the subchains still meeting every level. Thus the face poset is

$$C_\sigma = \{\kappa \subseteq P : \kappa \text{ a chain}, \ f\kappa = \sigma\}, \quad \text{ordered by inclusion.}$$

The restriction of a minimum-constructible sheaf associated to $F : P^{\mathrm{op}} \to \mathcal{C}$ has value $F(\min \kappa)$ on this cell, with its prescribed structure maps. The exit category of this finite polytopal cell stratification is its face poset: barycentrically triangulate the cells, use HA A.6.10 on the triangulation, and coarsen by the maximum face using HPT 3.3.5 and the last-vertex localization, exactly as in G.10. Hence

$$\Gamma(Z_x; F|_{Z_x}) \simeq \lim_{\kappa \in C_\sigma} F(\min \kappa). \tag{D.11c}$$

This uses the coefficient classes in G.10, and applies to the three classes in the statement.

We now prove the required initiality, rather than assuming an exit-category equivalence for the whole fibre. The minimum functor

$$m : C_\sigma \longrightarrow (P_{q_0})^{\mathrm{op}}$$

is homotopy initial. For $p \in P_{q_0}$, its comma category m/p is the poset of chains in C_σ whose every vertex is at least p. We prove that this poset is weakly contractible by induction on n. Add p to every chain, then delete every other vertex at level q_0. The two operations are connected to the identity by the natural inclusions

$$\kappa \ \subseteq \ \kappa \cup \{p\} \ \supseteq \ \{p\} \cup \{r \in \kappa : f(r) > q_0\}.$$

They are retractions, up to these natural transformations, onto the subposet with exactly p at its first level. For $n = 0$ this subposet is the one-object poset $\{\{p\}\}$.

For $n > 0$, delete p from this subposet and send each remaining chain to its minimum at level q_1. This gives a functor to $L(p, q_1)^{\mathrm{op}}$. Its comma category over an object $r \in L(p, q_1)$ consists of the chains over $(q_1 < \cdots < q_n)$ all of whose vertices are at least r. It is weakly contractible by the induction hypothesis: all the further lifting posets $L(r, q_i)$ and those above their elements are exactly the posets covered by (D.11a). Quillen's Theorem A therefore makes this minimum functor a weak equivalence of nerves. Its target is weakly contractible by (D.11a), proving the induction. The same Theorem A, now applied to m, proves homotopy initiality. Consequently (D.11c) is canonically

$$\Gamma(Z_x; F|_{Z_x}) \simeq \lim_{(P_{q_0})^{\mathrm{op}}} F. \tag{D.11d}$$

If the fibre P_{q_0} is empty both indexing categories are empty and both limits are terminal; the argument covers that case too.

Categorical right Kan extension has the pointwise value

$$(f_* F)(q_0) = \lim_{(P_{\leq q_0})^{\mathrm{op}}} F, \qquad P_{\leq q_0} = \{p : f(p) \leq q_0\}.$$

The inclusion $(P_{q_0})^{\mathrm{op}} \to (P_{\leq q_0})^{\mathrm{op}}$ is homotopy initial: its comma category at p is $L(p, q_0)^{\mathrm{op}}$, contractible by (D.11a). Thus this Kan value also equals the right side of (D.11d).

It remains to identify the comparison and detect it globally. Let E_P, E_Q denote the fully faithful exodromy functors of G.10. Minimum labels commute with f, so $g^* E_Q \simeq E_P f^*$. The counit gives the canonical mate

$$E_Q f_* F \longrightarrow g_* E_P F. \tag{D.11e}$$

The map g is proper because it is a map of compact Hausdorff finite polyhedra. Proper base change identifies the target stalk at x with (D.11c). The source stalk is $(f_*F)(q_0)$. The stalk of (D.11e) is precisely restriction of the limit cone along $C_\sigma \xrightarrow{m} (P_{q_0})^{\mathrm{op}} \hookrightarrow (P_{\leq q_0})^{\mathrm{op}}$: the counit at a cell with minimum p is evaluation at that same object p of the Kan indexing category. Both functors are homotopy initial, so this specified map is an equivalence.

Finally sheaves of spaces on a finite polyhedron are hypercomplete: finite covering dimension and paracompactness give finite homotopy dimension by Lurie (2009, Corollary 7.2.3.7), and Lurie (2009, Corollary 7.2.1.12) then gives hypercompleteness. Stalks therefore detect equivalences. For spectra and modules the same conclusion follows by applying maps from shifts of the sphere or the free rank-one module; these conservative functors commute with filtered colimits and limits, so they commute with stalks and sheaf evaluation. Thus (D.11e) is an equivalence of sheaves. Its source is constructible, proving preservation, and its construction as a mate gives the asserted compatibility with units and counits. □

D.12. [Open] The unrestricted topological realization and its proposed dual. For a locally compact Hausdorff base Y, is the functor C_Y in (D.1), with all topological total spaces allowed, an equivalence? If so, what precise localization of spaces over the same Y presents $\mathrm{Shv}(Y;\mathcal{S})$ and is compatible with the proposed dual class of base-change maps? Neither statement is established in this audit for a general Y; the first unproved test base in our argument is $\mathbb{R}$.

The construction in D.2 needs a smallest open neighbourhood for each point; a real line has none. D.1 gives a functor and detects exactly the original class on strict maps, but does not prove essential surjectivity or full faithfulness after localization. Testing every closed subset defines a legitimate different weak-equivalence class, as D.4 shows, but this fact alone supplies neither descent nor an equivalence with sheaves. We found no primary theorem in the searched sources establishing that unrestricted representation statement. Lurie's stable sheaf–cosheaf theorem D.8 makes no such assertion about Top/Y and cannot fill this gap.

D.13. [Proved] A specified stable cosheaf counterpart for smooth finite-poset maps. There is a precise geometric dual on finite coefficients which uses the maximum stratification and Verdier transport. Let $\mathcal{C}$ be spectra or modules over a commutative ring, let D be the coefficient duality on finite spectra or perfect modules, and let $E_P^{\max} : \mathrm{Fun}(P, \mathcal{C}) \to \mathrm{Shv}(|NP|;\mathcal{C})$ be G.10 for P^{op}. Define the contravariant fully faithful functor

$$\Phi_P(F) = \mathbb{V}_{|NP|}\, E_P^{\max}(DF), \qquad F \in \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{C}^{\mathrm{perf}}). \tag{D.13a}$$

If $f : P \to Q$ is categorically smooth, then its realization g satisfies

$$\Phi_Q(f_! F) \simeq g_+ \Phi_P(F). \tag{D.13b}$$

Thus ordinary cosheaf pushforward realizes this specified dual of left Kan extension. The cosheaf subcategories in (D.13a) are defined as Verdier images of maximum-constructible perfect sheaves; they are not being identified with literal minimum-constructible cosheaves. For a noncommutative ring the target coefficients are modules over its opposite ring.

Proof. Smoothness of f is properness of f^{op}. Apply D.11 to $f^{\mathrm{op}} : P^{\mathrm{op}} \to Q^{\mathrm{op}}$. Its minimum stratification is the maximum stratification on the same underlying polyhedron. Coefficient

duality on finite diagrams gives $D(f_!F) \simeq (f^{\mathrm{op}})_*(DF)$: all comma categories are finite direct, so the relevant limits and colimits remain perfect, and the duality in G.6 exchanges them. Consequently D.11 gives

$$E_Q^{\max} D(f_!F) \simeq g_* E_P^{\max}(DF).$$

The map g is proper, and D.8 identifies $\mathbb{V}_{|NQ|} g_* \simeq g_+ \mathbb{V}_{|NP|}$. Applying this equivalence proves (D.13b), including the canonical maps obtained from these adjunctions. Full faithfulness follows from coefficient biduality, G.10, and the equivalence D.8. The description of the image is part of the construction, so this does not make an unproved constructibility assertion about cosheaves for a coarser minimum stratification; G.11 illustrates why that assertion would require a separate argument. □

5 An abstract theorem and its actual hypotheses

E.1. [Proved] The formal slice construction. Let $\mathcal{M}$ be a $\mathbb{U}$-small category with pullbacks, let W contain the isomorphisms and satisfy two-out-of-three, and let $\mathcal{L}$ contain the isomorphisms and be stable under composition and pullback. The same assertion holds for a larger category when all the localizations below exist in the chosen larger universe. For $S \in \mathcal{M}$, put

$$W_S^{\mathcal{L}} = \{u : X \to Y \text{ in } \mathcal{M}/S : u \times_S T \in W \text{ for every } (T \to S) \in \mathcal{L}\}.$$

This class contains isomorphisms, has two-out-of-three, and is contained in W_S, the class detected on total objects. If W is closed under retracts, so is $W_S^{\mathcal{L}}$. Every postcomposition functor $\Sigma_f : \mathcal{M}/S \to \mathcal{M}/T$, for $f : S \to T$, preserves these classes. If $f \in \mathcal{L}$, strict pullback $f^{-1} : \mathcal{M}/T \to \mathcal{M}/S$ also preserves them and the ordinary adjunction $\Sigma_f \dashv f^{-1}$ descends to both the ordinary and enhanced localizations.

Proof. For each $\ell : U \to S$ in $\mathcal{L}$, pullback is a functor on slices. The inverse image under a functor of a class with two-out-of-three has two-out-of-three; intersections preserve this property. The identity test gives containment in W_S. Retracts are handled by the same argument. For $u \in W_S^{\mathcal{L}}$ and $\ell : U \to T$ in $\mathcal{L}$, the map testing $\Sigma_f u$ is canonically isomorphic to the pullback of u along $S \times_T U \to S$. That last map is in $\mathcal{L}$. For $f \in \mathcal{L}$ and $v \in W_T^{\mathcal{L}}$, the test of $f^{-1}v$ along $U \to S$ is the test of v along $U \to S \xrightarrow{f} T$, also in $\mathcal{L}$. Finally, the unit and counit of the slice adjunction are natural transformations between relative functors. Localization sends them to natural transformations satisfying the same triangle identities. This proves the descended adjunction, not merely the existence of two descended functors. □

E.2. [Refuted] Arbitrary raw pullback does not follow. The hypotheses of E.1 do not imply that raw pullback preserves the fine classes for every f. In the presheaf fine theory of $\mathrm{Cat}/[1]$, the inclusion $u : e_1 \to [1]$ over $[1]$ is a fine equivalence, whereas its pullback along $e_0 \to [1]$ is $\varnothing \to e$, which is not. Consequently the inverse image on the fine localization, when it exists, cannot in general be computed by raw pullback.

Proof. The category $[1]$ has objects $0, 1$, their identity arrows, and one further arrow $0 \to 1$. For $s = 0$ and $s = 1$, the comma category $s\backslash e_1$ is terminal, and $s\backslash[1]$ has an initial object. Thus $s\backslash u$ is a weak homotopy equivalence for both objects. The strict inverse image of e_1 at 0 is empty and that of $[1]$ is terminal. Their nerves have different sets of components. The program `code/ef_finite_checks.py` verifies the complete finite category and comma data. In the notation of Part B, the presheaf class is $W_S^{\mathrm{Prop}} = W_S^{\mathrm{pre}}$; the dual covariant class is $W_S^{\mathrm{Liss}} = W_S^{\mathrm{co}}$. Thus this example also fixes which pole enters E.1. □

E.3. [Proved] A recognition theorem giving all three Kan functors. Suppose that the enhanced localizations $\mathscr{E}(S) = (\mathcal{M}/S)[(W_S^{\mathcal{L}})^{-1}]_\infty$ exist for bases S in a category $\mathcal{B}$. Suppose we are given a functor $C : \mathcal{B} \to \mathrm{Cat}_\infty$ with small values and fully faithful functors $h_S : C(S) \to \mathscr{E}(S)$ satisfying the following precise presentation hypotheses. The category $\mathscr{E}(S)$ has all small colimits. Each functor $\mathrm{Map}(h_S(c), -) : \mathscr{E}(S) \to \mathcal{S}$ preserves all small colimits. These functors jointly detect equivalences. For each $f : S \to T$, postcomposition descends to a colimit-preserving functor $F_f : \mathscr{E}(S) \to \mathscr{E}(T)$ and there is a coherent equivalence $F_f h_S \simeq h_T C(f)$.

Then the restricted Yoneda functors give equivalences

$$H_S : \mathscr{E}(S) \simeq \mathrm{Fun}(C(S)^{\mathrm{op}}, \mathcal{S}), \qquad H_S(X)(c) = \mathrm{Map}(h_S(c), X).$$

Under these equivalences, F_f is left Kan extension along $C(f)^{\mathrm{op}}$; it has right adjoint f^*, given by restriction, and this restriction has right adjoint f_*. These are coherent functors of the base, hence yield the requested pseudofunctors on homotopy categories. The locally constant subcategory has both adjoints to its inclusion. None of the presentation hypotheses is claimed to follow from E.1 or from factorization through intrinsic fibrations.

Proof. Let $P(C) = \mathrm{Fun}(C^{\mathrm{op}}, \mathcal{S})$. The colimit-preserving extension of h_S along the Yoneda functor exists: explicitly, it sends a presheaf F to the colimit of $h_S(c)$ indexed by the right fibration of elements of F. Denote it by $L_S : P(C(S)) \to \mathscr{E}(S)$. The universal property of that colimit and the infinity-categorical Yoneda lemma give $L_S \dashv H_S$. By full faithfulness of h_S, the unit is an equivalence on representables. Both L_S and H_S preserve colimits by hypothesis; representables generate $P(C(S))$ under small colimits. The unit is therefore an equivalence everywhere. For $X \in \mathscr{E}(S)$, applying H_S to the counit $L_S H_S X \to X$ gives an equivalence by the triangle identity. Joint detection by the mapping functors makes the counit an equivalence. This proves the recognition assertion, including essential surjectivity.

The two colimit-preserving functors $H_T F_f L_S$ and $\mathrm{Lan}_{C(f)^{\mathrm{op}}}$ agree on every representable, by the specified equivalence $F_f h_S \simeq h_T C(f)$. The colimit presentation makes their extension and coherence unique. Restriction on functor categories has both Kan adjoints because spaces are complete and cocomplete. Adjunctions transport through H_S, H_T, and their composition constraints follow from uniqueness of adjoints.

Write $q_S : C(S) \to BC(S)$ for groupoid completion. Its universal property identifies $q_S^* \mathrm{Fun}(BC(S)^{\mathrm{op}}, \mathcal{S})$ with the full subcategory of presheaves carrying every arrow to an equivalence. Both Kan extensions along q_S^{op} exist. Full faithfulness of restriction gives a reflection and a coreflection. This also covers the empty base, since its functor category is terminal; disconnected bases and infinite families are included in the small-colimit hypothesis. □

E.4. [Proved] Coefficient localization and exact squares. In E.3 one may replace the coefficient category of spaces by any presentable infinity-category $\mathcal{V}$, once the slice presentation with those coefficients has been verified. In particular, an accessible reflective localization $L : \mathcal{S} \to \mathcal{V}$ gives the objectwise localization

$$\mathrm{Fun}(C(S)^{\mathrm{op}}, \mathcal{S}) \longrightarrow \mathrm{Fun}(C(S)^{\mathrm{op}}, \mathcal{V}).$$

Restriction still has both Kan adjoints, and the locally constant subcategory is reflective and coreflective. For a strictly commuting square of ordinary indexing categories, base change for these Kan functors follows whenever its relevant comma-category comparison is homotopy final for left Kan extension, or homotopy initial for right Kan extension. In particular the presheaf left base-change map is invertible if the vertical map is W_∞-smooth or the horizontal map is W_∞-proper; the right map is invertible with these two roles reversed.

Proof. The objectwise adjunction $L \dashv \iota$ gives a reflective localization of functor categories: its unit is an equivalence exactly on diagrams with local values. Limits in $\mathcal{V}$ are formed in the ambient category; colimits are obtained by applying L to ambient colimits. Kan extension and restriction in $\mathcal{V}$ therefore exist for every small indexing functor. Groupoid completion has the same universal property for functors into any infinity-category, so the argument for locally constant objects is unchanged. For the base-change assertion, evaluate at an object of the lower-left indexing category. The pointwise Kan formulas express the mate as a colimit or limit comparison along an explicitly induced comma functor. The homotopy-final or homotopy-initial hypothesis is exactly the cofinality theorem for that comparison. The lifting-category proof, with the precise orientation of both maps, is given in F.2. It applies to every derivator and hence also to every such $\mathcal{V}$. □

E.5. [Proved] Instances and the extra coarse comparison. For W_∞, the presheaf fine localization of Cat/S satisfies E.3 with $C(S) = NS$ and $h_S(s)$ the point $e_s \to S$; the dual localization uses NS^{op}. The contravariant and covariant simplicial model structures give the same instances over quasicategorical bases, with the proviso in Part B about raw strict slice tests over arbitrary simplicial sets. For finite Alexandrov posets with downsets open and unrestricted topological total spaces, the open-test localization is the covariant instance $C(P) = NP^{\mathrm{op}}$ proved in Part D. Discrete topological bases give the product of copies of spaces directly.

The additional assertion that the coarse localization agrees with the locally constant part requires the coarse comparison

$$(\mathcal{M}/S)[W_S^{-1}]_\infty \simeq \mathcal{S}_{/BC(S)} \simeq \mathrm{Fun}(BC(S)^{\mathrm{op}}, \mathcal{S})$$

and compatibility of its localization functor with $\mathrm{Lan}_{q_S^{\mathrm{op}}}$. For (Cat, W_∞) this comparison holds. It is not part of the abstract recognition hypotheses.

Proof. Part B constructs the enhanced equivalence using nerves and simplex categories, under which e_s is the representable presheaf $t \mapsto \mathrm{Hom}_S(t, s)$. Evaluation at s preserves all colimits and detects equivalences jointly in s; postcomposition carries a point to its image. These facts verify every presentation hypothesis of E.3. The proof for simplicial sets is the straightening comparison recorded there. The topological equivalence, including the topology on its realization functor and its unit and counit, is proved in D; for a discrete base it reduces to the homeomorphism $X \cong \coprod_{s \in S} X_s$.

For the coarse comparison in Cat, use the Thomason Quillen equivalence with spaces and its right proper slice comparison, or the functorial smooth-and-proper replacement of Part B. The latter replaces $X \to S$ by $E \to S$ through a total weak equivalence, with strict fibres the homotopy fibres of $BX \to BS$. These fibres form the corresponding locally constant diagram, by straightening over the space BS. The total space of a presheaf F is its Grothendieck homotopy colimit, so the canonical functor from the fine theory to the coarse one is the left Kan extension of F along $S \to BS$. Equivalently, it is characterized by the adjunction against every locally constant coefficient diagram. This verifies compatibility of the specified comparison, not only an abstract equivalence of categories. The reflector and coreflector for an arbitrary derivator are also the precisely stated content of Cisinski (2009, Theorem 3.13). □

E.6. [Refuted] The coarse theory is not the locally constant fine theory for every localizer. Replacing W_∞ by an arbitrary fundamental localizer in the coarse comparison of E.5 is false. Already for W_0, the component equivalences, and $S = BC_2$, the canonical coarse localization identifies two distinct locally constant objects of the fine theory.

Proof. Let $C_2 = \{1, g\}$ with $g^2 = 1$, and regard S as its one-object category. The functor $u : e \to S$ sends the sole identity to 1 and is considered as a morphism from $e \to S$ to id_S in Cat/S. Both total categories are connected, so $u \in W_0$. At the sole object s, the comma category $s\backslash e$ is the discrete set C_2: an arrow in it must map under u to the identity, so distinct group elements cannot be connected. The category $s\backslash S$ is the translation groupoid, with exactly one arrow between any ordered pair of objects; it is contractible. The induced map is not a component equivalence. Both associated diagrams are locally constant because all arrows of S are invertible. Thus the fine localization does not invert u, whereas the coarse one does. In the objectwise-localized description these are the regular C_2-set and the one-point C_2-set. The complete multiplication and comma calculations are included in `code/ef_finite_checks.py`. □

E.7. [Proved] What enough intrinsic fibrations supplies. Let $\mathcal{M}$ have a terminal object and pullbacks, and let W contain isomorphisms and satisfy two-out-of-three. Define Fib_W by universal preservation of W under strict pullback. Assume that every morphism factors as a map in W followed by a map in Fib_W. Then every map in $W \cap \mathrm{Fib}_W$ remains in W after arbitrary pullback. Consequently the full subcategory of objects whose terminal map lies in Fib_W, with these weak equivalences and fibrations, is a category of fibrant objects and presents the same enhanced localization as $\mathcal{M}$. It supplies finite homotopy limits and the usual homotopy-fibre construction. This does not identify the fine slice theories, and does not imply E.3.

Proof. Let $p : T \to B$ lie in $W \cap \mathrm{Fib}_W$, and let $g : X \to B$ be arbitrary. Factor $g = qi$ with $i : X \to Z$ in W and $q : Z \to B$ in Fib_W. The map $p_Z : T \times_B Z \to Z$ is in W because it is obtained by pulling back p along q. It is also in Fib_W, since that class is stable under pullback. The comparison $T \times_B X \to T \times_B Z$ lies in W, since it is the pullback of i along p_Z. The equality of composites

$$i\, p_X = p_Z\, (T \times_B i)$$

and two-out-of-three imply $p_X \in W$. This is the non-formal Brown axiom.

Composition and pullback stability of Fib_W follow by iterating the defining test. If X, Y have fibrant terminal maps, a factorization $X \to Z \to Y$ as above has Z fibrant. A path object for a fibrant X is obtained by factoring $X \to X \times X$; products and fibrant pullbacks exist by pullback stability. Factoring $X \to e$ supplies a weak-equivalence fibrant replacement for every object. The Brown localization theorem therefore identifies the localization of the fibrant subcategory with that of $\mathcal{M}$ and gives finite homotopy limits. In the infinity-categorical form, the finite inverse diagram comparison is Cisinski (2019, Theorems 7.5.18 and 7.6.17); the latter is stated for I^{op} with I finite direct. No infinite diagram rectification is inferred. The counterexamples in E.2 and E.6 show why this construction alone does not supply the fine presentation or its locally constant interpretation. □

E.8. [Known] The accessible localizers satisfying the intrinsic factorization axiom. For an accessible fundamental localizer W of Cat, factorization through Fib_W is equivalent to aspherical generation of W, to generation by a set of maps $A \to e$, and to right properness of the associated Thomason model structure. In the corresponding accessible localization of spaces it is precisely nullification. Thus truncation localizers satisfy this factorization axiom; the nonzero-spectrum homology localizers in Part B do not. The right class of the associated modality is not the class Fib_W: every $X \to e$ belongs to Fib_W, whereas a terminal map is in that right class only when its fibre is local.

Proof. The equivalence of aspherical generation, generation by a set of categories, and properness is Cisinski (2006, Theorem 6.1.11). The comparison with right properness of the Thomason model is his Theorem 5.2.15, and Example 6.1.14 makes every model fibration intrinsic in the proper case. Model factorization therefore gives the required factorization.

For the converse, E.7 applies, and all terminal maps are intrinsic because fundamental localizers are stable under products. Let W_{asph} be the fundamental localizer generated by the maps $A \to e$ already in W. A trivial intrinsic fibration $p : T \to B$ is universally in W by E.7. Pullback along $B/b \to B$ shows that T/b is W-aspherical for each b. Hence $T/b \to e$ is one of the generators for W_{asph}, and relative Theorem A gives $p \in W_{\mathrm{asph}}$. The Brown lemma applied to the identity functor, with target weak equivalences W_{asph}, now gives $W \subseteq W_{\mathrm{asph}}$. The reverse inclusion is immediate. This direction needs no accessibility; accessibility enters when Theorem 6.1.11 turns it into set generation and model properness. Locality with respect to $A_i \to e$ means exactly that $Z \to \mathrm{Map}(BA_i, Z)$ is an equivalence, which is the definition of the corresponding nullification. We use no identification of the modality's right maps with intrinsic strict pullback maps. The product statement already separates them whenever a nonlocal object exists. □

E.9. [Proved] What a path object must supply. Suppose $\mathcal{M}$ has finite limits, W has two-out-of-three, and $\mathcal{A} = \mathrm{Liss} \cap \mathrm{Prop}$ is stable under pullback and composition and contains product projections. Suppose functorially for each S there is a path object

$$S \xrightarrow{c_S} P(S) \xrightarrow{(s_S, t_S)} S \times S, \qquad s_S c_S = t_S c_S = \mathrm{id}_S,$$

whose endpoint map belongs to $\mathcal{A}$ and whose source evaluation s_S is universally in W. Then every $f : X \to S$ factors functorially as

$$X \xrightarrow{i_f} X \times_{S, s_S} P(S) \xrightarrow{p_f} S$$

with $i_f(x) = (x, c_S f(x))$ in W and $p_f = t_S \mathrm{pr}_2$ in $\mathcal{A}$. Merely requiring $c_S \in W$ and $(s_S, t_S) \in \mathcal{A}$ does not prove the required assertion about i_f; the transfer hypothesis must be supplied.

Proof. The projection $r_f : X \times_S P(S) \to X$ is a pullback of s_S, hence lies in W, and $r_f i_f = \mathrm{id}_X$. Two-out-of-three gives $i_f \in W$. The map (r_f, p_f) to $X \times S$ is the pullback of (s_S, t_S) along $f \times \mathrm{id}_S$, so belongs to $\mathcal{A}$. Composing with the product projection gives the assertion about p_f. All formulas respect strictly commuting squares, which gives the stated functoriality.

In Cat, the fence-path replacement of Part B supplies the conclusion by explicit contractions and lifting categories. In simplicial sets, functorially factor the diagonal in the Kan–Quillen structure as a weak equivalence followed by a Kan fibration. Kan fibrations and product projections lie in both simplicial poles; right properness and E.7 make s_S, which has the weak-equivalence section c_S, universally weak. Thus the displayed construction applies. In compactly generated spaces, the usual path space $P(S) = S^{[0,1]}$ has a Hurewicz-fibration endpoint map and a source evaluation with an explicit fibrewise contraction; it supplies intrinsic fibration replacement. This fact by itself does not identify Hurewicz fibrations with topological sheaf-proper or shape-submersive maps. The two latter classes belong to the additional sheaf theory of Part D. □

E.10. [Proved] The groupoid test and the meaning of collapse. Let S be a small groupoid and let $\mathcal{M} = \mathrm{Gpd}$ with categorical equivalences. For functors of groupoids over S, both comma-test classes agree with the total equivalences. Their enhanced localizations consequently agree with the coarse theory, which is the infinity-category of families of 1-types over NS. Both variances are identified by inversion in S. This does not mean that every artificially chosen class $\mathcal{L}$ gives the coarse localization.

Proof. If $u : X \to Y$ is an equivalence of groupoids over S, the induced functor $s\backslash X \to s\backslash Y$ is fully faithful. Given $(y, \alpha : s \to p(y))$, choose x and an isomorphism $\beta : u(x) \to y$. Then $(x, p(\beta)^{-1}\alpha)$ maps isomorphically to (y, α), proving essential surjectivity. Conversely, equivalence on these comma groupoids implies equivalence on total classifying spaces by their Grothendieck homotopy colimits, hence equivalence of groupoids because their nerves are 1-types. Replacing every arrow of S by its inverse identifies the two comma constructions. An isofibration replacement $X \to E \to S$, where E has objects $(x, s, \alpha : p(x) \xrightarrow{\sim} s)$, is a total equivalence and has groupoid fibres. Straightening of this explicit groupoid fibration gives precisely families of groupoids, or 1-types, over NS.

For the last assertion let S be the two-object groupoid with exactly one arrow between each ordered pair, and let $u : e_0 \to S$ be its full inclusion, regarded over S. This is a categorical equivalence. Its pullback at object 1 is $\varnothing \to e$. If $\mathcal{L}$ consists of all maps, this is not in $W_S^{\mathcal{L}}$. Fibre evaluation preserves that latter weak-equivalence class and separates the two objects after localization, so the distinction is genuine. The natural invariant comma theories collapse; arbitrary strict tests can retain presentation data. No assertion about weak globular infinity-groupoids or the unrefereed 2026 claim is needed. □

E.11. [Proved] The intermediate-subcategory assertion under replacement data. Let $\mathcal{M}$ have pullbacks, let W contain isomorphisms and satisfy two-out-of-three, and suppose that every arrow factors as a W-map followed by an intrinsic W-fibration. Suppose in addition that for every base S there is a functor

$$R_S : \mathcal{M}/S \longrightarrow \mathrm{Fib}_W\, S$$

and a natural transformation $\eta_S : 1_{\mathcal{M}/S} \to i_S R_S$ whose components are in W_S, where i_S denotes the full inclusion. Then XII's axioms (a), (b), and (c), in their exact formulations A.10, hold. In particular, for every full subcategory $\mathrm{Fib}_W\, S \subseteq \mathcal{H} \subseteq \mathcal{M}/S$, both inclusions induce equivalences at the restricted W_S.

Proof. The factorization argument of E.7, applied to a W-map between two intrinsic fibrations over S, proves axiom (b) as follows. Factor an arbitrary $T \to S$ as $T \xrightarrow{i} T_0 \xrightarrow{p} S$ with $i \in W$ and p intrinsic. The pullback of the map along p is weak. Both comparison maps from pullback over T to pullback over T_0 are weak, because the two structural fibrations preserve i. Two-out-of-three gives weakness after pullback along $T \to S$.

For $u \in W_S$, naturality gives $R_S(u)\eta_X = \eta_Y u$, so $R_S(u) \in W_S$. Hence R_S descends to localizations. Its restriction to $\mathcal{H}$ has values in $\mathrm{Fib}_W\, S \subseteq \mathcal{H}$. Since $\mathcal{H}$ is full, every comparison $\eta_X : X \to R_S X$ for $X \in \mathcal{H}$ is a morphism inside $\mathcal{H}$. After localization these comparisons give natural isomorphisms between the identity and inclusion followed by R_S, both on $\mathcal{H}$ and on the fibrant subcategory. They exhibit the first inclusion as an equivalence. The same argument with $\mathcal{M}/S$ exhibits the composite inclusion as an equivalence, hence also the second. This proves (c). Finally (b) identifies the two restricted weak-equivalence classes on fibrant objects, so (c) gives (a). No compatibility of the chosen R_S for distinct bases is required for this assertion. □

E.12. [Refuted] A six-object counterexample to XIIc B062. The implication (a)+(b) ⇒ (c) of A.10 is false, even for a finite lattice and a strongly saturated class W. Both premises can hold at every base while the conclusion fails at one base. The same example refutes the assertion with infinity-categorical localizations in place of ordinary localizations.

Proof. Let $\mathcal{M}$ be the poset category on $\{0,1,2,3,4,5\}$ with Hasse covers

$$0<1, \quad 0<2, \quad 1<3, \quad 2<3, \quad 1<4, \quad 3<5, \quad 4<5.$$

There is exactly one morphism $a \to b$ when $a \le b$, and no morphism otherwise; composition is the unique morphism $a \to c$. Take

$$W = \{1_x \mid 0 \le x \le 5\} \ \cup \ \{1 \to 3,\ 1 \to 4,\ 2 \to 3\}.$$

This specifies all objects, arrows, compositions and weak equivalences. The object 0 is initial and 5 terminal. Pullbacks are meets. For comparable elements the meet is the smaller element; the remaining unordered pairs are $\{1,2\}$, $\{2,4\}$ and $\{3,4\}$, whose meets are respectively $0,0,1$. Thus all pullbacks exist; the finite meet semilattice with maximum 5 is a lattice.

Write $\mathcal{F}_s$ for the full subcategory of intrinsic fibrations over s. The following table gives every slice and every $\mathcal{F}_s$:

s	$\mathrm{Ob}(\mathcal{M}/s)$	$\mathrm{Ob}(\mathcal{F}_s)$
0	0	0
1	0, 1	0, 1
2	0, 2	0, 2
3	0, 1, 2, 3	0, 3
4	0, 1, 4	0, 1, 4
5	0, 1, 2, 3, 4, 5	0, 3, 5

Indeed $x \le s$ is intrinsic precisely when

$$a \wedge x \longrightarrow b \wedge x \ \in W \qquad \text{for every } (a \to b) \in W \text{ with } b \le s.$$

Over 3, the candidate 1 fails on $2 \to 3$, giving $0 \to 1$, and the candidate 2 fails on $1 \to 3$, giving $0 \to 2$. Over 5 these failures persist, and the candidate 4 fails on $2 \to 3$, giving $0 \to 1$. For $x = 3$ over 5, the three nonidentity weak arrows pull back respectively to $1 \to 3$, 1_1 and $2 \to 3$, all weak. Pullback along 0 or along the identity of a base also preserves W. Over 4 the only nonidentity weak arrow is $1 \to 4$; its pullbacks along $0, 1, 4$ are respectively $1_0, 1_1, 1 \to 4$. The slices over $0, 1, 2$ have only identity weak arrows. These observations prove every entry of the table.

We next check axiom (b) at every base. For $s \neq 4$, the restriction of W_s to $\mathcal{F}_s$ consists solely of identities, which are universal weak equivalences. For $s = 4$, the additional arrow $1 \to 4$ is universally weak in that slice by the three pullbacks just computed. Therefore $W_s^u = W_s$ on $\mathcal{F}_s$ for all six bases.

For (a), the bases $0, 1, 2, 4$ need no argument beyond (b), since $\mathcal{F}_s = \mathcal{M}/s$. At base 3, define

$$R_3(0) = 0, \qquad R_3(1) = R_3(2) = R_3(3) = 3.$$

This is a relative functor $\mathcal{M}/3 \to \mathcal{F}_3$, is the identity on $\mathcal{F}_3$, and has a natural transformation $1_{\mathcal{M}/3} \to i_3 R_3$ with weak components $1 \to 3$, $2 \to 3$, and identities. It therefore induces an inverse to the localized inclusion.

At base 5, let $i : \mathcal{F}_5 \hookrightarrow \mathcal{M}$ be the inclusion and define functors $R : \mathcal{M} \to \mathcal{F}_5$ and $T : \mathcal{M} \to \mathcal{M}$ by

x	0	1	2	3	4	5
$R(x)$	0	3	3	3	3	5
$T(x)$	0	1	2	3	1	5.

Monotonicity is checked on the seven displayed Hasse covers. Both functors preserve W: R sends the three nonidentity weak arrows to 1_3, whereas T sends them to $1 \to 3$, 1_1, and $2 \to 3$, respectively. Since $T(x) \le x$ and $T(x) \le R(x)$ for every x, there are natural transformations

$$1_{\mathcal{M}} \ \longleftarrow \ T \ \longrightarrow \ iR.$$

All components are weak: the only nonidentity component of the left transformation is $1 \to 4$ at $x = 4$; those of the right transformation are $1 \to 3$ at $x = 1, 4$ and $2 \to 3$ at $x = 2$. In a poset every such naturality square commutes by uniqueness of its arrows. Also $Ri = 1_{\mathcal{F}_5}$, and both transformations restrict to identities on $\mathcal{F}_5$. Localizing the two natural transformations gives a natural isomorphism $1 \cong iR$. Together with $Ri = 1$ this proves that the inclusion induces an equivalence. By (b), this is exactly the comparison in (a), including its prescribed class W_5^u. Thus (a) holds at every base.

For completeness, W satisfies all the stated saturation requirements. Inspection of the arrows gives

$$(a \to b) \in W \quad \Longleftrightarrow \quad R(a) = R(b).$$

The inverse image of identities under a functor to a poset satisfies two-out-of-three. Since R inverts W, it factors through $\mathcal{M}[W^{-1}]$. Any original arrow becoming invertible there must map to an isomorphism in the chain $\mathcal{F}_5 = \{0 < 3 < 5\}$, hence to an identity, and hence belongs to W. This proves strong saturation as well as the isomorphism requirements.

Finally take the full intermediate subcategory

$$\mathcal{F}_5 = \{0, 3, 5\} \ \subseteq \ \mathcal{H} = \{0, 3, 4, 5\} \ \subseteq \ \mathcal{M}/5.$$

It is the diamond with $0 < 3 < 5$ and $0 < 4 < 5$. The restriction $W_5 \cap \mathrm{Mor}(\mathcal{H})$ consists only of identities, as does its restriction to $\mathcal{F}_5$. Consequently the first localized inclusion is the ordinary inclusion of a three-object chain into this four-object diamond. It is not essentially surjective: 4 is not isomorphic to any object of $\mathcal{F}_5$. The second localized inclusion is not full, since $\mathrm{Hom}_{\mathcal{H}}(3, 4)$ is empty, whereas $3 \leftarrow 1 \to 4$ gives an isomorphism from 3 to 4 in $\mathcal{M}[W^{-1}]$. Both required equivalences in (c) therefore fail.

The enhanced assertion follows from the same certificate, without any assumption about discrete mapping spaces of a general localization. The relative functors and componentwise weak natural transformations above descend under infinity-categorical localization to functors and natural equivalences, exhibiting the localized inclusions in (a) as equivalences. The localizations of $\mathcal{F}_5$ and $\mathcal{H}$ at identity arrows are their nerves; the failure of essential surjectivity persists there. Thus the example also meets the stronger enhanced premises and violates the enhanced conclusion. □

The independent program `code/a_b062_verify.py` verifies every arrow, every composition, every meet, every intrinsic-fibration test, every universal-equivalence test, and the displayed functors and natural transformations. Its certificate does not use the search program or a bounded-path approximation to localization. The example does not have enough intrinsic fibrations: $4 \to 5$ cannot factor as a weak arrow followed by an intrinsic fibration, since the only weak arrow with source 4 is its identity. Nor does it have a coaugmented replacement as in E.11. The weak natural zigzag at base 5 passes through $T(4) = 1$, an object excluded from $\mathcal{H}$; this is exactly why it proves (a) without proving (c).

E.13. [Refuted] Ordinary homotopy categories do not supply coherent gluing. A gluing theorem for the enhanced fine theories cannot in general be replaced by the same pullback or lax-pullback construction on their homotopy categories. The comparison $\mathrm{HoFun}([1], \mathcal{V}) \to \mathrm{Fun}([1], \mathrm{Ho}\mathcal{V})$ need not be faithful, even for $\mathcal{V} = D(\mathbb{Z})$.

Proof. Put $X = \mathbb{Z}[0]$ and $Y = \Sigma X$ in the stable infinity-category $D(\mathbb{Z})$. Consider the coherent arrows $a = (X \to 0)$ and $b = (0 \to Y)$. Their mapping space in the coherent arrow category is the homotopy pullback

$$\mathrm{Map}(X, 0) \times_{\mathrm{Map}(X,Y)} \mathrm{Map}(0, Y) \simeq \Omega \mathrm{Map}(X, Y).$$

Its set of components is $[\Sigma X, Y] \cong \mathbb{Z}$. In the arrow category of Ho$D(\mathbb{Z})$, both vertical components of a morphism $a \to b$ must be zero, so there is exactly one morphism. The comparison maps all these integer classes to that single morphism. Thus the homotopy category has forgotten precisely the coherent square data needed for the gluing statement. A derivator retains coherent diagram categories at additional shapes; E.3 instead uses an infinity-categorical enhancement. The finite inverse diagram consequence of E.7 is not an assertion that all strict diagrams of arbitrary small shape compute that enhancement. □

E.14. [Proved] The precise modality and locally cartesian closed comparison. Let $L : \mathcal{S} \to \mathcal{V}$ be an accessible reflective localization of spaces. The following conditions are equivalent: L is a nullification; its local objects are closed under dependent sums over local bases, hence form a modality; the canonical map

$$L(X \times_Z A) \longrightarrow LX \times_Z A$$

is an equivalence for every $X \to Z \leftarrow A$ with A, Z local; and $\mathcal{V}$ is locally cartesian closed. Consequently these are additional equivalent conditions in E.8. The modality's left class consists of maps with L-contractible homotopy fibres and can be strictly smaller than the class inverted by L.

Proof. Suppose the generators are $K_i \to *$. For a family $Y \to Z$ whose base and fibres are local, the mapping-space fibration gives

$$\mathrm{Map}(K_i, Y) \simeq \int_{f \in \mathrm{Map}(K_i, Z)} \mathrm{Map}_{/Z}(K_i, Y) \simeq \int_{z \in Z} \mathrm{Map}(K_i, Y_z) \simeq \int_{z \in Z} Y_z = Y.$$

Here the middle equivalence is induced by the constant-map equivalence $Z \to \mathrm{Map}(K_i, Z)$; over the constant at z the mapping space is $\mathrm{Map}(K_i, Y_z)$. All equivalences are inverse to constant maps, so Y is local. The equivalence between this closure condition and the modality factorization system is Rijke, Shulman, and Spitters (2020, Theorems 1.34 and 1.54); the nullification statement is Theorem 2.19 there.

Conversely suppose this is an accessible modality generated as a localization by $u_i : A_i \to B_i$. Take the set of homotopy fibres of the units of all A_i, B_i. They are L-contractible. Every local space is null for them. If Y is null for them and $U \to LU$ is one of those units, its fibre diagram gives

$$\mathrm{Map}(U, Y) = \lim_{z \in LU} \mathrm{Map}(U_z, Y) \simeq \lim_{z \in LU} Y = \mathrm{Map}(LU, Y).$$

The limits are over the full homotopy type LU, with its coherent monodromy. Naturality of the units and invertibility of Lu_i then imply u_i-locality of Y. This proves equality of the local objects, hence nullification; compare Lemma 2.20 of the same source.

In a modality every unit has L-contractible fibres. Base change of $X \to LX$ along $LX \times_Z A \to LX$ retains that property and is inverted by L. Its target is local, since local objects are closed under limits. This proves the displayed locally cartesian identity. Conversely that identity applied to a point of LX makes every unit fibre L-contractible. For a family over a local Z with local fibres, the same identity identifies the fibres of its unit with the localizations of those fibres, hence makes its unit an equivalence. This recovers closure under dependent sums.

The locally cartesian identity makes pullback along $A \to Z$ preserve colimits in the local slices: compute each colimit in spaces, apply L, and use universality of colimits in spaces. The adjoint functor theorem gives the dependent product, so $\mathcal{V}$ is locally cartesian closed. For the converse, every space $X \to Z$ is the colimit, indexed by the infinity-groupoid X, of its

points $* \to Z$. These are local objects of the slice when Z is local. Their colimit in $\mathcal{V}/Z$ is $LX \to Z$. If pullback to local $A \to Z$ has a right adjoint, it preserves this colimit; its values on the point diagram are the local fibres A_x. Their colimit in spaces is $X \times_Z A$. Localizing that colimit yields exactly the displayed identity. This last implication uses the fact that the ambient category is spaces, generated by its point in this manner; it is not asserted for an arbitrary ambient topos. For $L = \tau_{\leq 0}$, a map between connected spaces is inverted by L, but may have disconnected homotopy fibres, for example $* \to BC_2$. Thus the modality's left class and the class inverted by L are indeed different. □

6 The six-operations shape of a derivator

F.1. [Known] The operations supplied by Der 1–4. Let $\mathbb{D}$ be a derivator on all $\mathbb{U}$-small categories, with presheaf variance, satisfying Der 1–4: coproducts of indexing categories give products, evaluations jointly detect isomorphisms, restriction has both Kan adjoints, and the pointwise comma formulas hold. Then every $f : X \to Y$ has

$$f_! \dashv f^* \dashv f_* : \mathbb{D}(X) \rightleftarrows \mathbb{D}(Y),$$

where the display denotes the three functors with their usual alternating directions. For $A \in \mathbb{D}(X)$ and $y \in Y$ the formulas are

$$(f_! A)_y \cong \operatorname*{colim}_{(y\backslash X)^{\mathrm{op}}} A, \qquad (f_* A)_y \cong \lim_{(X/y)^{\mathrm{op}}} A.$$

The diagram under each limit is restriction along the forgetful functor to X, interpreted as a coherent diagram in the derivator. Comma squares are homotopy exact. Exact squares are closed under horizontal and vertical pasting, with the direction of their two-cell fixed. Strongness, pointedness and stability are not needed for these statements.

The adjoint and pointwise assertions are the axioms, with variance converted from Groth (2013, Definition 1.5 and Section 1.2). Pasting is the mate calculus: substituting the unit–counit formula for each mate and cancelling the two adjacent unit–counit pairs by the triangle identities gives the mate of the pasted two-cell. The universal criterion for exactness is Groth, Ponto, and Shulman (2014b, Theorem 3.16); its equivalent factorization-category test is also developed in Maltsiniotis (2012, Section 2). These are statements about the canonical comparisons, not about an unspecified isomorphism of their source and target.

F.2. [Proved] The two strict base-change theorems and their converses. Consider a strict pullback square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y. \end{array}$$

The canonical maps, with source category $\mathbb{D}(X)$ and target category $\mathbb{D}(Y')$, are

$$f'_! g'^* \longrightarrow g^* f_!, \qquad g^* f_* \longrightarrow f'_* g'^*.$$

The first is invertible if f is W_∞-smooth or g is W_∞-proper. The second is invertible if f is W_∞-proper or g is W_∞-smooth. Conversely, f is proper if its right comparison is invertible for every such square with space-valued presheaves and all coefficients; the dual assertion characterizes smoothness by universal left base change.

Proof. For proper f, the inclusion $X_y \to X/y$ is homotopy final. Its opposite is homotopy initial, so the pointwise right Kan formula in F.1 becomes

$$(f_*A)_y \cong \lim_{X_y^{\mathrm{op}}} A|_{X_y}.$$

The analogous inclusion $X_y \to y\backslash X$ for smooth f is homotopy initial; its opposite is homotopy final, giving the left fibre formula. These cofinality statements hold in every derivator by Groth et al. (2014b, Theorems 3.8 and 3.16). The base-changed functor has the same property, and $(X')_{y'}$ is canonically isomorphic, as a category over X, to $X_{g(y')}$. The comparisons on these identical fibre diagrams are exactly the pointwise Beck–Chevalley maps. Der 2 gives their invertibility. Taking adjoint mates in the transposed square proves the alternative hypotheses on g.

For the converse, take $Y' = e$, selecting $y \in Y$, and write $i : X_y \to X$ for g'. The left adjoint mate of the right comparison is

$$i_! p_y^* \longrightarrow f^* y_! : \mathbb{D}(e) \longrightarrow \mathbb{D}(X), \qquad p_y : X_y \to e.$$

Evaluate this mate on the terminal space and at $x \in X$. Its target is the discrete space $\mathrm{Hom}_Y(f(x), y)$. Its source is the classifying space of $x\backslash X_y$, which is the disjoint union, indexed by $\alpha : f(x) \to y$, of the outgoing lifting categories

$$L_f^+(x, \alpha) = \{v : x \to z \mid f(z) = y,\ f(v) = \alpha\}.$$

A morphism $v \to v'$ is a $w : z \to z'$ with $f(w) = \mathrm{id}_y$ and $wv = v'$. The mate collapses each such summand to its index. Its invertibility forces every summand to have weakly contractible nerve, including the nonemptiness requirement. That is the properness criterion. The argument for smoothness is obtained by opposition, with incoming lifts. The terminal-space test occurs on this *left mate*; a right Kan comparison evaluated only on the terminal presheaf is automatically invertible for every square and would detect nothing. This clarifies, and agrees with, the actual proof of STRATA D.4. □

F.3. [Refuted] Unrestricted pullback base change. The left or right Kan half is not a formalism satisfying its corresponding base-change identity for all strict pullback squares of Cat. Failure already occurs for endpoint inclusions into [1], with discrete finite coefficients. It also occurs with stable coefficients.

Proof. For the left comparison take $f : e_1 \to [1]$ and $g : e_0 \to [1]$. The pullback is empty. If A is a two-point space on e_1, then $(f_!A)_0 = A$, while left Kan extension from the empty pullback has initial value. The comparison is $\varnothing \to A$. For the right comparison interchange 0 and 1. Then $(f_*A)_1 = A$ and right Kan extension from the empty pullback is terminal, giving $A \to *$. In a nonzero stable derivator replace A by a nonzero object and the initial and terminal values by zero. Neither comparison is invertible. The program `code/ef_finite_checks.py` enumerates the indexing categories, their complete composition tables, and the cardinalities of these comparisons. □

F.4. [Proved] The projection formula on the smooth half. Let $\mathbb{D}$ be a symmetric monoidal derivator whose external tensor preserves homotopy colimits separately, in the sense of Groth, Ponto, and Shulman (2014a, Definitions 3.23 and 3.25). If $f : X \to Y$ is W_∞-smooth, then for $A \in \mathbb{D}(X)$ and $B \in \mathbb{D}(Y)$ the canonical projection map

$$f_!(A \otimes f^*B) \longrightarrow f_!A \otimes B$$

is invertible. More generally this map is invertible for an arbitrary f when B is locally constant. The external formula

$$(f \times \mathrm{id}_Z)_!(A \boxtimes C) \cong f_!A \boxtimes C$$

holds for every f, every small Z, and every $C \in \mathbb{D}(Z)$.

Proof. For smooth f, evaluate at y. F.2 computes both sides by the coherent diagram on X_y^{op}. The restriction of f^*B to this fibre is the constant diagram of value B_y. Hence the comparison is

$$\operatorname*{colim}_{X_y^{\mathrm{op}}}(A|_{X_y} \otimes B_y) \longrightarrow \big(\operatorname*{colim}_{X_y^{\mathrm{op}}} A|_{X_y}\big) \otimes B_y,$$

which is invertible by separate cocontinuity. Der 2 proves the result. For a locally constant B, the natural transformation on $y\backslash X$ induced by the arrows $y \to f(x)$ identifies the restriction of f^*B with the constant coefficient B_y, since its components are invertible. The same argument uses the full comma formula instead of the fibre formula. This proves the larger coefficientwise assertion. The external formula is exactly separate cocontinuity of the external tensor, and uses no assertion that the internal tensor of two variable diagrams commutes with arbitrary Kan extension. □

F.5. [Refuted] Monoidality alone does not give an internal projection formula for all maps. Even for space-valued or stable presheaves the projection formula of F.4 fails for a general functor. Thus the unqualified monoidality assertion in Lead 3 needs its map or coefficient hypothesis.

Proof. Take $f : e_1 \to [1]$, and let $A = *$ and $B = (B_1 \to B_0) = (\varnothing \to *)$ be a presheaf of spaces. At 0 the two sides of the projection map are respectively $\varnothing$ and $*$. In spectra, or in $D(k)$ for a nonzero ring k, use $A = \mathbb{1}$ and $B = (0 \to \mathbb{1})$; the comparison at 0 is $0 \to \mathbb{1}$. All arrows, values, and composites are specified, and the set-valued version is checked in the finite program. The separate-colimit hypothesis in the definition of a monoidal derivator concerns the external tensor. It does not identify B_1 with B_0 in this example. □

F.6. [Proved] The right projection formula and its precise restrictions. For any functor f and any dualizable object B of the monoidal category $\mathbb{D}(Y)$, the right projection comparison

$$f_*A \otimes B \longrightarrow f_*(A \otimes f^*B)$$

is invertible. For proper f it is also invertible whenever tensoring with B_y preserves the homotopy limits indexed by X_y^{op}, for every y. In a stable monoidal derivator this includes homotopy-finite fibres, for arbitrary B. Without such restrictions the right projection formula is false, even for a proper and smooth projection.

Proof. If B has a dual $B^\vee$, tensoring with B has both adjoints given by tensoring with $B^\vee$. Apply the adjunction $f^* \dashv f_*$ and the strong monoidality of f^* to obtain, naturally in $M \in \mathbb{D}(Y)$,

$$\begin{aligned}
\mathrm{Hom}(M, f_*A \otimes B) &\cong \mathrm{Hom}(M \otimes B^\vee, f_*A) \\
&\cong \mathrm{Hom}(f^*M \otimes f^*B^\vee, A) \\
&\cong \mathrm{Hom}(f^*M, A \otimes f^*B) \\
&\cong \mathrm{Hom}(M, f_*(A \otimes f^*B)).
\end{aligned}$$

These adjunctions identify the resulting isomorphism with the projection comparison. For proper f, use the fibre-limit formula in F.2. Separate cocontinuity makes tensor exact in a stable derivator; it therefore preserves homotopy-finite limits, giving the stated finite case.

For failure without the restrictions, let $f : \{0, 1\} \to e$, take terminal coefficients on both points, and take B to be a two-point space. The comparison is the diagonal $B \to B \times B$, with cardinalities 2 and 4. Product projections are proper and smooth; the finite computation verifies this obstruction. In a stable example with infinitely many points, tensor does not in general preserve the relevant products. No dual assertion about monoidal derivators may therefore be obtained merely by reversing every arrow. □

F.7. [Proved] Exchange formulas obtained by taking mates. Suppose $\mathbb{D}$ is closed symmetric monoidal. For every $f : X \to Y$ there is a canonical isomorphism

$$f_*\underline{\mathrm{Hom}}_X(f^*B, A) \cong \underline{\mathrm{Hom}}_Y(B, f_*A).$$

For smooth f, put $f^! = f^*$ as the right adjoint of $f_!$. The projection formula then gives

$$f_*\underline{\mathrm{Hom}}_X(A, f^!B) \cong \underline{\mathrm{Hom}}_Y(f_!A, B),$$
$$f^!\underline{\mathrm{Hom}}_Y(B, C) \cong \underline{\mathrm{Hom}}_X(f^*B, f^!C).$$

The right adjoint mate of the left base-change isomorphism in F.2 is

$$f^*g_* \cong g'_*f'^*.$$

All these isomorphisms have the coherence inherited from their specified adjunctions.

Proof. For the first formula, test against $M \in \mathbb{D}(Y)$ and successively use $f^* \dashv f_*$, closedness, and $f^*(M \otimes B) \cong f^*M \otimes f^*B$. The resulting represented functor is $\mathrm{Hom}(M, \underline{\mathrm{Hom}}_Y(B, f_*A))$. For the second formula, the same calculation reduces the comparison to $f_!(A \otimes f^*M) \cong f_!A \otimes M$, which is F.4. For the third, test against $N \in \mathbb{D}(X)$, use $f_! \dashv f^!$, and apply the projection formula to $N \otimes f^*B$. These Yoneda calculations identify the canonical mates rather than arbitrary isomorphisms. Finally the right adjoints of $f'_!g'^*$ and $g^*f_!$ are respectively $g'_*f'^*$ and f^*g_*. Taking the mate reverses the displayed arrow. Triangle identities ensure that iterated mates agree with horizontal and vertical composition; this is also the calculation underlying F.1. □

F.8. [Proved] Why the adjunction directions reverse. Define the opposite derivator by $\mathbb{D}^\vee(S) = \mathbb{D}(S^{\mathrm{op}})^{\mathrm{op}}$. Then

$$f_!^{\mathbb{D}^\vee} = ((f^{\mathrm{op}})_*^{\mathbb{D}})^{\mathrm{op}}, \qquad f_*^{\mathbb{D}^\vee} = ((f^{\mathrm{op}})_!^{\mathbb{D}})^{\mathrm{op}}.$$

Opposition of indexing categories exchanges incoming and outgoing lifting categories, hence exchanges smooth and proper maps. The reversal of the adjunction arrows additionally uses opposition of the coefficient categories. Base opposition alone supplies a comparison of the two fine theories over opposite bases; it is not a duality of coefficients over a fixed base. Nor need $\mathbb{D}^\vee$ be monoidal with a separately cocontinuous tensor.

Proof. An adjunction $L \dashv R$ is, by definition, a natural bijection of the relevant Hom sets; reversing both categories changes it into $R^{\mathrm{op}} \dashv L^{\mathrm{op}}$. Apply this to $(f^{\mathrm{op}})_! \dashv (f^{\mathrm{op}})^* \dashv (f^{\mathrm{op}})_*$. An arrow $f(x) \to y$ becomes $y \to f(x)$, and the category of its outgoing lifts becomes the opposite of the incoming lifting category. Weak contractibility is unchanged by opposition. This proves the exchange of the two exactness tests and explains the opposite adjunction directions at VI ms. 84. Separate cocontinuity becomes a limit-preservation requirement after opposition; it is not automatic. This last point is explicitly recorded in Groth et al. (2014a, Example 3.28). The failures in F.6 and Part G exhibit what cannot be inferred from this formal duality. □

F.9. [Known] Open–closed extensions and stable recollement. Let $j : A \hookrightarrow X$ be a sieve and $i : B \hookrightarrow X$ its complementary cosieve. For a pointed derivator there are adjunctions

$$j^? \dashv j_! \dashv j^* \dashv j_*, \qquad i_! \dashv i^* \dashv i_* \dashv i^!.$$

All four Kan extensions are fully faithful; $j_!$ and i_* are extension by zero. In a stable derivator there are coherent fibre–cofibre sequences

$$j_! j^* A \longrightarrow A \longrightarrow i_* i^* A, \qquad i_* i^! A \longrightarrow A \longrightarrow j_* j^* A.$$

Strongness is not needed for these coherent constructions. A triangulated recollement on the ordinary values follows when the stable derivator is strong.

These statements, in the opposite diagram convention, are Groth (2013, Lemma 3.7, Corollary 3.8, Theorem 4.16, and the recollement discussion in Section 4). To verify the hypotheses and variance directly, the outside comma categories for $j_!$ and i_* are empty, so their values are respectively initial and terminal, which agree in a pointed derivator. Their restrictions to their source are the identity, because the inclusions are fully faithful. The adjoints outside the three Kan operations are obtained by the cylinder construction of the cited lemma; it applies after every shift. Taking the coherent cofibre of $j_! j^* A \to A$ is zero on A and restricts to $i^* A$ on B. The essential-image characterization of i_* identifies it with $i_* i^* A$. The dual argument gives the second sequence. The coherent construction uses Kan extensions on $X \times [1]^2$. Strongness enters only when arbitrary morphisms in $\mathbb{D}(X)$ are lifted to coherent arrows to define the triangulated structure.

F.10. [Proved] A correspondence formalism on the smooth class, with an enhancement. Let $\mathcal{V}$ be a $\mathbb{V}$-presentable closed symmetric monoidal infinity-category whose tensor preserves $\mathbb{U}$-small colimits separately. Regard the ordinary category $\mathrm{Cat}_{\mathbb{U}}$ as an infinity-category by its nerve, so its pullbacks are the strict categorical pullbacks. Put $E = \mathrm{Liss}_{W_\infty}$. There is a lax symmetric monoidal functor

$$\mathrm{Corr}(N\mathrm{Cat}_{\mathbb{U}})_{E,\mathrm{all}} \longrightarrow \mathrm{Cat}_{\infty,\mathbb{V}}, \qquad S \longmapsto \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{V}),$$

sending a span $X \xleftarrow{a} M \xrightarrow{b} Y$, $b \in E$, to $b_! a^*$. It is a pre-six-functor formalism in Mann (2022, Definitions A.5.1 and A.5.6) and a six-functor formalism in Definition A.5.7 there. The restriction to homotopy categories is the indicated half of the represented derivator. This assertion uses the given infinity-category, not merely its valuewise homotopy categories.

Proof. Identities, composites, and strict pullbacks of smooth functors are smooth, so E is a geometric setup in the cited 2022 convention. Restriction of presheaves is a coherent contravariant functor on $N\mathrm{Cat}$. Its left Kan adjoints exist, and F.2 and F.4 apply before taking homotopy categories: the same comma inclusions are final infinity-functors and the same colimits compute the adjoints. For spans $X \xleftarrow{a} M \xrightarrow{b} Y$ and $Y \xleftarrow{c} N \xrightarrow{d} Z$, put $P = M \times_Y N$, with projections $r : P \to M$ and $s : P \to N$. The composition comparison is the specified composite

$$d_! c^* b_! a^* \;\simeq\; d_! s_! r^* a^* \;\simeq\; (ds)_! (ar)^*.$$

The first equivalence is left base change for b; the second is the composition equivalence for left Kan extension. These maps define the composition on the span construction, including all higher simplices. More explicitly, for a finite string of spans take its iterated strict pullback. Every parenthesization is compared with the single left Kan extension from that pullback by the same adjunction universal property. The space of extensions with their units is contractible. The resulting comparisons agree on every face and degeneracy, because their adjuncts are the

composite restriction identifications. This gives coherent composition, not only an associativity assertion in a homotopy category.

The external tensor $A \boxtimes B$ supplies the lax monoidal map from the product of the two presheaf categories to presheaves on the product base. It commutes with restriction and the external left Kan operations, by separate cocontinuity; the preceding unit-based coherence argument applies to several inputs as well. This constructs the map of infinity-operads in the statement. Finally restriction has a right Kan adjoint, $b_!$ has right adjoint b^*, and the pointwise tensor on a presentable functor category is separately cocontinuous and therefore closed. These verify every additional adjoint in Definition A.5.7. The obstruction F.3 prevents extending this construction with the same Kan functors to $E = \mathrm{all}$. □

F.11. [Proved] The terminology of the resulting operations. In F.10 every allowed map has $f^! = f^*$ and $f^!\mathbb{1} = \mathbb{1}$, compatibly with base change. Thus all these maps satisfy the cohomological smoothness formula with the trivial invertible twist; in the usual terminology for this identity they are cohomologically *étale*. This does not say that a Grothendieck-proper functor has $f_! = f_*$. Even a functor belonging to both Grothendieck classes need not have this ambidexterity. For a discrete bifibration with finite fibres, however, a stable coefficient category has the canonical ambidexterity $f_! \simeq f_*$.

Proof. The first identity is the defining Kan adjunction, and its base-change compatibility is the identity between the corresponding restrictions. The map $f^*B \otimes f^!\mathbb{1} \to f^!B$ is consequently the unit isomorphism. For failure of ambidexterity take $p : [1] \to e$. Every map to e is smooth and proper: its relevant lifting category has an identity initial or terminal object. For a presheaf $A = (A_1 \to A_0)$, however, $p_!A = A_0$ and $p_*A = A_1$. Choosing $A = (0 \to \mathbb{1})$ in a nonzero stable coefficient category separates these functors. In the finite discrete bifibration case F.2 gives the sum and product of the values on the same discrete fibre. The stable finite biproduct isomorphism is canonical and respects the bijections of fibres induced by transport, so it gives a natural isomorphism before and after every base change. This is a sufficient class of cohomologically proper maps for the Kan formalism. An arbitrary Grothendieck-proper map is instead characterized by the right base-change theorem; no identification of its two direct images follows. □

F.12. [Refuted] The all-smooth class does not satisfy the later diagonal convention. The class $E = \mathrm{Liss}_{W_\infty}$ in F.10 is not closed under diagonals. Hence it is not a geometric setup under Heyer and Mann (2024, Definition 2.1.1 and Remark 2.1.2), which add this requirement to the earlier convention. Restricting E to discrete fibrations does satisfy the later convention and gives a six-functor formalism by the same construction. In that restriction the preceding cohomological *étale* description is unchanged.

Proof. The projection $[1] \to e$ is smooth, but its diagonal $\Delta : [1] \to [1] \times [1]$ is not. For $x = 1$ and the arrow $(0, 1) \to (1, 1) = \Delta(x)$, an incoming lift would require an object mapping to $(0, 1)$; there is none. This empty lifting category violates smoothness. The finite checker lists the objects, arrows and this missing lift. Discrete fibrations are closed under composition and pullback. Their diagonal is also a discrete fibration: a pair of arrows with the same image and common target has equal sources and equal arrows by uniqueness of a cartesian lift. Thus every incoming arrow to a diagonal object has a unique diagonal lift. Identity maps belong to this class. These facts verify the later geometric setup. F.10 restricts to it, and its adjoints remain the Kan adjoints. This comparison fixes the version of the definition; it does not identify strict pullbacks in $N\mathrm{Cat}$ with pullbacks in the infinity-category of all infinity-categories. □

F.13. [Known] Hörmann's diagram correspondences and the enhancement boundary. Fibered multiderivators encode the two halves and their exchange maps on the appropriate 2-multicategory of diagram correspondences. In Hörmann (2017, Theorem 6.12), for pre-multiderivators $\mathbb{E}, \mathbb{S}$ satisfying Der 1–2, a strict morphism $\mathbb{E} \to \mathbb{S}$ is a left, respectively right, fibered multiderivator precisely when the induced map $\mathrm{Dia}^{\mathrm{cor}}(\mathbb{E}) \to \mathrm{Dia}^{\mathrm{cor}}(\mathbb{S})$ is a 1-opfibration, respectively 1-fibration, of 2-multicategories. A closed monoidal derivator supplies the associated diagram-level six-operation structure. This does not turn all strict pullback squares of ordinary categories into exact squares and does not itself construct an infinity-categorical enhancement of an abstract derivator.

The theorem is used with its diagram-correspondence base, whose 2-morphisms and fibre replacements retain the comma-category information. It is not a theorem about the unmodified span category of Cat with all strict pullbacks. Indeed any reading with that conclusion would imply the false comparison of F.3. The construction in F.10 has a different input: it starts with the enhanced coefficient category and restricts the class of spans so that strict base change is exact. Passing from it to its derivator loses higher mapping information; a reverse passage is not supplied by Der 1–4 or Der 5. E.13 gives a concrete obstruction to reconstructing even coherent arrows from the valuewise homotopy categories.

F.14. [Proved] Comparison with relative smoothness and properness. For the indexed enhanced category $S \mapsto \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S})$ on the ordinary category Cat with strict pullbacks, the smooth and proper calibrations of Anel and Weinberger (2025, Definition 2.3.1) are exactly Liss_{W_∞} and Prop_{W_∞} in our convention. This is a statement about that indexed category and that base category; replacing its base by Cat_∞ changes the pullback test.

Proof. All restrictions have both adjoints. The definition in the cited source requires the corresponding Beck–Chevalley property after every base change. Smooth and proper functors retain their lifting conditions under base change, and F.2 proves the required invertibility. Conversely that definition in particular gives the comparison for every point of the base. The left-adjoint-mate computation in F.2 recovers contractibility of each lifting category, and therefore recovers exactly the two Grothendieck classes. The extra word "stably" in the relative definition is satisfied by their pullback stability. This proof provides the comparison for a specific indexed category; it does not use an assertion that all indexed categories on a given base have the same calibration. □

F.15. [Known] The genus-zero category and its geometric coefficients. For $n \geq 3$, let Γ_n be the pure orientation-preserving mapping class group of the n-punctured sphere. Let $\mathcal{C}_n$ have as objects the isotopy classes of essential multicurves, including the empty multicurve, and put

$$\mathrm{Hom}_{\mathcal{C}_n}(\sigma, \tau) = \{g \in \Gamma_n : g\sigma \supseteq \tau\}/\Delta_\sigma, \qquad [h] \circ [g] = [hg],$$

where Δ_σ is generated by the Dehn twists in the curves of σ and acts on the right. Set $T_{0,n} = \mathcal{C}_n^{\mathrm{op}}$. The exit category of $\overline{\mathcal{M}}_{0,n}(\mathbb{C})$ with its dual-tree stratification is $N\mathcal{C}_n$. For every compactly generated presentable infinity-category $\mathcal{V}$,

$$\mathrm{Shv}_{\mathrm{cbl}}(\overline{\mathcal{M}}_{0,n}(\mathbb{C}); \mathcal{V}) \simeq \mathrm{Fun}(N\mathcal{C}_n, \mathcal{V}) = \mathrm{Fun}(NT_{0,n}^{\mathrm{op}}, \mathcal{V}).$$

Forgetting the last mark induces $Q : \mathcal{C}_{n+1} \to \mathcal{C}_n$, $\sigma \mapsto q\sigma$, $[g] \mapsto [\phi(g)]$, and $F = Q^{\mathrm{op}} : T_{0,n+1} \to T_{0,n}$. Here q erases curves which become peripheral and identifies repeated isotopy classes, and $\phi : \Gamma_{n+1} \twoheadrightarrow \Gamma_n$ is the forgetful homomorphism.

The exit-category and coefficient assertions are Jansen (2023, Construction 3.18, Proposition 3.23, Corollary 6.6, and Theorem 6.7), applied to a smooth pointed sphere. We use the general

stable-surface hypothesis $2g-2+n>0$; the second displayed inequality in the attached text of Theorem 6.7 has a sign slip and is not our hypothesis. For the explicit functor, a twist about a vanishing curve maps to the identity, and twists about curves with the same image map to the same twist. Hence $\phi(\Delta_\sigma)\subseteq\Delta_{q\sigma}$, which makes Q independent of representatives and compatible with multiplication. The operation on a degeneration is forgetting its last mark and stabilizing; on an automorphism it is ϕ. These are the generators in Construction 3.18, so this identifies Q with the functor induced by the stratified geometric forgetful map. Genus-zero labelled stable curves have no automorphisms: every component has at least three special points, fixed individually by an automorphism. Thus these examples are spaces rather than stacky quotients.

F.16. [Proved] Forgetting a mark is proper and need not be smooth. For every $n\geq 3$, $F:T_{0,n+1}\to T_{0,n}$ is W_∞-proper. For $n=4$ it is not W_∞-smooth: a specified lifting category is equivalent to $B\mathbb{Z}$.

Proof. We work first in exit variance. Fix an upstairs multicurve σ, set $\tau=q\sigma$, and fix $\beta:\eta\to\tau$ in $\mathcal{C}_n$. The incoming lifting category for Q has objects $(\rho,\alpha:\rho\to\sigma)$ such that $q\rho=\eta$ and $Q\alpha=\beta$; morphisms are triangles over the identity of η. Represent β by $b\in\Gamma_n$ and choose a lift $\widetilde{b}\in\Gamma_{n+1}$. Replacing ρ by $\widetilde{b}\rho$ and the representative g of α by $g\widetilde{b}^{-1}$ identifies this category with the one for $b\eta\supseteq\tau$ and $\beta=[1]$. The inverse uses $\widetilde{b}^{-1}$. We have used the surjectivity in the Birman exact sequence, valid also for the pure groups here; see Kent, Leininger, and Schleimer (2009, Section 2).

In the case $\beta=[1]$, an object represented by $g:\rho\to\sigma$ is isomorphic in the lifting category to $(g\rho,[1])$: the isomorphism is $[g]$, which lies over the identity since $\phi(g)\in\Delta_\eta$. The full subcategory of these normalized objects is the poset

$$\mathcal{P}(\eta;\sigma)=\{\rho\supseteq\sigma:q\rho=\eta\},\quad\text{ordered by reverse inclusion.}$$

Indeed a triangle between identity arrows forces its representative h to lie in Δ_ρ; it is then the unique identity coset, and exists exactly if $\rho\supseteq\rho'$. This proves an equivalence of categories, not merely a bijection on components.

We verify contractibility of this poset, including the cases affected by stabilization. First take $\sigma=\varnothing$. If η is nonempty, choose disjoint geodesic representatives on the downstairs punctured sphere. Their inverse image in its universal disc is a locally finite family of disjoint separating lines. The graph whose vertices are complementary regions and whose edges are these lines is a tree: a line separates, so every edge separates, and any two regions are joined by the finite sequence crossed by a compact transverse path. A lift of η with no repeated image records the region occupied by the extra puncture; a lift with one repeated image records the two curves bounding the annulus containing that puncture, and hence an edge. Deleting either of the repeated curves records the two incident vertices. No second repeated image is possible, since the two disjoint annuli would both have to contain the sole extra puncture. This identifies the nonperipheral part of $\mathcal{P}$ with the face poset, in reverse order, of that tree. Equivalently this is the simplicial fibre calculation of Kent et al. (2009, Theorem 7.1); the preceding description fixes which face poset is being used.

A curve which disappears on forgetting encloses just the extra puncture and one retained puncture. At most one can occur in a disjoint multicurve. It cannot coexist with the repeated-image annulus, since the retained puncture is outside that annulus. Its deletion therefore attaches one leaf to the tree just described. All its occurrences give leaves, possibly infinitely many; contracting the added edges, or using the unique paths in the enlarged tree, preserves contractibility. For $\eta=\varnothing$ the empty multicurve itself is a terminal object of $\mathcal{P}$, and the same description consists of a central vertex and these leaves.

For general σ, cut along its curves. Curves of $\eta \setminus q\sigma$ in every component not containing the extra puncture lift uniquely; there the forgetful operation is the identity, with boundary circles regarded as fixed punctures. In the component containing the extra puncture, if the remaining component is stable, the same tree-and-leaves argument applies to its required multicurve. The retained puncture in a peripheral leaf may also be a retained boundary circle: such a leaf is a duplicate of a curve already in σ. Boundary twists act trivially on curves disjoint from the boundary, so passing from boundary circles to punctures changes none of this poset. If that component becomes unstable, it was a pair of pants whose other two special points are either a retained mark and a boundary, or two boundaries. It admits no essential refinement. Keeping σ fixes respectively the peripheral bubble or the bridge; all refinements in the other components remain unique. In this case the remaining poset is a singleton. These descriptions also cover an empty required refinement. Thus every incoming lifting category for Q is weakly contractible. By opposition this proves properness of F.

For nonsmoothness take $n = 4$ and choose an upstairs maximally degenerate curve whose extra mark 5 is on the middle component of a three-component chain. The two curves c_1, c_2 separate 12 from 345 and 34 from 125. Their images are the same downstairs curve c, separating 12 from 34. Put $\sigma = \{c_1, c_2\}$. We test outgoing lifts for Q of $[1] : \{c\} \to \varnothing$. If $q\rho = \varnothing$ and $\sigma \to \rho$ is such a lift, then $\rho = \varnothing$: any nonempty such ρ encloses 5 and one retained mark, whereas neither curve in any pure mapping class translate of σ has that partition. The lifting category is therefore the action groupoid of

$$K = \ker \phi \quad \text{on} \quad \{g\Delta_\sigma : \phi(g) \in \langle t_c \rangle\}.$$

We have $\Delta_\sigma = \langle t_{c_1}, t_{c_2} \rangle \cong \mathbb{Z}^2$ and $\phi(t_{c_1}) = \phi(t_{c_2}) = t_c$. Independence of twists follows by choosing a transverse arc to each disjoint annulus and observing its winding number. The action is transitive because the image of Δ_σ is all of $\langle t_c \rangle$, and the stabilizer of its identity coset is

$$K \cap \Delta_\sigma = \ker(\mathbb{Z}^2 \xrightarrow{(a,b)\mapsto a+b} \mathbb{Z}) = \langle t_{c_1} t_{c_2}^{-1} \rangle \cong \mathbb{Z}.$$

Choosing one object in a transitive action groupoid gives a fully faithful, essentially surjective functor from its stabilizer group. Consequently this lifting category is equivalent to $B\mathbb{Z}$, whose nerve has nonzero H_1. It is not weakly contractible. This is the incoming test for F, so F is not smooth. The arithmetic in the last display is checked by the finite-support integer calculation in the supplementary code; the category itself is infinite, and no finite category is claimed here. □

F.17. [Proved] Geometric direct image and the boundary squares in this example. In the coefficient equivalence of F.15, the categorical right Kan functor F_* is the ordinary topological direct image of constructible sheaves for $\overline{\mathcal{M}}_{0,n+1}(\mathbb{C}) \to \overline{\mathcal{M}}_{0,n}(\mathbb{C})$. Categorical right base change agrees with topological proper base change for point fibres and for the squares obtained by a closed boundary inclusion. A boundary is represented by the replete full cosieve in the entrance category whose multicurves contain a curve with the specified label partition. The left Kan half supplies no universal smooth base-change statement for this forgetful functor.

Proof. The geometric forgetful map is the universal stable curve, hence is proper; the contraction and stabilization description is recalled in Keel (1992, pp. 546–548). We check the extra issue that its direct image preserves the specified constructibility. Let B be a sufficiently small contractible ball in a fixed dual-tree stratum of the base. Normalize its nodal curves. Each component is a sphere with finitely many labelled special points; over B choose three to rigidify it and disjoint discs about all the others. Isotopies supported in these discs move the special points to their central positions. Gluing back the specified pairs gives a trivialization

over B by a fixed marked nodal sphere. The strata in the universal curve distinguish its smooth unmarked points, marked points, and nodes; the latter two are respectively the bubble and bridge cases of stabilization. The trivialization preserves all these strata.

A constructible coefficient on this product is constant in the contractible B-direction: by exodromy, its diagram is indexed by the product of the fundamental infinity-groupoid of B with the exit category of the fixed stratified fibre, and the first factor is contractible. Proper base change therefore makes every stalk of its direct image on B the sections of the same fibre coefficient, with invertible transport. For the compactly generated coefficients in F.15 this base-change theorem follows from its space-valued version: write $\mathcal{V} \simeq \mathrm{Ind}(\mathcal{V}^{\omega})$ and describe its sheaves as finite-limit-preserving functors from $(\mathcal{V}^{\omega})^{\mathrm{op}}$ to space-valued sheaves. Inverse image and direct image give the same comparison at each compact test object, and both respect this finite-limit condition. This proves the required coefficient extension, not only its stable special case. It also proves constructibility on each base stratum. Inverse image already preserves constructibility, since the map sends each stratum into one base stratum. Its restriction corresponds to F^{*} by the naturality verified in F.15. Both its restricted topological direct image and F_{*} are right adjoint to this same functor between the full constructible categories. The contractible uniqueness of right adjoints identifies them, including their units and counits.

For clarity we also identify the indexing category of an actual fibre; a mere equivalence of its total homotopy type would not suffice for the base-change assertion. Fix a stable labelled nodal curve downstairs, represented by τ. In its inverse image an extra point may lie in a smooth unmarked component, at a retained marked point, or at a node. The last two give a bubble or a bridge. Their marked upstairs multicurves are precisely the σ with $q\sigma = \tau$. A morphism which fixes the downstairs point has representative g with $\phi(g) \in \Delta_{\tau}$, exactly the identity condition in the strict fibre of Q. Every smooth component minus its special points is a sphere minus at least three points, hence a $K(\pi, 1)$ with free fundamental group. The link of a marked point is a circle, and a node has one circle on each branch. Each peripheral $\mathbb{Z}$ injects into the corresponding free group: choose a free basis of all but one puncture loops, in which the remaining loop is their nontrivial product. The homotopy fibres computing exit paths from a singular point into a smooth component are therefore discrete coset sets. There are no higher exit mapping homotopies, and the closed-point automorphisms are trivial. Point-pushing gives these same free groups in the strict categorical fibre; quotienting by the added twists gives precisely these peripheral cyclic cosets. Composition is multiplication of their representatives, as in F.15. These descriptions are exhaustive because the component containing the forgotten point is either retained, a bubble, or a bridge. Thus the exit category of the fibre is the opposite of the strict fibre of F. In particular, for $n = 4$ the ordinary fibres have homotopy types S^{2} over a smooth curve and $S^{2} \vee S^{2}$ over a boundary curve; this does not contradict categorical properness.

For a fixed stable partition of the labels, degeneration can add curves and cannot remove the specified partition. In entrance variance its replete full subcategory is consequently a cosieve. Cutting along the unique curve of that partition identifies it with the product of the two smaller entrance categories: twists along the cut are divided out, and the remaining homeomorphisms are exactly those of its two labelled components. This is also the stratified category of the closed clutching boundary. The inverse-image boundary selects exactly the upstairs multicurves whose q-image has that partition. Hence the geometric boundary square induces the strict categorical pullback square. A subcategory requiring a single chosen isotopy representative, without taking its repletion, would not justify this assertion.

The two comparisons in these point and boundary squares are mates of the same restriction identity under the identified adjunctions. They therefore agree as canonical maps, and both are invertible by properness. No such identification has been asserted for an arbitrary geometric pullback: one would first have to prove the corresponding exit-category pullback comparison. Finally F.16 and the converse in F.2 exclude universal left Kan base change for this map. This

separates the topological exceptional direct image, which exists because the geometric map is proper, from categorical left Kan extension. □

F.18. [Open] Indexing an abstract derivator by strict higher categories. For a specified extension of an ordinary derivator to strict n-categorical indexing shapes, $2 \le n \le \omega$, with lax or colax comma objects and a chosen Street-nerve weak-equivalence class, does the corresponding universal smooth/proper lifting condition imply the two base-change isomorphisms of F.2, with coherent mates? We have not established this statement. Even $n = 2$ requires an explicit higher Der 4 and an invariance theorem for the chosen lax comma construction; neither is part of an ordinary derivator on Cat. The model-structure claim discussed in Part C supplies neither datum. No proof in Part F uses that unrefereed claim.

The attempted reduction would replace a higher comma object by a category of simplices and then apply F.2. Its missing step is a comparison respecting both the proposed higher Kan adjunction and the canonical mate; equality of total classifying-space types does not provide it. This is the precise remaining higher-indexing question, not an extension claimed from an ordinary Theorem A.

F.19. [Proved] An exact proper class in the diagonally closed setup. In the discrete-fibration setup of F.12, take $\mathcal{V} = \mathrm{Sp}$ or $D(k)$ for a nonzero commutative ring k. For a discrete fibration $p : E \to S$, the following are equivalent: p is cohomologically proper in Heyer and Mann (2024, Definition 4.6.1(b)); after every strict base change its left and right presheaf Kan extensions are naturally equivalent; and p is a discrete bifibration with finite fibres. Finiteness is pointwise and requires no uniform bound. For this class the canonical comparison $p_! \to p_*$ is the finite biproduct equivalence.

Proof. The cited definition requires a truncated map, cohomological primness, and properness of its diagonal, recursively. All maps in the nerve of an ordinary category are 0-truncated, since their maps between representable mapping spaces are maps of sets. Lemma 4.6.3 of the cited source gives base-change stability; Lemma 4.6.4(ii) constructs the canonical comparison when the diagonal is proper and says that its equivalence is equivalent to properness. Thus properness implies the second condition. We do not assert existence of that canonical comparison for an arbitrary discrete fibration before its diagonal has been checked.

Suppose the second condition holds. Pullback to a vertex gives a discrete set $I \to e$, whose Kan functors are direct sum and product. For infinite I, product does not preserve filtered colimits: for each finite $J \subset I$, take the diagram with value $M = \mathbb{S}$ or k on J and zero elsewhere. The filtered-colimit comparison is

$$\bigoplus_{i \in I} M \longrightarrow \prod_{i \in I} M,$$

whose degree-zero homotopy map is the non-surjective inclusion of the direct sum into the product of copies of $\mathbb{Z}$ or k. The sum functor preserves those colimits. Hence the two functors cannot be naturally equivalent, and every fibre of p is finite.

Pull back next along an arbitrary arrow $\alpha : [1] \to S$. The resulting category has a finite set A over 0, a finite set B over 1, and exactly one arrow $r(b) \to b$ for each $b \in B$, for the reindexing function $r : B \to A$. For $a \in A$ take the presheaf F with value M at a, zero elsewhere, and all nonidentity maps zero. The smooth fibre formula and the actual right comma formula give

$$(p_!F)(1) = 0, \qquad (p_*F)(1) = \mathrm{fib}\big(M \xrightarrow{\mathrm{diag}} M^{r^{-1}(a)}\big).$$

For $m = |r^{-1}(a)|$, the latter is M if $m = 0$, zero if $m = 1$, and $(\Omega M)^{m-1}$ if $m \geq 2$. Indeed subtracting the first coordinate from the others turns the diagonal into the first-coordinate inclusion; its cofiber is M^{m-1}, and its fibre is the loop of that cofiber. Taking the indicated nonzero M forces $m = 1$. Thus every reindexing function is bijective, exactly the additional discrete-opfibration condition.

Conversely let p be a finite discrete bifibration. Its diagonal is a discrete bifibration with empty or singleton fibres. The next diagonal is an isomorphism, so Lemma 4.6.4(ii), first applied to the diagonal, proves its properness: both direct images there are the identical extension by zero on a union of components. The canonical comparison for p consequently exists. F.2 computes it by the direct sums and products on the same finite fibres. Unwinding the mate in Lemma 4.6.4(ii) gives the matrix with identity on each matching summand and zero on distinct summands. It is the biproduct isomorphism, compatible with every fibre-transport bijection. The same lemma now proves properness of p. Every strict base change remains a finite discrete bifibration. Empty fibres cause no exception, since the stable empty sum and empty product are both zero. The finite star calculations for $m = 0, 1, 2$ are included in the supplementary checker. □

7 Finite bases: weights, duality, and the geometric comparison

Throughout this section the variance is fixed by $\mathbb{D}_{\mathcal{C}}(S) = \mathrm{Ho}\,\mathrm{Fun}(S^{\mathrm{op}}, \mathcal{C})$. Thus p^S_* is a limit over S^{op} and $p^S_!$ is a colimit over S^{op}. A homotopy finite category means a category equivalent to a finite skeletal category with no nonidentity endomorphisms. This is a condition on the indexing category, not merely on the weak homotopy type of its nerve. In particular a nontrivial finite group, regarded as a one-object category, is not homotopy finite.

G.1. [Known] The exact stability input and the available tensor action. In a stable derivator satisfying Der 1–4, homotopy finite limits commute with arbitrary homotopy colimits. Every such derivator has a canonical action by the derivator of spectra. Consequently a right homotopy finite right Kan extension can be expressed as a weighted homotopy colimit with a spectral profunctor as weight. No internal monoidal structure on the coefficient derivator and no strongness axiom are included in this assertion.

Proof. The commutation theorem is Groth and Shulman (2017, Theorem 2.14). The spectral action is the consequence of Cisinski and Tabuada (2011, Theorem A.11 and Appendix A.3): an object X determines the cocontinuous morphism from spectra sending the sphere to X. Their Definition A.1 separates "strong" from "triangulated or stable"; Theorem A.11 is stated for any triangulated derivator. The resulting coherent action, not an arbitrary choice of tensor product, is the one used here. The weighted Kan-extension statement is precisely Groth and Shulman (2017, Theorem 4.11(iv)), applied to this module structure; their Construction 4.8 and Lemma 4.10 give the coend and its Yoneda identity. Their convention is covariant diagrams, so it is applied to the opposite of the indexing functor when using the presheaf convention of this paper. □

G.2. [Proved] The canonical dualizing weight and its exact variance. Let S be homotopy finite. Write

$$R_s(t) = \Sigma^\infty_+ \mathrm{Hom}_S(t, s), \qquad w_S(s) = \lim_{t \in S^{\mathrm{op}}} R_s(t). \tag{G.2a}$$

Then $w_S : S \to \mathrm{Sp}$ is a covariant spectral diagram, with finite-spectrum values, and for every stable derivator there is a canonical natural isomorphism

$$p^S_* F \simeq w_S \otimes_{[S]} F = \int^{s \in S} w_S(s) \otimes F(s). \tag{G.2b}$$

The integral denotes the derived coend, with its coherent spectral action. It is not an objectwise tensor followed by the ordinary colimit over S^{op}.

Proof. The spectral Yoneda identity gives the coherent density expression

$$F(t) \simeq \int^{s \in S} R_s(t) \otimes F(s).$$

Apply the homotopy limit over $t \in S^{\mathrm{op}}$. By G.1 this limit commutes with the homotopy colimits defining the coend. It also commutes with the tensor action in its spectral variable: the action preserves colimits, and a cocontinuous functor between stable derivators preserves finite homotopy limits. The result is (G.2b). This is also the terminal-target instance of the weight in Groth–Shulman's Theorem 4.11(iv), so the calculation is coherent in every external diagram variable, not just an objectwise calculation in the underlying category.

Replace S by a finite skeletal representative. Every hom-set is finite, so every $R_s(t)$ is a finite wedge of sphere spectra. A limit over this finite direct shape is a finite homotopy limit; finite spectra are closed under those limits. This proves finiteness of $w_S(s)$. Equivalent indexing categories give canonically equivalent weights after transport. For $S = \varnothing$, the coend and the limit are both the zero object in a stable category. □

G.3. [Refuted] A pointwise dualizing-object twist already fails on one arrow. The assertion that every homotopy finite S admits $\omega \in \mathrm{Fun}(S^{\mathrm{op}}, \mathrm{Sp})$ with

$$p_*^S F \simeq p_!^S(\omega \otimes F) \quad \text{for all } F \tag{G.3}$$

where the tensor is pointwise, is false. It remains false for pointwise perfect complexes over a nonzero field.

Proof. Take $S = [1] = (0 \to 1)$. A presheaf is an arrow $F_1 \to F_0$. The initial object of S^{op} is 1, and its terminal object is 0; therefore $p_*F = F_1$ and $p_!F = F_0$. Choose $F = (M \to 0)$, with M any nonzero finite spectrum, or the field concentrated in degree zero. The right side of (G.3) is $\omega_0 \otimes 0 = 0$, while the left side is M. This contradiction does not depend on the value, dualizability, or invertibility of ω. In contrast, the actual weight in (G.2a) is $w_{[1]} = (0 \to \mathbb{S})$, a diagram on S, and its coend with F is F_1. The endpoint and weight calculations are included in `code/dg_finite_checks.py`. □

G.4. [Proved] Transitivity and base change belong to the kernels. For a functor $u : R \to S$ between homotopy finite categories, set

$$K_u(s, r) = (u_* R_r)(s), \qquad K_u \in \mathrm{Fun}(S^{\mathrm{op}} \times R, \mathrm{Sp}).$$

Then $u_*F \simeq K_u \otimes_{[R]} F$. For $v : S \to T$ there are canonical composition and transitivity identities

$$K_{vu}(t, r) \simeq \int^{s \in S} K_v(t, s) \wedge K_u(s, r), \qquad w_R(r) \simeq \int^{s \in S} w_S(s) \wedge K_u(s, r). \tag{G.4}$$

A homotopy exact square gives the corresponding base-change equivalence of these kernels and of their weighted functors. No claim that every cartesian square is homotopy exact is needed.

Proof. The comma categories computing u_* are finite and have no nonidentity endomorphisms after replacing the indexing categories by finite skeletal representatives. Hence u_* preserves homotopy colimits by G.1. Apply u_* to the density expression for F in G.2. This gives the first formula and proves its naturality. Apply the formula twice to v_*u_*F, and use Fubini for homotopy coends. The canonical isomorphism $(vu)_* \simeq v_*u_*$ identifies the resulting kernel, proving the first identity in (G.4). Set $T = e$ to obtain the second identity. Associativity of these isomorphisms is the associativity of Kan extension and of coends; therefore this is a coherent transitivity formula.

For a square whose specified Beck–Chevalley mate is invertible, evaluate that mate on all the spectral representables R_r. This is the asserted kernel comparison. Conversely, when the functors concerned preserve colimits, their density expressions show that equivalence on these representables gives the equivalence on every diagram. Thus the compatibility is with the actual base-change mate and its pasted composites. This argument does not replace that mate by an unrelated isomorphism of functors. □

G.5. [Refuted] The weight is not invariant under unrestricted restriction. The formula $i^*w_S \simeq w_R$ for every functor $i : R \to S$ is false, even for finite posets. Consequently no base-change statement for a "dualizing object" may be supplied by ordinary restriction without an additional hypothesis or an extraordinary pullback.

Proof. Let $i : e \to [1]$ select 0. G.3 computes $i^*w_{[1]} = 0$, whereas $w_e = \mathbb{S}$. The nonzero sphere cannot be equivalent to zero. The same example shows that the transform in G.9 below does not commute with this restriction: its value at 0 is always zero, whereas the transform on e is the identity. The kernel transitivity formula G.4 is the valid replacement. □

G.6. [Proved] A coefficient duality on precisely specified finite objects. For every small S, pointwise Spanier–Whitehead duality gives

$$\mathrm{Fun}(S^{\mathrm{op}}, \mathrm{Sp}^{\mathrm{fin}})^{\mathrm{op}} \simeq \mathrm{Fun}(S, \mathrm{Sp}^{\mathrm{fin}}). \tag{G.6a}$$

For a ring A, derived A-linear duality similarly gives

$$\mathrm{Fun}(S^{\mathrm{op}}, \mathrm{Perf}(A))^{\mathrm{op}} \simeq \mathrm{Fun}(S, \mathrm{Perf}(A^{\mathrm{op}})). \tag{G.6b}$$

For homotopy finite S, all limits and colimits in these formulas remain perfect and $D(p^S_* F) \simeq p_!^{S^{\mathrm{op}}}(DF)$. This is a contravariant duality. In (G.6b) the coefficient ring can be kept unchanged when it is commutative.

Proof. For a finite spectrum E, the evaluation map $E \to \mathrm{Map}(\mathrm{Map}(E, \mathbb{S}), \mathbb{S})$ is an equivalence: it holds for the sphere and is preserved by finite cofibres, suspensions, and retracts. This proves the equivalence on finite spectra. Applying it to values and reversing structure maps gives (G.6a), including all coherences. For perfect A-modules the same proof starts from finite free modules and uses $\mathrm{RHom}_A(-, A)$; the dual is a module on the opposite side, which accounts for A^{op}.

A limit over a homotopy finite shape is constructed by finite homotopy limits, and finite spectra and perfect modules are closed under those operations. An equivalence to an opposite category exchanges their limits and colimits. Applying this to the diagram gives the last formula. No assertion is made that arbitrary spectra or arbitrary modules are reflexive under this duality. □

G.7. [Refuted] Finite EI is not a coefficient-independent extension of homotopy finite. The weighted-colimit assertion for all stable coefficients does not extend from homotopy finite categories to all finite EI categories. For $S = BC_2$ and coefficients $k = \mathbb{F}_2$, the functor p_*^S does not preserve countable coproducts, and hence cannot be any fixed weighted homotopy colimit on all diagrams.

Proof. The category has one object and exactly the arrows $1, g$, with $g^2 = 1$. It is finite and every endomorphism is invertible. Write $A = k[C_2] = k[\epsilon]/(\epsilon^2)$, where $\epsilon = g - 1$. The trivial module k has the bounded-above projective resolution

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \longrightarrow k \longrightarrow 0,$$

with A in every degree $0, -1, -2, \ldots$. Exactness follows from $\ker(\epsilon : A \to A) = \epsilon A = \operatorname{im}(\epsilon)$ and $\ker(A \to k) = \epsilon A$. A bounded-above complex of projectives computes derived Hom into any complex.

Let $M_n = k[n]$ for $n \geq 0$ in cohomological notation, with trivial action, and let $M = \bigoplus_{n\geq 0} M_n$. In degree zero the Hom complex from the displayed resolution to M is $\prod_{n\geq 0} k$. Every differential is zero, since ϵ acts trivially on the target. Thus

$$H^0 \operatorname{RHom}_A(k, M) = \prod_{n\geq 0} k, \qquad \bigoplus_{n\geq 0} H^0 \operatorname{RHom}_A(k, M_n) = \bigoplus_{n\geq 0} k.$$

The comparison is the usual inclusion of the direct sum into the product and omits the sequence $(1, 1, \ldots)$. Since $p_*^S = \operatorname{RHom}_A(k, -)$, it fails to preserve this coproduct. A fixed weighted coend preserves all colimits in its input, so none can represent this functor. The multiplication table, associativity, and the resolution's differential matrices are checked in `code/dg_finite_checks.py`; the infinite product argument above, rather than a finite computation, proves the obstruction. □

G.8. [Proved] A finite-group case with its necessary coefficient hypothesis. Let G be a finite group and k a commutative ring in which $|G|$ is invertible. For the derivator of k-modules and $S = BG$, the norm gives a natural equivalence $p_!^S \simeq p_*^S$ on all unbounded complexes. In particular this finite EI case has ambidexterity, although G.7 rules out a uniform assertion without the coefficient hypothesis.

Proof. For any $k[G]$-module M, the map $[m] \mapsto \sum_{g\in G} gm$ from coinvariants to invariants has inverse $x \mapsto |G|^{-1}[x]$. The averaging idempotent $|G|^{-1}\sum_g g$ makes invariants an exact direct summand functor. The trivial $k[G]$-module is projective on either side, so coinvariants and invariants compute their derived functors degree by degree on all complexes. The displayed natural inverse therefore supplies the derived norm equivalence. This is a proved class of finite EI examples, not a classification of all finite EI categories or all coefficient systems. □

G.9. [Known] The exact canonical equivalence criterion for finite posets. For a finite poset P define the colimit-preserving transform

$$(V_P F)(s) = \lim_{t\in P^{\mathrm{op}}} \big(\Sigma_+^\infty \operatorname{Hom}_P(t, s) \otimes F(t)\big), \qquad V_P : \mathbb{D}(P) \longrightarrow \mathbb{D}(P^{\mathrm{op}}). \tag{G.9a}$$

There is always a canonical formula

$$p_*^P \simeq p_!^{P^{\mathrm{op}}} V_P. \tag{G.9b}$$

For spectra the transform (G.9a) is an equivalence if and only if each strict upper interval $P_{>p}$ is Gorenstein* over $\mathbb{Z}$. Here a finite poset L is Gorenstein* when every open interval in L with

a newly adjoined bottom and top has the integral homology of a sphere; the empty interval is allowed as the (-1)-sphere. When this condition holds, (G.9a) gives an equivalence for every stable derivator, and it restricts to pointwise finite spectra or perfect complexes. This is a condition on this canonical transform; it is not a classification of every possible equivalence between the two diagram categories.

Proof. Apply Aoki (2022, Theorem A and Proposition 5.12) to $Q = P^{\mathrm{op}}$. His diagram category is $\mathrm{Fun}(Q, \mathrm{Sp})$, so his lower-interval condition becomes the upper-interval condition just stated. His Definition 3.9 and Proposition 3.11 give the precise Gorenstein* convention, including the empty case. This proves the if-and-only-if assertion with its exact variance.

For completeness, formula (G.9b) itself follows without that condition. Commute the finite limit over t with the colimit over s in (G.9a). For fixed t, the covariant diagram $s \mapsto \mathrm{Hom}_P(t, s)$ has category of elements t/P, which has an initial object. Its homotopy colimit is contractible, so the colimit of its suspension spectra is the sphere. The result is $\lim_t F(t)$, as required.

The extension of the equivalence to stable derivators can be made without an enhancement assumption on that derivator. By G.1 and the density argument of G.4, V_P is tensoring with a spectral bimodule. An inverse to its colimit-preserving equivalence for spectral diagrams is also colimit preserving, and density represents that inverse by a spectral bimodule. The two composites are the identity bimodules: evaluate the two equivalences on representables. Tensoring these bimodule identities with any stable derivator proves the assertion. For a finite direct shape, an objectwise perfect diagram is compact, by its finite normalized bar resolution in representables, and every compact diagram is objectwise perfect. The equivalence preserves compact objects; hence so does its inverse. This proves the stated restriction to finite objects. □

G.10. [Proved] Verification of the finite least-vertex geometric dictionary. Let P be a finite poset and $Z = |NP|$. Label a point by the least vertex with positive barycentric coordinate. With downsets open on P, this is a continuous map $s : Z \to P$; its strata are contractible, the stratification is conical, and its exit-path infinity-category is P^{op}. Consequently

$$\mathrm{Shv}_{P\text{-cons}}(Z; \mathcal{C}) \simeq \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{C}) \tag{G.10}$$

for spaces, for every compactly assembled coefficient infinity-category, and for every stable presentable coefficient infinity-category. The inverse-image functor from sheaves on the finite downset Alexandrov space P is the fully faithful functor onto the subcategory in (G.10).

Proof. For a downset U, the complement of $s^{-1}U$ is the closed subcomplex on $P \setminus U$: a chain has minimum outside U exactly when all its vertices lie outside U. This proves continuity. The stratum labelled p consists of chains with minimum p and positive coordinate at p; it is the open cone on $|N(P_{>p})|$ and is contractible.

Here is an explicit local chart, which also checks the geometric hypothesis instead of assuming it from the attached STRATA package. The open star $t_p > 0$ of a vertex p has the homeomorphism

$$\{t_p > 0\} \cong C_\infty |N(P_{<p})| \;\times\; C_\infty |N(P_{>p})|. \tag{G.10a}$$

The two radial coordinates are $\sum_{q<p} t_q/t_p$ and $\sum_{q>p} t_q/t_p$; the remaining coordinates are normalized on their respective chains. The inverse divides all coordinates by $1 + a + b$. If a radial coordinate is zero, its normalized coordinates are collapsed to the cone vertex, so these formulas extend continuously there. The second factor is a factor inside the p-stratum. The first factor is the normal cone, whose nonvertex strata have their original minimum labels below p. Taking neighbourhoods in the second factor gives conical charts at every point of the p-stratum. These are finite polyhedral charts, hence locally singular and paracompact.

Refine this stratification to the open-simplex stratification of the finite order complex. Its exit category is the face poset $\mathrm{Ch}(P)$ of nonempty chains, ordered by inclusion, by Lurie (2017, Theorem A.6.10). The coarsening map is $m : \mathrm{Ch}(P) \to P^{\mathrm{op}}$, $m(\sigma) = \min \sigma$. The coarsening theorem Haine, Porta, and Teyssier (2026, Theorem 3.3.5) identifies the new exit category with the localization of $\mathrm{Ch}(P)$ at inclusions preserving the minimum. Its hypotheses hold for this finite conically stratified polyhedron, by Lurie (2017, Theorems A.6.4 and A.9.3). Finally $N\,\mathrm{Ch}(P) = \mathrm{Sd}(NP^{\mathrm{op}})$, and Nm is the last-vertex map for P^{op}. Lurie (2026, Proposition 6.3.6.2) says exactly that this map is universally localizing at those edges. Thus the localized exit category is P^{op}.

Space-valued exodromy is now HA, Theorem A.9.3. The coefficient extension in the precise two classes stated above is Haine et al. (2026, Corollary 4.1.15); the finite indexing poset is noetherian, as required for its stable-coefficient case. Finally sheaves on a finite downset Alexandrov space are diagrams on P^{op}, by finite gluing on principal opens, and the map on exits induced by s is the just computed equivalence. Naturality of exodromy therefore identifies s^* with (G.10). This verifies the specific dictionary needed here and does not assert its arbitrary-presentable-coefficient extension. □

G.11. [Refuted] Minimum constructibility is not automatically preserved by Verdier duality. Even for $P = [1]$, usual contravariant Verdier duality on $|NP|$ does not preserve the least-vertex constructible sheaves with perfect stalks. Also, the canonical covariant transform V_P in G.9 is not an equivalence for this P.

Proof. The strata of $[0,1]$ are $[0,1)$ and $\{1\}$. The constant sheaf k is constructible for this stratification. Its Verdier dual is the dualizing sheaf of the closed interval. At the endpoint 0 the local costalk of k is

$$\mathrm{fib}\big(R\Gamma([0,\epsilon); k) \to R\Gamma((0,\epsilon); k)\big) = 0.$$

At an interior point the punctured interval has two components, and the corresponding restriction is the diagonal $k \to k \oplus k$. Its fibre is $k[-1]$. Dualizing these finite costalks shows that the stalk of the dualizing sheaf is zero at the endpoint and $k[1]$ at an interior point; the other endpoint is the same as 0. Equivalently it is $j_! k_{(0,1)}[1]$. It is not locally constant on $[0,1)$, so the proposed preservation fails even for a finite, compact, contractible realization. The local relative-cochain ranks are checked in `code/dg_finite_checks.py`.

Algebraically, for $F = (F_1 \to F_0)$, (G.9a) gives $V_{[1]}F = (0 \to F_1)$. The nonzero diagram $(0 \to k)$ is sent to zero. This agrees with Aoki's obstruction: the strict upper interval of 0 is a singleton, whose reduced homology is zero and is not the homology of any sphere. This refutes the claimed canonical identification on every homotopy finite base. It does not refute an equivalence obtained by other operations: [1] is isomorphic to its opposite, and G.14 gives further such examples. □

G.12. [Known] A correct geometric Verdier identification with a checked hypothesis. Suppose the finite poset P satisfies the upper-interval criterion of G.9. For $Z = |NP|$ and its minimum map $s : Z \to P$, the spectral transform has the identification

$$V_P \simeq s_+ \, \mathbb{V}_Z \, s^*. \tag{G.12}$$

Here $\mathbb{V}_Z$ is Lurie's covariant sheaf–cosheaf duality and s_+ is cosheaf pushforward, with value on U given by the value on $s^{-1}U$. The same identification holds for derived coefficients over a ring. Thus it includes an explicit pushforward of cosheaves; it does not require the ordinary sheaf-valued Verdier dual to stay constructible for the same minimum strata.

A concrete class satisfying the criterion is $P = \mathrm{Face}(K)^{\mathrm{op}}$ for any finite simplicial complex K, with nonempty faces. In this case the minimum strata on the barycentric subdivision are precisely the open simplices of K, so the identification is the usual cellular geometric one.

Proof. Use Aoki (2022, Theorem D, or Theorem 6.13 with Remark 6.14) with $Q = P^{\mathrm{op}}$. Its hypotheses are: Q is Verdier, the source topos is proper and separated, and the spectral inverse-image functor is fully faithful. G.9 verifies the first; the compact Hausdorff finite polyhedron Z verifies the second; G.10 verifies the third. Aoki uses the upsets-open convention on Q, which is the downsets-open convention on P. His identification is exactly (G.12), with these variances. The ring-coefficient assertion follows by tensoring the colimit-preserving spectral constructions with the module category; equivalently the same spectral kernels act on that stable coefficient category.

For the stated simplicial example and a simplex σ, the strict upper interval at σ in P is the opposite of the proper nonempty face poset of σ. Its order complex is the barycentric subdivision of the boundary sphere of σ. All augmented open intervals are boundaries of simplex faces, so the precise Gorenstein* condition holds, including an empty interval at a vertex. The usual barycentric homeomorphism $|N\,\mathrm{Face}(K)| \cong |K|$ sends a point with maximal face σ in its supporting chain to the relative interior of σ: its coefficient on the barycenter of σ is positive, giving positive coordinate at every vertex of σ. Conversely every point of that relative interior has that maximal face in its supporting barycentric simplex. Minimum in the opposite face poset is this maximum, proving the asserted match of strata. □

G.13. [Refuted] The weighted-limit argument does not persist unstably. There is no fixed-colimit-weight description of all finite limits of spaces. Consequently neither the stability argument of G.1 nor its duality conclusions may be exported to unstable fine theories merely because the base is finite.

Proof. Let S have two objects and only identity arrows. Its limit is the product. Put $F = (*, \varnothing)$ and $G = (\varnothing, *)$. Then

$$(\lim F) \amalg (\lim G) = \varnothing, \qquad \lim(F \amalg G) = *.$$

Thus this finite limit functor fails to preserve a binary coproduct. A weighted colimit with fixed weight preserves coproducts of the input, so it cannot be this functor. All four values are finite discrete spaces and their cardinalities are recorded by the finite-check program. Independently, D.3 gives a finite poset whose two unstable fine categories are not even equivalent. For a discrete finite base the two diagram categories do coincide by opposition, but their product and coproduct operations still differ as this example shows. □

G.14. [Known] Tilting gives additional equivalences with different data. If T is the path category of a finite oriented tree, then $\mathbb{D}(T) \simeq \mathbb{D}(T^{\mathrm{op}})$ for every stable derivator, pseudonaturally in exact morphisms of derivators. More generally any two orientations of the same finite tree are strongly stably equivalent. These are reflection-functor equivalences and are not the assertion that the transform V_T in G.9 is always an equivalence.

Proof. This is Groth and Šťovíček (2016, Theorem 9.19 and Corollary 9.23). The supplied source `tilting2.tex` was checked at the reflection theorem and its tree corollary. A source reflection replaces the value there by the cofiber of the map to the finite biproduct of its outgoing neighbours; the inverse uses a fibre. The published proof constructs these as coherent finite Kan extensions and proves inverse equivalences. Successive source and sink reflections realize any reorientation of a tree, including complete reversal. Only finite limits and colimits

occur, which explains the asserted compatibility with exact morphisms of stable derivators and, for spectra or rings, preservation of pointwise perfect diagrams. These operations depend on the chosen reflection sequence. In particular their existence does not supply a dimension, an orientation, or a pointwise dualizing-object formula such as the refuted (G.3). □

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